

Supplemental Notes for "Engineering Quantum Interactions: From Effective Models to Validated Physical Predictions"

Sinara S. Dourado^{1,†} , Ciro Micheletti Diniz¹ , Gabriel P. L. M. Fernandes¹,
Rogério J. de Assis¹, G. D. de Moraes Neto^{2,*} , and Celso J. Villas-Boas¹

¹Departamento de Física, Universidade Federal de São Carlos, 13565-905 São Carlos, São Paulo, Brazil

²Department of Fundamental Sciences, Hainan Bielefeld University of Applied Sciences, Danzhou, Hainan 578101, China

[†]sinara.dourado@estudante.ufscar.br ^{*}gdmneto@gmail.com

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Abstract

These notes develop calculations and conventions that complement the main article. They begin with exact changes of representation and the algebraic structures used by small rotations, then give a detailed comparison of seven derivations of the three-photon Rabi resonance. The open-system part connects quantum Langevin equations, Liouville-space propagation, and finite-time map tests. A guide to the companion notebooks, a glossary, and compact summary sheets provide routes for calculation, teaching, and numerical checks.

How to use these notes

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Section S1 is a primer on exact changes of frame and on the endpoint maps that return a transformed calculation to laboratory observables. Section S2 is a recognition guide for the polynomial algebras used by the method of small rotations. Section S3 is the longest worked example: the same three-photon resonance is derived by projection, Schrieffer–Wolff theory, small rotations, Magnus and James–Jerke constructions, adiabatic elimination, and Floquet theory. Section S4 collects the open-system conventions and shows how a damped mediator is eliminated in quantum Langevin and Liouville-space language. Section S5 describes the companion notebooks, Sec. S6 defines terms used across several methods, and Sec. S7 gathers the equations and diagnostics most often needed in practice.

We set $\hbar = 1$. Hats denote operators on Hilbert space. Calligraphic letters denote superoperators or maps unless their meaning is stated explicitly. The dissipator is

$$\mathcal{D}[\hat{L}]\hat{\rho} = \hat{L}\hat{\rho}\hat{L}^\dagger - \frac{1}{2}\left\{\hat{L}^\dagger\hat{L}, \hat{\rho}\right\}.$$

Rates may be written outside the dissipator or absorbed into a physical jump operator $\hat{J} =$

$\sqrt{\gamma}\hat{L}$.

S1 Unitary Representation Changes and Sequential Frames

Throughout the main text, we move between representations, or *frames*, in order to expose the dynamical structures on which effective-model constructions act. Examples include the interaction picture used in the rotating-wave approximation, the rotating frames used for driven systems, and the dressed frames employed in Schrieffer–Wolff, polaron, and small-rotation methods. These changes of representation are implemented by unitary transformations.

A unitary transformation is exact: by itself, it does not discard terms or reduce the Hilbert space. An approximation arises only when the transformed Hamiltonian is truncated, projected, averaged, or subjected to an additional timescale-separation argument. This distinction underlies the transformation–evolution–reconstruction framework introduced in the article. This section derives the basic transformation formulas and illustrates them through the interaction picture, a magnetic-resonance rotating frame, and a sequence of transformations used for Hamiltonian engineering.

S1.1 General theory of time-dependent unitary transformations

Consider a state obeying the Schrödinger equation

$$i\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}(t)|\psi(t)\rangle,$$

where we set $\hbar = 1$. We define the transformed state by

$$\begin{aligned} |\tilde{\psi}(t)\rangle &= \hat{U}^\dagger(t)|\psi(t)\rangle \\ &\iff \\ |\psi(t)\rangle &= \hat{U}(t)|\tilde{\psi}(t)\rangle, \end{aligned} \quad (\text{S1})$$

with

$$\hat{U}^\dagger(t)\hat{U}(t) = \hat{U}(t)\hat{U}^\dagger(t) = \hat{I}.$$

Taking the time derivative of Eq. (S1) gives

$$\begin{aligned} i\frac{\partial}{\partial t}|\tilde{\psi}(t)\rangle &= i\dot{\hat{U}}^\dagger(t)|\psi(t)\rangle + i\hat{U}^\dagger(t)\frac{\partial}{\partial t}|\psi(t)\rangle \\ &= i\dot{\hat{U}}^\dagger(t)\hat{U}(t)|\tilde{\psi}(t)\rangle \\ &\quad + \hat{U}^\dagger(t)\hat{H}(t)\hat{U}(t)|\tilde{\psi}(t)\rangle. \end{aligned}$$

The transformed state therefore obeys

$$i\frac{\partial}{\partial t}|\tilde{\psi}(t)\rangle = \tilde{\hat{H}}(t)|\tilde{\psi}(t)\rangle,$$

where

$$\begin{aligned} \tilde{\hat{H}}(t) &= \hat{U}^\dagger(t)\hat{H}(t)\hat{U}(t) + i\dot{\hat{U}}^\dagger(t)\hat{U}(t) \\ &= \hat{U}^\dagger(t)\hat{H}(t)\hat{U}(t) - i\hat{U}^\dagger(t)\dot{\hat{U}}(t). \end{aligned} \quad (\text{S2})$$

The equivalence of the two lines follows by differentiating the unitarity condition,

$$\dot{\hat{U}}^\dagger\hat{U} + \hat{U}^\dagger\dot{\hat{U}} = 0.$$

Equation (S2) is the transformation rule used throughout Supplemental Sec. S1. The first contribution is the original Hamiltonian represented in the new frame. The second is an inertial, or connection, term generated by the time dependence of the transformation itself [1]. For a time-independent unitary, $\dot{\hat{U}} = 0$, and the familiar similarity transformation $\tilde{\hat{H}} = \hat{U}^\dagger\hat{H}\hat{U}$ is recovered.

The tilde indicates that, at this stage, Eq. (S2) defines the exact Hamiltonian in another representation, not yet an effective Hamiltonian. An effective generator is obtained only after an additional approximation, such as neglecting rapidly oscillating components, truncating a perturbative series, or projecting onto a retained manifold.

S1.1.1 Density operators and observables

The density operator transforms according to

$$\tilde{\hat{\rho}}(t) = \hat{U}^\dagger(t)\hat{\rho}(t)\hat{U}(t),$$

while a laboratory observable $\hat{O}$ is represented in the transformed frame as

$$\tilde{\hat{O}}(t) = \hat{U}^\dagger(t)\hat{O}\hat{U}(t).$$

Their expectation value is independent of the representation,

$$\begin{aligned} \text{Tr}[\hat{O}\hat{\rho}(t)] &= \text{Tr}[\hat{U}^\dagger(t)\hat{O}\hat{U}(t)\hat{U}^\dagger(t)\hat{\rho}(t)\hat{U}(t)] \\ &= \text{Tr}[\tilde{\hat{O}}(t)\tilde{\hat{\rho}}(t)]. \end{aligned}$$

Thus, transforming the state while continuing to evaluate an untransformed operator generally defines a different quantity. This is the origin of the reconstruction requirement emphasized in the article’s common framework.

For an open system with jump operators $\hat{L}_k(t)$, the same exact change of frame gives

$$\tilde{\hat{L}}_k(t) = \hat{U}^\dagger(t) \hat{L}_k(t) \hat{U}(t).$$

Consequently, an exact unitary transformation must be applied to the Hamiltonian, states, observables, and dissipative channels consistently. Replacing only the Hamiltonian by its transformed expression generally does not describe the same physical open-system dynamics.

S1.1.2 Relation to light–matter transformations

The freedom to choose $\hat{U}(t)$ in Eq. (S2) is used throughout quantum optics. Beyond interaction and rotating frames, related transformations connect different representations of light–matter coupling. An important example is the Power–Zienau–Woolley transformation between minimal-coupling descriptions involving $\mathbf{p} \cdot \mathbf{A}$ and multipolar descriptions involving $\mathbf{d} \cdot \mathbf{E}$. Before Hilbert-space truncation, such representations are unitarily related. After a finite-level truncation, however, the transformation, retained self-energy terms, and observables must be handled consistently, as discussed in the main text.

Santos *et al.* [2] developed a systematic time-dependent transformation approach for molecular QED in which $\hat{U}(t)$ is chosen using the evolution operator associated with a selected subsystem, while part of the radiation field is treated as an environment. An appropriately designed transformation can reorganize otherwise complicated higher-order light–matter processes into simpler interaction structures in the transformed representation. The physical content of such a description nevertheless depends on the subsequent approximation and on a consistent reconstruction of states and observables.

S1.2 The interaction, or Dyson, picture

Suppose that the Hamiltonian can be separated as

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t),$$

where $\hat{H}_0$ is exactly solvable and $\hat{H}_1(t)$ contains the remaining interaction or perturbation. The interaction picture factors the evolution generated by $\hat{H}_0$ out of the state, leaving the transformed dynamics governed by the interaction represented in that frame.

We choose

$$\hat{U}_0(t) = e^{-i\hat{H}_0 t}.$$

The first contribution to Eq. (S2) is

$$\begin{aligned} \hat{U}_0^\dagger(t) \hat{H}(t) \hat{U}_0(t) &= \hat{U}_0^\dagger(t) \hat{H}_0 \hat{U}_0(t) \\ &\quad + \hat{U}_0^\dagger(t) \hat{H}_1(t) \hat{U}_0(t) \\ &= \hat{H}_0 + \hat{H}_I(t), \end{aligned}$$

where

$$\hat{H}_I(t) = \hat{U}_0^\dagger(t) \hat{H}_1(t) \hat{U}_0(t).$$

The inertial contribution is

$$-i\hat{U}_0^\dagger(t) \dot{\hat{U}}_0(t) = -\hat{H}_0.$$

The two copies of $\hat{H}_0$ cancel, leaving

$$\tilde{\hat{H}}(t) = \hat{H}_I(t) = \hat{U}_0^\dagger(t) \hat{H}_1(t) \hat{U}_0(t). \quad (\text{S3})$$

Equation (S3) is an exact result. The interaction picture does not neglect $\hat{H}_0$ physically; its effect has been transferred into the time dependence of the transformed states and operators. Any subsequent rotating-wave, secular, or perturbative truncation is a separate approximation.

S1.2.1 Example: the quantum Rabi model

As discussed in the article, the Rabi Hamiltonian is

$$\hat{H}_{\text{RM}} = \omega_a \hat{a}^\dagger \hat{a} + \frac{\omega_0}{2} \hat{\sigma}_z + g \left(\hat{a} + \hat{a}^\dagger \right) \hat{\sigma}_x.$$

We choose

$$\begin{aligned} \hat{H}_0 &= \omega_a \hat{a}^\dagger \hat{a} + \frac{\omega_0}{2} \hat{\sigma}_z, \\ \hat{H}_1 &= g \left(\hat{a} + \hat{a}^\dagger \right) \hat{\sigma}_x, \end{aligned}$$

and

$$\hat{U}_0(t) = e^{-i\hat{H}_0 t}.$$

The Baker–Campbell–Hausdorff identity,

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!}[\hat{A}, [\hat{A}, \hat{B}]] + \dots,$$

gives

$$\begin{aligned}\hat{U}_0^\dagger(t)\hat{a}\hat{U}_0(t) &= \hat{a}e^{-i\omega_a t}, \\ \hat{U}_0^\dagger(t)\hat{a}^\dagger\hat{U}_0(t) &= \hat{a}^\dagger e^{i\omega_a t}, \\ \hat{U}_0^\dagger(t)\hat{\sigma}_\pm\hat{U}_0(t) &= \hat{\sigma}_\pm e^{\pm i\omega_0 t}.\end{aligned}$$

Using $\hat{\sigma}_x = \hat{\sigma}_+ + \hat{\sigma}_-$, the interaction-picture Hamiltonian becomes

$$\begin{aligned}\hat{H}_I(t) &= g \left[\hat{a}\hat{\sigma}_+ e^{i(\omega_0 - \omega_a)t} + \hat{a}^\dagger\hat{\sigma}_- e^{-i(\omega_0 - \omega_a)t} \right] \\ &\quad + g \left[\hat{a}\hat{\sigma}_- e^{-i(\omega_0 + \omega_a)t} + \hat{a}^\dagger\hat{\sigma}_+ e^{i(\omega_0 + \omega_a)t} \right].\end{aligned}\tag{S4}$$

With the article's notation, the first line oscillates at the detuning $\Delta = \omega_0 - \omega_a$, whereas the second oscillates at the sum frequency $\Sigma = \omega_0 + \omega_a$.

Equation (S4) is the exact Rabi interaction in the interaction picture. It becomes the starting point for the rotating-wave approximation only after the rapidly oscillating counter-rotating terms in the second line are neglected under the conditions stated in the article's rotating-wave analysis. The change of picture and the RWA must therefore be understood as two separate steps.

S1.3 The rotating-frame transformation

A rotating frame is particularly useful when a Hamiltonian contains an external drive with a well-defined carrier frequency. The transformation is chosen so that the reference frame rotates with the drive, converting its explicit oscillation into a static or slowly varying interaction.

The difference between an interaction picture and a rotating frame is primarily one of purpose rather than mathematical structure. Both are generated by Eq. (S2). In an interaction picture, $\hat{U}(t)$ is usually chosen from a solvable part of the Hamiltonian. In a rotating frame, it is commonly chosen from the drive frequency or a control-defined reference frequency.

S1.3.1 Example: magnetic resonance

A canonical example is a spin $\hat{\mathbf{I}}$ in a static magnetic field $B_0\hat{\mathbf{z}}$ and a transverse field of

magnitude B_1 rotating in the xy plane [1]. To avoid confusion with the light–matter coupling g used elsewhere in the Tutorial, we denote the spin gyromagnetic ratio by γ_s . The Hamiltonian is

$$\hat{H}(t) = -\gamma_s B_0 \hat{I}_z - \gamma_s B_1 \left(\hat{I}_x \cos \omega t + \hat{I}_y \sin \omega t \right).\tag{S5}$$

We choose a frame rotating around the z axis at angular frequency ω ,

$$\hat{U}_r(t) = e^{-i\omega t \hat{I}_z}.$$

The longitudinal operator is unchanged,

$$\hat{U}_r^\dagger(t)\hat{I}_z\hat{U}_r(t) = \hat{I}_z,$$

whereas the transverse operators transform as

$$\hat{U}_r^\dagger(t)\hat{I}_x\hat{U}_r(t) = \hat{I}_x \cos \omega t - \hat{I}_y \sin \omega t,\tag{S6}$$

$$\hat{U}_r^\dagger(t)\hat{I}_y\hat{U}_r(t) = \hat{I}_y \cos \omega t + \hat{I}_x \sin \omega t.\tag{S7}$$

Substituting Eqs. (S6) and (S7) into the transverse part gives

$$\begin{aligned}\hat{U}_r^\dagger(t) \left[\hat{I}_x \cos \omega t + \hat{I}_y \sin \omega t \right] \hat{U}_r(t) \\ = \hat{I}_x.\end{aligned}$$

Thus,

$$\hat{U}_r^\dagger(t)\hat{H}(t)\hat{U}_r(t) = -\gamma_s B_0 \hat{I}_z - \gamma_s B_1 \hat{I}_x.$$

The inertial term is

$$\begin{aligned}-i\hat{U}_r^\dagger(t)\dot{\hat{U}}_r(t) &= -i\hat{U}_r^\dagger(t) \left(-i\omega \hat{I}_z \hat{U}_r(t) \right) \\ &= -\omega \hat{I}_z.\end{aligned}$$

The rotating-frame Hamiltonian is therefore

$$\tilde{\hat{H}}_r = -(\gamma_s B_0 + \omega) \hat{I}_z - \gamma_s B_1 \hat{I}_x.\tag{S8}$$

For the sign convention in Eq. (S5), define the signed Larmor frequency $\omega_L = -\gamma_s B_0$ and the rotating-frame detuning $\delta = \omega - \omega_L$. Equation (S8) then becomes

$$\tilde{\hat{H}}_r = -\delta \hat{I}_z - \gamma_s B_1 \hat{I}_x.$$

At resonance, $\delta = 0$, and the spin evolves under a static transverse Hamiltonian. The explicit time dependence of the drive has disappeared because the description now rotates with the field. Alternative conventions for the sense of rotation or the sign of the gyromagnetic ratio produce the familiar equivalent form with $\delta = \omega_L - \omega$.

S1.4 Case study: sequential transformations for Hamiltonian engineering

In advanced quantum-control protocols, several transformations are often applied sequentially to isolate or engineer a particular interaction. The following example, adapted from Ref. [3], illustrates how two classical drives and a quantized cavity mode can be combined to produce a JC-like coupling in a doubly dressed atomic basis. This interaction is the coherent building block of a nonadiabatic, decoherence-free protocol based on reservoir engineering.

S1.4.1 Physical context

The broader objective of Ref. [3] is to engineer a reservoir that protects a nonstationary superposition of atomic electronic states from decoherence. Before constructing the dissipative protocol, one must engineer the time-dependent coherent interaction that drives the desired evolution. The target interaction has the structure

$$\hat{H}_{\text{target}} \propto \hat{a}^\dagger \hat{\sigma}_{\uparrow\downarrow} + \hat{a} \hat{\sigma}_{\downarrow\uparrow},$$

where $|\uparrow\rangle$ and $|\downarrow\rangle$ are dressed atomic states defined below.

S1.4.2 Initial Hamiltonian

The system consists of an atomic transition driven by two classical fields with Rabi frequencies Ω_1 and Ω_2 and coupled to a single quantized cavity mode with vacuum Rabi coupling g . In a standard interaction picture, the Hamiltonian is

$$\begin{aligned} \hat{H}_1(t) = & \left[g e^{-i\Delta_a t} \hat{a} + \Omega_1 e^{i(\phi_1 - \Delta_1 t)} \right. \\ & \left. + \Omega_2 e^{i(\phi_2 - \Delta_2 t)} \right] \hat{\sigma}_{eg} + \text{H.c.}, \end{aligned}$$

where

$$\hat{\sigma}_{eg} = |e\rangle\langle g|.$$

The states $|g\rangle$ and $|e\rangle$ are the bare atomic ground and excited states, while $\hat{a}$ and $\hat{a}^\dagger$ annihilate and create excitations of the quantized mode. The phases ϕ_1 and ϕ_2 determine the relative phases of the classical fields. Their detunings are $\Delta_1 = \omega_1 - \omega_0$, $\Delta_2 = \omega_2 - \omega_0$, where $\omega_{1,2}$ are the classical drive frequencies and ω_0 is the bare atomic transition frequency. The cavity detuning is denoted by Δ_a .

S1.4.3 First resonant drive and dressed basis

The first classical field is chosen to be resonant with the atomic transition, $\Delta_1 = 0$. The Hamiltonian then contains the time-independent drive

$$\hat{H}_{\Omega_1} = \Omega_1 \left(e^{i\phi_1} \hat{\sigma}_{eg} + e^{-i\phi_1} \hat{\sigma}_{ge} \right).$$

We move to the frame generated by this drive using

$$\hat{U}_1(t) = \exp \left[-i\hat{H}_{\Omega_1} t \right].$$

The eigenstates of $\hat{H}_{\Omega_1}$ are the dressed states

$$|\pm\rangle = \frac{1}{\sqrt{2}} \left(|e\rangle \pm e^{-i\phi_1} |g\rangle \right),$$

with an energy splitting proportional to $2\Omega_1$. In this first dressed frame, the second drive decomposes into components oscillating at frequencies shifted by the dressed-state splitting, $\Delta_2 \pm 2\Omega_1$.

S1.4.4 Second resonance and doubly dressed basis

A resonant contribution is selected by imposing $\Delta_2 = -2\Omega_1$, which makes one component of the second drive stationary in the first dressed frame [3]. Denoting

$$\hat{\sigma}_{+-} = |+\rangle\langle -|,$$

the second transformation is generated by

$$\hat{U}_2(t) = \exp \left[-i\frac{\Omega_2}{2} \left(e^{i\varphi} \hat{\sigma}_{+-} + e^{-i\varphi} \hat{\sigma}_{-+} \right) t \right],$$

where

$$\varphi = \phi_1 - \phi_2.$$

The resulting doubly dressed states are

$$|\uparrow\rangle = \frac{1}{\sqrt{2}} \left(|+\rangle + e^{-i\varphi} |-\rangle \right),$$

$$|\downarrow\rangle = \frac{1}{\sqrt{2}} \left(|+\rangle - e^{-i\varphi} |-\rangle \right).$$

S1.4.5 Cavity resonance and the final rotating-wave approximation

In the doubly dressed frame, the cavity interaction

$$g e^{-i\Delta_a t} \hat{a} \hat{\sigma}_{eg} + \text{H.c.}$$

decomposes into several terms oscillating at different combinations of Δ_a , Ω_1 , and Ω_2 . The resonance condition $\Delta_a = -\Omega_2$ makes one of these contributions stationary. A final rotating-wave approximation discards the remaining rapidly oscillating components, provided that their frequency separations are large compared with the corresponding transformed coupling matrix elements.

The resulting Hamiltonian in the doubly dressed frame is

$$\hat{H}_2 = \frac{g}{2} \left(e^{i\phi_1} \hat{a}^\dagger \hat{\sigma}_{\uparrow\downarrow} + e^{-i\phi_1} \hat{a} \hat{\sigma}_{\downarrow\uparrow} \right), \quad (\text{S9})$$

where

$$\hat{\sigma}_{\uparrow\downarrow} = |\uparrow\rangle\langle\downarrow|, \quad \hat{\sigma}_{\downarrow\uparrow} = |\downarrow\rangle\langle\uparrow|.$$

This is the target JC-like interaction in the final dressed basis [3].

S1.4.6 Composition and reconstruction of sequential frames

The two transformations must be applied in the correct order. Starting from the laboratory or initial interaction-picture state,

$$|\tilde{\psi}_1\rangle = \hat{U}_1^\dagger |\psi\rangle,$$

and then applying the second transformation gives

$$\begin{aligned} |\tilde{\psi}_2\rangle &= \hat{U}_2^\dagger |\tilde{\psi}_1\rangle \\ &= \hat{U}_2^\dagger \hat{U}_1^\dagger |\psi\rangle. \end{aligned}$$

The total unitary entering the convention of Eq. (S1) is therefore

$$\hat{U}_{\text{tot}}(t) = \hat{U}_1(t) \hat{U}_2(t),$$

because

$$|\tilde{\psi}_2\rangle = \hat{U}_{\text{tot}}^\dagger |\psi\rangle.$$

A state obtained from the engineered Hamiltonian Eq. (S9) is reconstructed as

$$\hat{\rho}(t) = \hat{U}_1(t) \hat{U}_2(t) \tilde{\hat{\rho}}_2(t) \hat{U}_2^\dagger(t) \hat{U}_1^\dagger(t).$$

Likewise, a laboratory observable is represented in the final frame by

$$\tilde{\hat{O}}_2(t) = \hat{U}_2^\dagger(t) \hat{U}_1^\dagger(t) \hat{O} \hat{U}_1(t) \hat{U}_2(t).$$

These relations are needed when the experimentally measured quantity is defined in the bare atomic basis rather than in the doubly dressed basis.

This case study illustrates the complete logic of Hamiltonian engineering. Exact transformations first reorganize the description, resonance conditions identify the desired slowly varying process, and a final rotating-wave approximation removes the remaining off-resonant terms. The resulting Hamiltonian is simple only in the doubly dressed frame; laboratory predictions require the inverse sequence of transformations. It therefore provides a concrete realization of the encoding–effective-evolution–reconstruction map introduced in the article.

Further reading. Cohen-Tannoudji, Dupont-Roc, and Grynberg give a quantum-optics treatment of interaction pictures, dressed states, and atom–field transformations [4]. Time-dependent effective transformations for light–matter models are developed in Ref. [2]. The sign of the derivative term depends on the direction in which the state is transformed, so it is safest to derive that term from the convention used in the calculation.

S2 Algebraic Structures for the Method of Small Rotations

The method of small rotations introduced in the article is most effective when the interaction Hamiltonian can be written using a small collection of ladder operators that close under commutation. The examples below form a representative catalog of such structures and, equally importantly, identify several cases that require a generalization of the simplest one-triplet construction.

The strict single-triplet MSR form is

$$\hat{H} = \hat{H}_c \left(\{\hat{N}_i\} \right) + \Delta \hat{X}_3 + g \left(\hat{X}_+ + \hat{X}_- \right),$$

with

$$\hat{X}_- = \hat{X}_+^\dagger, \quad (\text{S10a})$$

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm, \quad (\text{S10b})$$

$$[\hat{X}_+, \hat{X}_-] = P \left(\hat{X}_3, \{\hat{N}_i\} \right), \quad (\text{S10c})$$

$$[\hat{N}_i, \hat{X}_3] = [\hat{N}_i, \hat{X}_\pm] = 0. \quad (\text{S10d})$$

Here, P is a polynomial or another diagonal function of mutually commuting operators. The

first small rotation then produces

$$\begin{aligned}\tilde{H}^{(2)} &= \hat{H}_c(\{\hat{N}_i\}) + \Delta \hat{X}_3 \\ &+ \frac{g^2}{\Delta} P(\hat{X}_3, \{\hat{N}_i\}) + \mathcal{O}\left(\frac{g^3}{\Delta^2}\right).\end{aligned}$$

Within a fixed invariant sector, the commutator may be represented by a polynomial in $\hat{X}_3$, even when its expression in the original microscopic operators is less compact. The result must be diagonal in the same algebraic basis, and the successive commutators remain controllable.

S2.1 Practical checks for an MSR realization

Before applying the standard formula, five checks should be performed.

Hermiticity. For real g , the interaction must contain

$$g(\hat{X}_+ + \hat{X}_-),$$

whereas the anti-Hermitian combination $\hat{X}_+ - \hat{X}_-$ belongs in the rotation generator.

Normalization of $\hat{X}_3$. The coefficient multiplying $\hat{X}_3$ must be identified after enforcing

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm.$$

Changing the normalization of $\hat{X}_3$ changes the detuning appearing in the small parameter.

Closure. The commutator $[\hat{X}_+, \hat{X}_-]$ must be expressible using diagonal operators and integrals of motion. If it generates new off-diagonal operators, the single-triplet algebra is not closed.

Invariant sectors. The proposed $\hat{N}_i$ must commute with the complete interaction algebra. These operators determine the sectors in which the local matrix-element-to-detuning ratio is evaluated.

Perturbative control. The matrix-element condition is

$$\frac{|g| |\langle \nu | \hat{X}_+ + \hat{X}_- | \mu \rangle|}{|\Delta|} \ll 1$$

for all appreciably populated transitions.

S2.2 Exact and polynomially deformed realizations

S2.2.1 Two-mode beam-splitter interaction: $\mathfrak{su}(2)$

Consider two linearly coupled bosonic modes in a frame that removes their mean frequency,

$$\hat{H}_{\text{BS}} = \frac{\Delta}{2} (\hat{a}^\dagger \hat{a} - \hat{b}^\dagger \hat{b}) + g (\hat{a}^\dagger \hat{b} + \hat{b}^\dagger \hat{a}).$$

Define

$$\hat{X}_3 = \frac{1}{2} (\hat{a}^\dagger \hat{a} - \hat{b}^\dagger \hat{b}), \quad \hat{X}_+ = \hat{a}^\dagger \hat{b}, \quad \hat{X}_- = \hat{b}^\dagger \hat{a}.$$

Then

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm, \quad (\text{S11a})$$

$$[\hat{X}_+, \hat{X}_-] = 2\hat{X}_3. \quad (\text{S11b})$$

The total occupation

$$\hat{N} = \hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b}$$

is conserved. The second-order MSR Hamiltonian is

$$\tilde{H}_{\text{BS}}^{(2)} = \left(\Delta + \frac{2g^2}{\Delta} \right) \hat{X}_3 + \mathcal{O}\left(\frac{g^3}{\Delta^2}\right).$$

Because this is an undeformed $\mathfrak{su}(2)$ algebra, the Hamiltonian can also be diagonalized exactly by a finite beam-splitter rotation. The MSR expression is the small- g/Δ expansion of that exact rotation.

S2.2.2 Jaynes–Cummings model: bosonically deformed $\mathfrak{su}(2)$

For

$$\hat{H}_{\text{JC,int}} = \frac{\Delta}{2} \hat{\sigma}_z + g (\hat{a} \hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_-),$$

choose

$$\hat{X}_3 = \frac{\hat{\sigma}_z}{2}, \quad \hat{X}_+ = \hat{a} \hat{\sigma}_+, \quad \hat{X}_- = \hat{a}^\dagger \hat{\sigma}_-.$$

The excitation integral

$$\hat{N}_{\text{JC}} = \hat{a}^\dagger \hat{a} + \frac{\hat{\sigma}_z}{2}$$

commutes with the algebra. The commutation relations are

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm, \quad (\text{S12a})$$

$$[\hat{X}_+, \hat{X}_-] = \hat{a}^\dagger \hat{a} \hat{\sigma}_z + \hat{\sigma}_+ \hat{\sigma}_-. \quad (\text{S12b})$$

Consequently,

$$\begin{aligned}\tilde{H}_{\text{JC,int}}^{(2)} &= \frac{\Delta}{2} \hat{\sigma}_z \\ &+ \frac{g^2}{\Delta} \left(\hat{a}^\dagger \hat{a} \hat{\sigma}_z + \hat{\sigma}_+ \hat{\sigma}_- \right).\end{aligned}$$

This is the dispersive result derived explicitly in the article's Jaynes–Cummings example.

S2.2.3 Tavis–Cummings model: collective polynomial deformation

For N identical two-level systems coupled collectively to one cavity mode, consider

$$\hat{H}_{\text{TC,int}} = \Delta \hat{S}_z + g \left(\hat{a} \hat{S}_+ + \hat{a}^\dagger \hat{S}_- \right),$$

where

$$\hat{S}_\alpha = \frac{1}{2} \sum_{j=1}^N \hat{\sigma}_\alpha^{(j)}.$$

Choose

$$\hat{X}_3 = \hat{S}_z, \quad \hat{X}_+ = \hat{a} \hat{S}_+, \quad \hat{X}_- = \hat{a}^\dagger \hat{S}_-.$$

The operators

$$\hat{N}_{\text{TC}} = \hat{a}^\dagger \hat{a} + \hat{S}_z, \quad \hat{\mathbf{S}}^2$$

are conserved. The deformed commutator is

$$\begin{aligned}[\hat{X}_+, \hat{X}_-] &= 2\hat{a}^\dagger \hat{a} \hat{S}_z \\ &+ \hat{\mathbf{S}}^2 - \hat{S}_z^2 + \hat{S}_z.\end{aligned}$$

It is diagonal in the joint basis of $\hat{a}^\dagger \hat{a}$, $\hat{S}_z$, and $\hat{\mathbf{S}}^2$. The second-order collective dispersive Hamiltonian is therefore

$$\begin{aligned}\tilde{H}_{\text{TC,int}}^{(2)} &= \Delta \hat{S}_z \\ &+ \frac{g^2}{\Delta} \left(2\hat{a}^\dagger \hat{a} \hat{S}_z + \hat{\mathbf{S}}^2 - \hat{S}_z^2 + \hat{S}_z \right).\end{aligned}$$

The matrix elements of $\hat{a} \hat{S}_+$ grow with both photon number and collective spin, so the validity condition must be imposed separately in each populated Dicke manifold.

S2.2.4 Two-mode parametric amplifier: $\mathfrak{su}(1, 1)$

Consider

$$\hat{H}_{\text{PA}} = \frac{\Delta}{2} \left(\hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b} + 1 \right) + g \left(\hat{a}^\dagger \hat{b}^\dagger + \hat{a} \hat{b} \right).$$

Define

$$\begin{aligned}\hat{X}_3 &= \frac{1}{2} \left(\hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b} + 1 \right), \\ \hat{X}_+ &= \hat{a}^\dagger \hat{b}^\dagger, \\ \hat{X}_- &= \hat{a} \hat{b}.\end{aligned}$$

Then

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm, \quad (\text{S13a})$$

$$[\hat{X}_+, \hat{X}_-] = -2\hat{X}_3. \quad (\text{S13b})$$

The number difference

$$\hat{N}_- = \hat{a}^\dagger \hat{a} - \hat{b}^\dagger \hat{b}$$

is conserved. The second-order result is

$$\tilde{H}_{\text{PA}}^{(2)} = \left(\Delta - \frac{2g^2}{\Delta} \right) \hat{X}_3 + \mathcal{O} \left(\frac{g^3}{\Delta^2} \right).$$

This undeformed $\mathfrak{su}(1, 1)$ problem can also be diagonalized by an exact two-mode squeezing transformation in its dynamically stable regime. The small rotation gives the perturbative expansion of that finite squeezing transformation.

S2.2.5 Three-wave mixing: quadratic polynomial deformation

Consider the three-mode interaction

$$\hat{H}_{3\text{wm}} = \Delta \hat{X}_3 + g \left(\hat{a}^\dagger \hat{b} \hat{c} + \hat{a} \hat{b}^\dagger \hat{c}^\dagger \right),$$

with

$$\begin{aligned}\hat{X}_3 &= \frac{1}{2} \left(\hat{a}^\dagger \hat{a} - \hat{b}^\dagger \hat{b} \right), \\ \hat{X}_+ &= \hat{a}^\dagger \hat{b} \hat{c}, \\ \hat{X}_- &= \hat{a} \hat{b}^\dagger \hat{c}^\dagger.\end{aligned}$$

Two independent Manley–Rowe invariants are

$$\hat{N}_1 = \hat{a}^\dagger \hat{a} + \hat{b}^\dagger \hat{b}, \quad \hat{N}_2 = \hat{a}^\dagger \hat{a} + \hat{c}^\dagger \hat{c}.$$

The ladder relation is

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm,$$

and direct evaluation gives

$$\begin{aligned}[\hat{X}_+, \hat{X}_-] &= -3\hat{X}_3^2 \\ &+ \left(2\hat{N}_2 - \hat{N}_1 + 1 \right) \hat{X}_3 \\ &+ \frac{\hat{N}_1^2}{4} + \frac{\hat{N}_1}{2}.\end{aligned} \quad (\text{S14})$$

This is a quadratic deformation of $\mathfrak{su}(2)$. The corresponding second-order diagonal correction is obtained by multiplying the right-hand side of Eq. (S14) by g^2/Δ .

Different choices of $\hat{X}_3$ are possible, but they need not produce an equally simple polynomial. The usefulness of an algebraic representation depends partly on selecting generators adapted to the available invariants.

S2.2.6 Detuned spin exchange: bond $\mathfrak{su}(2)$

For two spins with a frequency difference, write

$$\hat{H}_{\text{ex}} = \frac{\delta}{4} \left(\hat{\sigma}_z^{(1)} - \hat{\sigma}_z^{(2)} \right) + J \left(\hat{\sigma}_+^{(1)} \hat{\sigma}_-^{(2)} + \hat{\sigma}_-^{(1)} \hat{\sigma}_+^{(2)} \right).$$

Define

$$\hat{X}_3 = \frac{1}{4} \left(\hat{\sigma}_z^{(1)} - \hat{\sigma}_z^{(2)} \right),$$

and

$$\hat{X}_+ = \hat{\sigma}_+^{(1)} \hat{\sigma}_-^{(2)}, \quad \hat{X}_- = \hat{\sigma}_-^{(1)} \hat{\sigma}_+^{(2)}.$$

Then

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm, \quad (\text{S15a})$$

$$[\hat{X}_+, \hat{X}_-] = 2\hat{X}_3. \quad (\text{S15b})$$

The total magnetization

$$\hat{M}_z = \frac{1}{2} \left(\hat{\sigma}_z^{(1)} + \hat{\sigma}_z^{(2)} \right)$$

is conserved.

The detuning relevant to the small rotation is the local frequency difference δ , not a spatially uniform Zeeman field. A uniform term proportional to $\hat{\sigma}_z^{(1)} + \hat{\sigma}_z^{(2)}$ commutes with the exchange operator and belongs in $\hat{H}_c$.

For a chain with nonuniform local frequencies, similar bond rotations may be applied perturbatively. Rotations on overlapping bonds do not generally commute, however, and their nested commutators generate longer-range exchange and multispin operators. The full chain is therefore not simply a collection of independent two-spin $\mathfrak{su}(2)$ algebras.

S2.3 Multigenerator and higher-rank structures

S2.3.1 Three-level Λ system: deformed $\mathfrak{su}(3)$

Consider a three-level atom with lower states $|g\rangle$ and $|f\rangle$, an excited state $|e\rangle$, and two quantized

fields. For a common one-photon detuning, a convenient rotating frame gives

$$\begin{aligned} \hat{H}_\Lambda &= \Delta \hat{\sigma}_{ee} \\ &+ g_1 \left(\hat{a}_1 \hat{\sigma}_{eg} + \hat{a}_1^\dagger \hat{\sigma}_{ge} \right) \\ &+ g_2 \left(\hat{a}_2 \hat{\sigma}_{ef} + \hat{a}_2^\dagger \hat{\sigma}_{fe} \right). \end{aligned}$$

Each optical transition defines a JC-like triplet,

$$\hat{X}_+^{(g)} = \hat{a}_1 \hat{\sigma}_{eg}, \quad \hat{X}_-^{(g)} = \hat{a}_1^\dagger \hat{\sigma}_{ge}, \quad (\text{S16a})$$

$$\hat{X}_+^{(f)} = \hat{a}_2 \hat{\sigma}_{ef}, \quad \hat{X}_-^{(f)} = \hat{a}_2^\dagger \hat{\sigma}_{fe}, \quad (\text{S16b})$$

with

$$\hat{X}_3^{(g)} = \frac{\hat{\sigma}_{ee} - \hat{\sigma}_{gg}}{2}, \quad \hat{X}_3^{(f)} = \frac{\hat{\sigma}_{ee} - \hat{\sigma}_{ff}}{2}.$$

Within each transition,

$$[\hat{X}_3^{(k)}, \hat{X}_\pm^{(k)}] = \pm \hat{X}_\pm^{(k)}.$$

The individual deformation is

$$[\hat{X}_+^{(k)}, \hat{X}_-^{(k)}] = 2\hat{a}_k^\dagger \hat{a}_k \hat{X}_3^{(k)} + \hat{\sigma}_{ee}.$$

The two triplets are not independent. Their cross-commutators generate the Raman transition operators,

$$[\hat{X}_-^{(g)}, \hat{X}_+^{(f)}] = \hat{a}_1^\dagger \hat{a}_2 \hat{\sigma}_{gf}, \quad (\text{S17a})$$

$$[\hat{X}_-^{(f)}, \hat{X}_+^{(g)}] = \hat{a}_2^\dagger \hat{a}_1 \hat{\sigma}_{fg}. \quad (\text{S17b})$$

These new operators are part of the higher-rank $\mathfrak{su}(3)$ closure. The system therefore requires a multigenerator or successive-rotation construction rather than two independent applications of the single-triplet formula.

For $|\Delta| \gg |g_k| \sqrt{n_k + 1}$, elimination of the excited state gives the ground-manifold Hamiltonian

$$\begin{aligned} \hat{H}_{\Lambda, \text{eff}} &= -\frac{g_1^2}{\Delta} \hat{a}_1^\dagger \hat{a}_1 \hat{\sigma}_{gg} \\ &- \frac{g_2^2}{\Delta} \hat{a}_2^\dagger \hat{a}_2 \hat{\sigma}_{ff} \\ &- \frac{g_1 g_2}{\Delta} \left(\hat{a}_1^\dagger \hat{a}_2 \hat{\sigma}_{gf} + \hat{a}_2^\dagger \hat{a}_1 \hat{\sigma}_{fg} \right) \end{aligned}$$

up to the chosen rotating-frame energies and higher-order corrections. The first two terms are AC Stark shifts, and the last line is the effective Raman coupling. Unequal one-photon detunings require the appropriate symmetrized denominators and cannot be represented by a single scalar Δ .

S2.4 Generalized cases and boundaries of the simple construction

S2.4.1 Kerr-shifted Jaynes–Cummings interaction

Consider a diagonal Hamiltonian

$$\hat{H}_d = \frac{\Delta}{2} \hat{\sigma}_z + \frac{K}{2} \hat{n} (\hat{n} - 1), \quad \hat{n} = \hat{a}^\dagger \hat{a},$$

and the transition operator

$$\hat{X}_+ = \hat{a} \hat{\sigma}_+.$$

The commutator is

$$[\hat{H}_d, \hat{X}_+] = \hat{X}_+ [\Delta - K (\hat{n} - 1)].$$

The transition energy is therefore occupation dependent. The simple constant-angle generator $(g/\Delta)(\hat{X}_- - \hat{X}_+)$ does not cancel the interaction uniformly over the Hilbert space.

A generalized rotation can instead use the operator-valued denominator

$$[\Delta - K (\hat{n} - 1)]^{-1},$$

with the ordering rules discussed in the article. This case is naturally handled by the operator-basis Schrieffer–Wolff construction rather than by the strict single-parameter MSR formula. Occupations at which the denominator approaches zero must be retained as resonant sectors.

S2.4.2 Intensity-dependent transition operators

Let

$$\hat{X}_+ = \hat{f} (\hat{a}, \hat{a}^\dagger) \hat{\sigma}_+, \quad \hat{X}_- = \hat{f}^\dagger (\hat{a}, \hat{a}^\dagger) \hat{\sigma}_-,$$

and

$$\hat{X}_3 = \frac{\hat{\sigma}_z}{2}.$$

The ladder relation

$$[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm$$

holds because the bosonic and spin operators commute. However,

$$[\hat{X}_+, \hat{X}_-] = \hat{f} \hat{f}^\dagger \hat{\sigma}_+ \hat{\sigma}_- - \hat{f}^\dagger \hat{f} \hat{\sigma}_- \hat{\sigma}_+.$$

This commutator is diagonal only when $\hat{f} \hat{f}^\dagger$ and $\hat{f}^\dagger \hat{f}$ are functions of a common set of commuting number operators or invariants.

For example,

$$\hat{f} = \hat{a}^k F(\hat{n})$$

normally produces a number-diagonal deformation and can fit the MSR framework after the number-shift identities are applied. A completely arbitrary function of $\hat{a}$ and $\hat{a}^\dagger$ need not close and should not be classified automatically as an MSR-solvable interaction.

S2.4.3 Many-body lattices

A local bond interaction can possess an exact $\mathfrak{su}(2)$ or polynomially deformed algebra even when the full many-body Hamiltonian does not. For a lattice containing overlapping bonds, the generators associated with neighboring bonds generally fail to commute. Their nested commutators can generate next-nearest-neighbor, density-assisted, or multispin interactions.

MSR may still be useful in such systems, but the algebra must be enlarged to include the generated operators, or the calculation must be organized as a many-body Schrieffer–Wolff expansion. A successful two-site realization is therefore evidence for a local building block, not proof that the complete lattice closes under the same three generators.

S2.5 Practical recognition guide

A Hamiltonian is a strong candidate for the standard method of small rotations when the following statements are all true:

1. The interaction can be written as $g(\hat{X}_+ + \hat{X}_-)$ with $\hat{X}_- = \hat{X}_+^\dagger$.
2. A diagonal operator $\hat{X}_3$ can be normalized so that $[\hat{X}_3, \hat{X}_\pm] = \pm \hat{X}_\pm$.
3. The commutator $[\hat{X}_+, \hat{X}_-]$ is diagonal in a set of mutually commuting operators.
4. The integrals of motion decompose the Hilbert space into manageable invariant sectors.
5. The coupling matrix elements are small relative to the detuning throughout every populated sector.
6. The successive commutators generated by the rotation remain simpler than a direct matrix or numerical calculation.

Failure of one of these conditions does not imply that no effective Hamiltonian exists. It indicates instead that the simple single-rotation MSR formula should be replaced by a multigenerator construction, an operator-valued rotation, a general Schrieffer–Wolff transformation, a projection method, or direct numerical block diagonalization.

Algebraic scope of the method of small rotations

The strict MSR template requires more than the visual appearance of a diagonal term plus a pair of ladder operators. The normalization of $\hat{X}_3$, Hermiticity of the interaction, conserved-sector structure, closure of $[\hat{X}_+, \hat{X}_-]$, and occupation-dependent perturbative ratio must all be verified. Ordinary $\mathfrak{su}(2)$ and $\mathfrak{su}(1,1)$ examples admit exact finite rotations, polynomial deformations admit perturbative nonlinear rotations, and multilevel or occupation-dependent systems generally require enlarged or operator-valued generators.

Further reading. The original small-rotation construction and its systematic extension are given in Refs. [5–7]. Karassiov’s work develops the polynomial Lie-algebra structures that underlie many of the examples in this section [8–10]. For a broader tutorial on organizing quantum dynamics through Lie-algebra closure, see Ref. [11]; that method is related in spirit but is not the origin of MSR.

S3 Three-Photon Resonance Across Methods: Derivations and Reconstruction

The article develops the physical mechanism of the three-photon interaction and uses it to compare the methods. Here the emphasis is on the algebra: projection theory, Schrieffer–Wolff block diagonalization, the method of small rotations, the Magnus expansion, James–Jerke temporal coarse graining, systematic adiabatic elimination, and Floquet–Sambe perturbation theory are carried to the same retained-space invariants and reconstruction maps.

The derivations retain the intermediate

denominators, generators, commutators, resolvent expansions, model-space metrics, kick operators, slaving matrices, and Floquet sideband formulas. The static three-photon resonance and its cubic scaling were previously analyzed in Ref. [12]; the derivations below use the notation, retained-space convention, and reconstruction framework adopted throughout the article.

Readers primarily interested in the physical result can begin with Secs. S3.1 and S3.2, then read the final alignment in Sec. S3.10. The seven method-specific derivations in between are independent routes through the same retained two-state problem.

S3.1 Model, retained space, and conventions

We consider the quantum Rabi Hamiltonian

$$\hat{H}_R = \hat{H}_0 + \hat{V},$$

with

$$\hat{H}_0 = \omega_a \hat{a}^\dagger \hat{a} + \frac{\omega_0}{2} \hat{\sigma}_z, \quad (\text{S18a})$$

$$\hat{V} = g \left(\hat{a} + \hat{a}^\dagger \right) (\hat{\sigma}_+ + \hat{\sigma}_-). \quad (\text{S18b})$$

The detuning and sum frequency are $\Delta = \omega_0 - \omega_a$, $\Sigma = \omega_0 + \omega_a$. The unperturbed energies are

$$E_{e,n}^{(0)} = n\omega_a + \frac{\omega_0}{2}, \quad E_{g,n}^{(0)} = n\omega_a - \frac{\omega_0}{2}.$$

Near the bare three-photon condition $\omega_0 \simeq 3\omega_a$, the relevant pair is

$$|A\rangle \equiv |e, 0\rangle, \quad |B\rangle \equiv |g, 3\rangle,$$

with unperturbed mismatch $\delta_0 \equiv E_A^{(0)} - E_B^{(0)} = \omega_0 - 3\omega_a$. The retained and eliminated projectors are

$$\hat{P}_3 = |A\rangle\langle A| + |B\rangle\langle B|, \quad \hat{Q}_3 = \hat{I} - \hat{P}_3.$$

The Rabi parity operator is

$$\hat{\Pi} = \exp \left[i\pi \left(\hat{a}^\dagger \hat{a} + \hat{\sigma}_+ \hat{\sigma}_- \right) \right].$$

Its action on the bare basis is

$$\hat{\Pi} |e, n\rangle = (-1)^{n+1} |e, n\rangle,$$

$$\hat{\Pi} |g, n\rangle = (-1)^n |g, n\rangle.$$

Consequently,

$$\Pi_A = \Pi_B = -1,$$

so the target states may hybridize without breaking the exact parity symmetry. By contrast, $|e, 0\rangle$ and $|g, 4\rangle$ have opposite parities and cannot be connected by a parity-preserving effective Hamiltonian.

The unique shortest virtual chain connecting the retained states is

$$|A\rangle \xrightarrow{g} |g, 1\rangle \xrightarrow{\sqrt{2}g} |e, 2\rangle \xrightarrow{\sqrt{3}g} |B\rangle. \quad (\text{S19})$$

The middle vertex is generated by $\hat{a}^\dagger \hat{\sigma}_+$, and is therefore removed by the ordinary rotating-wave approximation. The same RWA that accurately describes near-resonant one-photon exchange thus gives $G_3^{\text{RWA}} = 0$ for the target process. The counter-rotating interaction is necessary for the third-order coupling to exist.

The nonzero matrix elements required through third order are

$$\langle g, 1 | \hat{V} | A \rangle = g, \quad (\text{S20a})$$

$$\langle e, 2 | \hat{V} | g, 1 \rangle = \sqrt{2}g, \quad (\text{S20b})$$

$$\langle B | \hat{V} | e, 2 \rangle = \sqrt{3}g, \quad (\text{S20c})$$

$$\langle e, 4 | \hat{V} | B \rangle = 2g. \quad (\text{S20d})$$

The last matrix element does not belong to the three-step path, but it contributes to the second-order shift of $|B\rangle$ and must therefore be retained in an order-consistent resonance condition. The corresponding energy differences are

$$E_A^{(0)} - E_{g,1}^{(0)} = \Delta, \quad (\text{S21a})$$

$$E_A^{(0)} - E_{e,2}^{(0)} = -2\omega_a, \quad (\text{S21b})$$

$$E_B^{(0)} - E_{e,2}^{(0)} = -\Delta, \quad (\text{S21c})$$

$$E_B^{(0)} - E_{g,1}^{(0)} = 2\omega_a, \quad (\text{S21d})$$

$$E_B^{(0)} - E_{e,4}^{(0)} = -\Sigma. \quad (\text{S21e})$$

Introduce a formal bookkeeping parameter $g \mapsto \lambda g$. The resonant perturbative organization is

$$\delta_0 = \lambda^2 \delta_0^{(2)} + \lambda^3 \delta_0^{(3)} + \dots, \quad (\text{S22})$$

because the bare mismatch must first compensate the second-order level shifts and must then be tuned within the third-order avoided-crossing window. Equivalently, in dimensional form,

$$\delta_0 = \mathcal{O}\left(\frac{g^2}{\omega_a}\right), \quad \delta_3 = \mathcal{O}\left(\frac{g^3}{\omega_a^2}\right) \quad (\text{S23})$$

near the resolved resonance.

Define pseudospin operators in the retained space by

$$\begin{aligned} \hat{\tau}_z &= |A\rangle\langle A| - |B\rangle\langle B|, \\ \hat{\tau}_+ &= |B\rangle\langle A|, \quad \hat{\tau}_- = |A\rangle\langle B|. \end{aligned}$$

All method-specific effective Hamiltonians are aligned with the canonical retained-space form

$$\hat{H}_{\text{eff},3} = \bar{E}_3 \hat{I}_3 + \frac{\delta_3}{2} \hat{\tau}_z + G_3 \hat{\tau}_+ + G_3^* \hat{\tau}_- + \dots. \quad (\text{S24})$$

Our convention is therefore $G_3 = \langle B | \hat{H}_{\text{eff},3} | A \rangle$. The instantaneous retained-space splitting is

$$\Omega_3 = \sqrt{\delta_3^2 + 4|G_3|^2},$$

and the minimum avoided-crossing gap is $\Delta_{\text{gap}}^{(3\gamma)} = 2|G_3|$. The matrix element $|G_3|$, the avoided-crossing gap $2|G_3|$, and the population-oscillation period $T_{\text{pop}} = \frac{\pi}{|G_3|}$ are therefore distinct quantities. This convention removes the factor-of-two ambiguity that arises when the matrix element itself is called the ‘‘Rabi frequency.’’

S3.2 Reference quasi-degenerate perturbative derivation

We first derive the canonical result directly from quasi-degenerate perturbation theory, without using the method-specific operator organizations introduced later. It fixes the physical invariants to be matched after the retained-space gauge and reconstruction convention are aligned.

S3.2.1 Second-order diagonal shifts

Because $\hat{P}_3 \hat{V} \hat{P}_3 = 0$, there is no first-order retained-space term. The second-order shift of $|A\rangle$ receives one contribution,

$$\begin{aligned} \delta E_A^{(2)} &= \frac{|\langle g, 1 | \hat{V} | A \rangle|^2}{E_A^{(0)} - E_{g,1}^{(0)}} \\ &= \frac{g^2}{\Delta}. \end{aligned} \quad (\text{S25})$$

The state $|B\rangle$ couples virtually to both $|e, 2\rangle$ and $|e, 4\rangle$, so

$$\begin{aligned} \delta E_B^{(2)} &= \frac{|\langle e, 2 | \hat{V} | B \rangle|^2}{E_B^{(0)} - E_{e,2}^{(0)}} \\ &\quad + \frac{|\langle e, 4 | \hat{V} | B \rangle|^2}{E_B^{(0)} - E_{e,4}^{(0)}} \\ &= -\frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma}. \end{aligned} \quad (\text{S26})$$

There is no second-order off-diagonal matrix element between $|A\rangle$ and $|B\rangle$: each Rabi vertex flips the qubit, so an even number of vertices cannot connect an excited-qubit state to a ground-qubit state.

The retained-space mean energy through second order is

$$\begin{aligned}\bar{E}_3 &= \frac{1}{2} [E_A^{(0)} + E_B^{(0)} + \delta E_A^{(2)} + \delta E_B^{(2)}] \\ &= \frac{3\omega_a}{2} - \frac{g^2}{\Delta} - \frac{2g^2}{\Sigma} + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right).\end{aligned}$$

More importantly, the effective detuning is

$$\begin{aligned}\delta_3 &= [E_A^{(0)} + \delta E_A^{(2)}] - [E_B^{(0)} + \delta E_B^{(2)}] + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right) \\ &= \omega_0 - 3\omega_a + 4g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma}\right) + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right).\end{aligned}\tag{S27}$$

The leading corrected resonance follows from $\delta_3 = 0$. At fixed ω_0 , write

$$\omega_a^{\text{res}} = \frac{\omega_0}{3} + c \frac{g^2}{\omega_0} + \mathcal{O}\left(\frac{g^4}{\omega_0^3}\right).$$

Substitution into Eq. (S27) gives $c = 3$, and therefore

$$\omega_a^{\text{res}} = \frac{\omega_0}{3} + \frac{3g^2}{\omega_0} + \mathcal{O}\left(\frac{g^4}{\omega_0^3}\right).$$

The resonance displacement has the frequency scale $\mathcal{O}(g^2/\omega_0)$. It appears one perturbative order earlier than, and is parametrically larger than, the target off-diagonal scale $\mathcal{O}(g^3/\omega_0^2)$. Using only the bare condition $\omega_0 = 3\omega_a$ while retaining the third-order interaction is therefore not an order-consistent approximation.

S3.2.2 Third-order resonant coupling

At third order the unique retained-space path is the chain in Eq. (S19). Since $\hat{P}_3 \hat{V} \hat{P}_3 = 0$, the folded terms proportional to a first-order retained-space interaction vanish. The off-diagonal matrix element is

$$\begin{aligned}G_3 &= \frac{\langle B | \hat{V} | e, 2 \rangle \langle e, 2 | \hat{V} | g, 1 \rangle \langle g, 1 | \hat{V} | A \rangle}{(E_A^{(0)} - E_{e,2}^{(0)}) (E_A^{(0)} - E_{g,1}^{(0)})} \\ &\quad + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right) \\ &= -\frac{\sqrt{6}g^3}{2\omega_a\Delta} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right).\end{aligned}\tag{S28}$$

Using the denominators referenced to $E_B^{(0)}$ gives the same result,

$$(E_B^{(0)} - E_{e,2}^{(0)}) (E_B^{(0)} - E_{g,1}^{(0)}) = -2\omega_a\Delta,$$

so the leading matrix element is Hermitian without an additional symmetrization. More generally, replacing the two quasi-degenerate reference energies by a common energy changes Eq. (S28) only by $\mathcal{O}(g^5/\omega_a^4)$ under the power counting of Eq. (S22).

At the leading three-photon resonance, $\Delta = 2\omega_a + \mathcal{O}(g^2/\omega_a)$, and hence

$$G_3 = -\frac{\sqrt{6}g^3}{4\omega_a^2} = -\frac{9\sqrt{6}g^3}{4\omega_0^2} + \mathcal{O}\left(\frac{g^5}{\omega_0^4}\right).\tag{S29}$$

With the real bare-state phase convention used here, the coupling is negative. The redefinition $|B\rangle \mapsto -|B\rangle$ reverses its sign, so the invariant predictions are $|G_3|$, the gap $2|G_3|$, and the reconstructed physical dynamics.

Within the retained manifold, the same result can be represented by

$$\begin{aligned}\hat{P}_3 \hat{H}_{\text{eff}}^{(3)} \hat{P}_3 &= -\frac{g^3}{2\omega_a\Delta} \hat{P}_3 \left[(\hat{a}^\dagger)^3 \hat{\sigma}_- + \hat{a}^3 \hat{\sigma}_+ \right] \hat{P}_3 \\ &\quad + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right).\end{aligned}$$

At $\omega_0 = 3\omega_a$, the operator coefficient is therefore $-g^3/(4\omega_a^2) = -9g^3/(4\omega_0^2)$. The factor $\sqrt{6}$ in Eq. (S29) is the matrix element of $(\hat{a}^\dagger)^3$ between $|0\rangle$ and $|3\rangle$.

Combining the second- and third-order terms gives the reference effective Hamiltonian

$$\begin{aligned}\hat{H}_{\text{eff},3}^{\text{ref}} &= \bar{E}_3 \hat{I}_3 + \frac{1}{2} \left[\omega_0 - 3\omega_a + 4g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) \right] \hat{\tau}_z \\ &\quad - \frac{\sqrt{6}g^3}{2\omega_a\Delta} (\hat{\tau}_+ + \hat{\tau}_-) \\ &\quad + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right)_{\text{diag}} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right)_{\text{off}}.\end{aligned}\tag{S30}$$

The order labels on the remainder distinguish the next diagonal and off-diagonal corrections. Even numbers of Rabi vertices preserve the qubit state, whereas odd numbers flip it; consequently, the next correction to the diagonal detuning appears at fourth order and the next correction to the three-photon matrix element appears at fifth order.

For an initial retained-space state $|A\rangle$, the transition probability generated by Eq. (S30) is

$$P_B^{\text{eff}}(t) = \frac{4|G_3|^2}{\delta_3^2 + 4|G_3|^2} \times \sin^2 \left[\frac{t}{2} \sqrt{\delta_3^2 + 4|G_3|^2} \right].$$

At $\delta_3 = 0$, this reduces to $P_B^{\text{eff}}(t) = \sin^2(|G_3|t)$. A necessary resonance-calibration condition for large-amplitude transfer is therefore

$$|\delta_3| \lesssim |G_3|.$$

This condition makes the hierarchy of orders operational: a residual resonance-position error of $\mathcal{O}(g^2/\omega_0)$ overwhelms an avoided crossing of $\mathcal{O}(g^3/\omega_0^2)$.

S3.2.3 Leading reconstruction and virtual populations

The two-state Hamiltonian describes the slow resonant amplitudes, but the physical states also contain virtual components in $\hat{Q}_3$. To first order, the wave operator that reconstructs the relevant bare-state admixtures is

$$\begin{aligned} \hat{\Omega}_3^{[1]} \hat{P}_3 = \hat{P}_3 + \frac{g}{\Delta} |g, 1\rangle \langle A| - \frac{\sqrt{3}g}{\Delta} |e, 2\rangle \langle B| \\ - \frac{2g}{\Sigma} |e, 4\rangle \langle B| + \mathcal{O}\left(\frac{g^2}{\omega_a^2}\right). \end{aligned} \quad (\text{S31})$$

For a retained-space state

$$|\psi_{\text{eff}}(t)\rangle = c_A(t) |A\rangle + c_B(t) |B\rangle,$$

define the leading normalized reconstructed basis states

$$\begin{aligned} |A\rangle_{\text{rec}} = \left(1 - \frac{g^2}{2\Delta^2}\right) |A\rangle \\ + \frac{g}{\Delta} |g, 1\rangle + \mathcal{O}\left(\frac{g^2}{\omega_a^2}\right)_Q, \end{aligned} \quad (\text{S32a})$$

$$\begin{aligned} |B\rangle_{\text{rec}} = \left(1 - \frac{3g^2}{2\Delta^2} - \frac{2g^2}{\Sigma^2}\right) |B\rangle \\ - \frac{\sqrt{3}g}{\Delta} |e, 2\rangle - \frac{2g}{\Sigma} |e, 4\rangle \\ + \mathcal{O}\left(\frac{g^2}{\omega_a^2}\right)_Q. \end{aligned} \quad (\text{S32b})$$

The reconstructed state is then

$$|\psi_{\text{rec}}(t)\rangle = c_A(t) |A\rangle_{\text{rec}} + c_B(t) |B\rangle_{\text{rec}}. \quad (\text{S33})$$

Here $\mathcal{O}(g^2/\omega_a^2)_Q$ denotes second-order amplitudes in the eliminated sector, not the normalization terms already displayed. The leading eliminated-sector population is

$$\begin{aligned} L_3(t) &\equiv \langle \psi_{\text{rec}}(t) | \hat{Q}_3 | \psi_{\text{rec}}(t) \rangle \\ &= |c_A(t)|^2 \frac{g^2}{\Delta^2} \\ &\quad + |c_B(t)|^2 \left(\frac{3g^2}{\Delta^2} + \frac{4g^2}{\Sigma^2} \right) + \mathcal{O}\left(\frac{g^3}{\omega_a^3}\right). \end{aligned}$$

Thus, even when the retained two-state model predicts complete transfer, the bare population of $|B\rangle$ need not reach unity. The missing weight is not automatically an error of the effective theory; part of it is the virtual dressing required to reconstruct the laboratory state. A fair comparison with the full Rabi dynamics must therefore distinguish true leakage from perturbatively reconstructed virtual population.

The same reconstruction affects observables. For example, the physical oscillator occupation is not obtained solely from $3|c_B(t)|^2$; the virtual states in Eq. (S33) contribute at the dimensionless relative scale $\mathcal{O}[(g/\omega_a)^2]$. Method-specific transformations may distribute these corrections differently between the state and the observable, but the complete prediction must agree after the same effective gauge is used.

Figure S1 tests this statement against the converged full Rabi calculation. The panels separate retained-space transfer, eliminated-sector population, and preparation transients so that virtual dressing is not misidentified as uncontrolled leakage.

S3.2.4 Reference invariants and local control

The eliminated-sector expansion is controlled locally by the largest matrix-element-to-gap ratio relevant to the displayed pathways,

$$\eta_{3\gamma} = \max \left\{ \frac{\sqrt{3}g}{|\Delta|}, \frac{2g}{\Sigma}, \frac{\sqrt{2}g}{\Sigma} \right\} \ll 1.$$

The resonant pair is excluded from this test because both endpoints have already been placed in $\hat{P}_3$. The method-specific derivations are

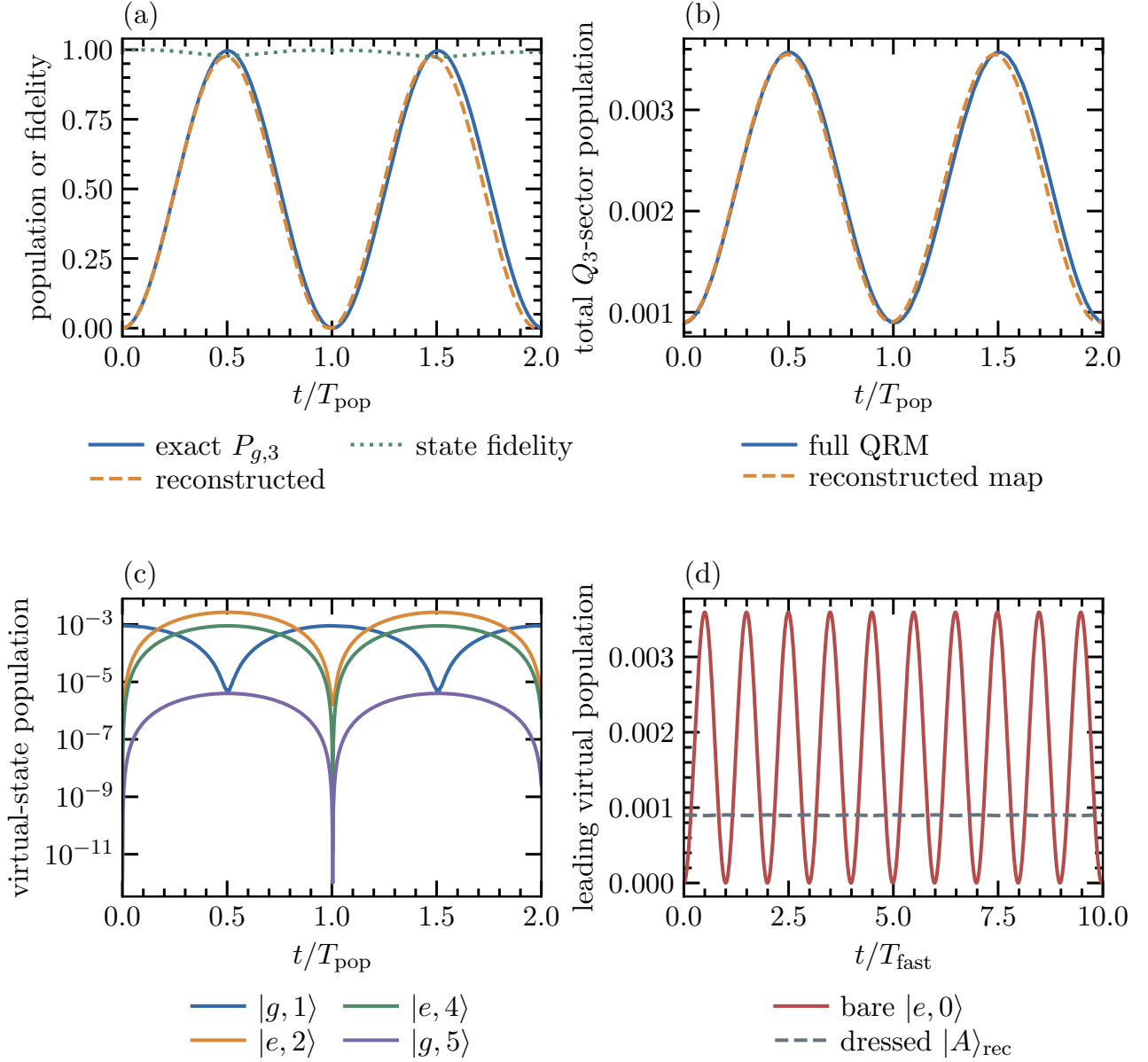


Figure S1: Reconstruction of the three-photon Rabi resonance. Panel (a) compares the exact and reconstructed $|g, 3\rangle$ populations for a dressed initial endpoint and includes their full-state fidelity. Panel (b) compares the total population carried by the leading eliminated states $|g, 1\rangle$, $|e, 2\rangle$, $|e, 4\rangle$, and $|g, 5\rangle$ with the prediction of the reconstruction map. Panel (c) resolves those exact virtual populations individually on a logarithmic scale. Panel (d) contrasts bare preparation, for which the virtual cloud forms through a fast initial transient, with preparation of the reconstructed endpoint, for which the leading virtual population is already present and nearly stationary. The parameters are $\omega_0 = 1$, $g/\omega_0 = 0.02$, and $N_{\text{cav}} = 16$. All curves are generated from the archived data and reconstruction code in companion Notebook N05.

aligned against

$$\begin{aligned}
 \delta E_A^{(2)} &= \frac{g^2}{\Delta}, & \delta E_B^{(2)} &= -\frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma}, \\
 \omega_a^{\text{res}} &= \frac{\omega_0}{3} + \frac{3g^2}{\omega_0} + \mathcal{O}\left(\frac{g^4}{\omega_0^3}\right), \\
 |G_3| &= \frac{9\sqrt{6}g^3}{4\omega_0^2} + \mathcal{O}\left(\frac{g^5}{\omega_0^4}\right), & \Delta_{\text{gap}}^{(3\gamma)} &= 2|G_3|.
 \end{aligned}
 \tag{S34}$$

The phase of G_3 is representation dependent; the resonance position, gap, and reconstructed observables are the comparison invariants.

S3.3 Projection and resolvent derivation

The projection construction makes the virtual-path interpretation of the three-photon interaction especially explicit. The retained amplitudes are not evolved by deleting the eliminated states. Instead, the eliminated sector is solved formally and returned to the retained problem through an energy-dependent self-energy. Diagonal self-energy insertions shift the resonance at second order, whereas propagation through two distinct eliminated states produces the third-order coupling between $|A\rangle$ and $|B\rangle$.

S3.3.1 Exact projected eigenvalue problem

Let $|\Psi\rangle$ satisfy

$$\hat{H}_R |\Psi\rangle = E |\Psi\rangle,$$

and decompose it as

$$|\Psi\rangle = |\psi_P\rangle + |\psi_Q\rangle,$$

$$|\psi_P\rangle = \hat{P}_3 |\Psi\rangle,$$

$$|\psi_Q\rangle = \hat{Q}_3 |\Psi\rangle.$$

The projected equations are

$$\hat{P}_3 \hat{H}_R \hat{P}_3 |\psi_P\rangle + \hat{P}_3 \hat{H}_R \hat{Q}_3 |\psi_Q\rangle = E |\psi_P\rangle, \quad (\text{S35a})$$

$$\hat{Q}_3 \hat{H}_R \hat{P}_3 |\psi_P\rangle + \hat{Q}_3 \hat{H}_R \hat{Q}_3 |\psi_Q\rangle = E |\psi_Q\rangle. \quad (\text{S35b})$$

Whenever E lies outside the relevant spectrum of $\hat{Q}_3 \hat{H}_R \hat{Q}_3$, the second equation gives

$$|\psi_Q\rangle = \hat{G}_Q(E) \hat{Q}_3 \hat{H}_R \hat{P}_3 |\psi_P\rangle, \\ \hat{G}_Q(E) = (E - \hat{Q}_3 \hat{H}_R \hat{Q}_3)^{-1}.$$

Substitution into the first equation produces the exact Bloch–Horowitz problem

$$\hat{H}_{\text{BH}}(E) |\psi_P\rangle = E |\psi_P\rangle,$$

with

$$\hat{H}_{\text{BH}}(E) = \hat{P}_3 \hat{H}_R \hat{P}_3 + \hat{P}_3 \hat{H}_R \hat{Q}_3 \hat{G}_Q(E) \hat{Q}_3 \hat{H}_R \hat{P}_3. \quad (\text{S36})$$

This equation is exact but nonlinear in the eigenvalue because the resolvent depends on E . For real E away from an eliminated-sector pole, $\hat{H}_{\text{BH}}(E)$ is Hermitian. Different exact roots are,

however, associated with different evaluations of the same energy-dependent operator. A single energy-independent effective Hamiltonian emerges only after a perturbative expansion and a choice of model-space normalization or gauge.

Since $\hat{H}_0$ is block diagonal with respect to $\hat{P}_3 \oplus \hat{Q}_3$, define

$$\hat{G}_0(E) = (E - \hat{Q}_3 \hat{H}_0 \hat{Q}_3)^{-1}.$$

The eliminated-sector resolvent has the Neumann expansion

$$\hat{G}_Q(E) = \hat{G}_0(E) + \hat{G}_0(E) \hat{Q}_3 \hat{V} \hat{Q}_3 \hat{G}_0(E) \\ + \hat{G}_0(E) \hat{Q}_3 \hat{V} \hat{Q}_3 \hat{G}_0(E) \hat{Q}_3 \hat{V} \hat{Q}_3 \hat{G}_0(E) \\ + \dots$$

Because $\hat{P}_3 \hat{V} \hat{P}_3 = 0$, insertion into Eq. (S36) gives

$$\hat{H}_{\text{BH}}(E) = \hat{P}_3 \hat{H}_0 \hat{P}_3 \\ + \hat{P}_3 \hat{V} \hat{Q}_3 \hat{G}_0(E) \hat{Q}_3 \hat{V} \hat{P}_3 \\ + \hat{P}_3 \hat{V} \hat{Q}_3 \hat{G}_0(E) \hat{Q}_3 \hat{V} \hat{Q}_3 \hat{G}_0(E) \\ \times \hat{Q}_3 \hat{V} \hat{P}_3 + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right). \quad (\text{S37})$$

The second line contains the second-order level shifts. The third line contains the first nonzero coupling between the two retained states.

S3.3.2 Second-order self-energies

In the ordered basis $\{|A\rangle, |B\rangle\}$, the diagonal self-energies obtained from the second line of Eq. (S37) are

$$\Sigma_A(E) = \frac{g^2}{E - E_{g,1}^{(0)}} + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right), \quad (\text{S38a})$$

$$\Sigma_B(E) = \frac{3g^2}{E - E_{e,2}^{(0)}} + \frac{4g^2}{E - E_{e,4}^{(0)}} + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right). \quad (\text{S38b})$$

The corresponding second-order off-diagonal term vanishes. Expanding the first expression at the root connected continuously to $E_A^{(0)}$, and the second at the root connected to $E_B^{(0)}$, gives

$$\Sigma_A(E) = \frac{g^2}{\Delta} + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right), \quad (\text{S39a})$$

$$\Sigma_B(E) = -\frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma} + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right). \quad (\text{S39b})$$

The energy dependence of the denominators contributes only at fourth order because each relevant eigenvalue differs from its bare value by $\mathcal{O}(g^2/\omega_a)$. Equations (S39) therefore recover the shifts in Eqs. (S25) and (S26). In particular, deleting $|e, 4\rangle$ from the eliminated sector removes the term $-4g^2/\Sigma$ and produces an incorrect resonance already at the first nontrivial order.

S3.3.3 Third-order propagation through the eliminated sector

The third line of Eq. (S37) contains an internal interaction within $\hat{Q}_3$. Its off-diagonal matrix element is

$$\begin{aligned}\Gamma_{BA}(E) &\equiv \langle B | \hat{H}_{\text{BH}}(E) | A \rangle \\ &= \frac{\langle B | \hat{V} | e, 2 \rangle \langle e, 2 | \hat{V} | g, 1 \rangle \langle g, 1 | \hat{V} | A \rangle}{(E - E_{e,2}^{(0)}) (E - E_{g,1}^{(0)})} \\ &\quad + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right) \\ &= \frac{\sqrt{6} g^3}{(E - E_{e,2}^{(0)}) (E - E_{g,1}^{(0)})} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right).\end{aligned}\tag{S40}$$

There is only one three-vertex path: the resolvent first propagates $|g, 1\rangle$, the internal $\hat{Q}_3 \hat{V} \hat{Q}_3$ vertex converts it into $|e, 2\rangle$, and the last vertex returns the amplitude to the retained state $|B\rangle$. This is the resolvent representation of the virtual chain in Eq. (S19).

Evaluating Eq. (S40) at either quasi-degenerate bare energy gives the same denominator product,

$$(E_A^{(0)} - E_{e,2}^{(0)}) (E_A^{(0)} - E_{g,1}^{(0)}) = -2\omega_a \Delta, \tag{S41a}$$

$$(E_B^{(0)} - E_{e,2}^{(0)}) (E_B^{(0)} - E_{g,1}^{(0)}) = -2\omega_a \Delta. \tag{S41b}$$

Hence

$$\Gamma_{BA}(E) = -\frac{\sqrt{6} g^3}{2\omega_a \Delta} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right),$$

which coincides with Eq. (S28). Replacing the bare energy in the resolvent by the exact quasi-degenerate root changes this coefficient only at fifth order. Thus, the energy dependence that is essential to the exact projected equation does not obstruct the construction of a unique cubic matrix element at the stated order.

To display the full projected problem, write

$$\begin{aligned}\hat{H}_{\text{BH}}(E) &= \begin{pmatrix} E_A^{(0)} + \Sigma_A(E) & \Gamma_{AB}(E) \\ \Gamma_{BA}(E) & E_B^{(0)} + \Sigma_B(E) \end{pmatrix} \\ &\quad + \mathcal{O}(g^4)_{\text{diag}} + \mathcal{O}(g^5)_{\text{off}}.\end{aligned}\tag{S42}$$

where $\Gamma_{AB}(E) = \Gamma_{BA}(E)^*$. The exact roots satisfy

$$0 = [E - E_A^{(0)} - \Sigma_A(E)] [E - E_B^{(0)} - \Sigma_B(E)] - |\Gamma_{BA}(E)|^2.$$

Expanding this equation with the power counting in Eq. (S22) yields the shifted resonance and the minimum gap stated in Eq. (S34).

S3.3.4 Energy-independent Hermitian model-space Hamiltonian

For comparison with the unitary block-diagonalization methods used later, it is convenient to replace the energy-dependent equation by one energy-independent Hermitian operator acting on a fixed, orthonormal model space. Through the required orders, symmetric quasi-degenerate partitioning gives

$$\begin{aligned}\langle i | \hat{H}_{\text{P,eff}}^{(2)} | j \rangle &= \frac{1}{2} \sum_{m \in Q_3} V_{im} V_{mj} \\ &\quad \times \left[\frac{1}{E_i^{(0)} - E_m^{(0)}} + \frac{1}{E_j^{(0)} - E_m^{(0)}} \right],\end{aligned}$$

where $i, j \in \{A, B\}$. Defining $D_{im} = E_i^{(0)} - E_m^{(0)}$, and using the absence of a first-order retained-space interaction, the relevant third-order off-diagonal term is

$$\begin{aligned}\langle i | \hat{H}_{\text{P,eff}}^{(3)} | j \rangle &= \frac{1}{2} \sum_{m,n \in Q_3} V_{im} V_{mn} V_{nj} \\ &\quad \times \left(\frac{1}{D_{im} D_{in}} + \frac{1}{D_{jm} D_{jn}} \right).\end{aligned}$$

Folded terms containing an intermediate retained-space interaction vanish here because $\hat{P}_3 \hat{V} \hat{P}_3 = 0$. For $i = B, j = A$, the only nonzero pair is $m = e, 2, n = g, 1$, together with the Hermitian ordering contained in the symmetric expression. The two denominator products are equal by Eq. (S41), and one recovers

$$\langle B | \hat{H}_{\text{P,eff}}^{(3)} | A \rangle = -\frac{\sqrt{6} g^3}{2\omega_a \Delta}.$$

The resulting energy-independent model-space Hamiltonian is therefore

$$\begin{aligned}\hat{H}_{\text{P,eff}} = & \left(E_A^{(0)} + \frac{g^2}{\Delta} \right) |A\rangle\langle A| \\ & + \left(E_B^{(0)} - \frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma} \right) |B\rangle\langle B| \\ & - \frac{\sqrt{6}g^3}{2\omega_a\Delta} (|B\rangle\langle A| + |A\rangle\langle B|) \\ & + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right)_{\text{diag}} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right)_{\text{off}}.\end{aligned}\quad (\text{S43})$$

After separating the mean energy, this is precisely the canonical Hamiltonian in Eq. (S30). Other energy-independent projection conventions can differ by a perturbative unitary transformation within $\hat{P}_3$, but they must give the same resonance position, avoided-crossing gap, and reconstructed observables.

S3.3.5 Wave operator, model-space metric, and reconstruction

The projection method also provides the reconstruction map directly. The exact energy-dependent wave operator is

$$\hat{\Omega}_{\text{BH}}(E)\hat{P}_3 = [\hat{P}_3 + \hat{G}_Q(E)\hat{Q}_3\hat{V}\hat{P}_3]\hat{P}_3, \quad (\text{S44})$$

so that

$$|\Psi\rangle = \hat{\Omega}_{\text{BH}}(E)|\psi_P\rangle.$$

Expanding the resolvent gives

$$\begin{aligned}\hat{\Omega}_{\text{BH}}(E)\hat{P}_3 = & \hat{P}_3 + \hat{G}_0(E)\hat{Q}_3\hat{V}\hat{P}_3 \\ & + \hat{G}_0(E)\hat{Q}_3\hat{V}\hat{Q}_3\hat{G}_0(E)\hat{Q}_3\hat{V}\hat{P}_3 \\ & + \mathcal{O}\left[(g/\omega_a)^3\right].\end{aligned}$$

At the respective quasi-degenerate bare energies, the components on the shortest virtual chain are

$$\begin{aligned}\hat{\Omega}_{\text{BH}}(E_A^{(0)})|A\rangle = & |A\rangle + \frac{g}{\Delta}|g, 1\rangle - \frac{\sqrt{2}g^2}{2\omega_a\Delta}|e, 2\rangle \\ & + \mathcal{O}\left[(g/\omega_a)^3\right],\end{aligned}\quad (\text{S45a})$$

$$\begin{aligned}\hat{\Omega}_{\text{BH}}(E_B^{(0)})|B\rangle = & |B\rangle - \frac{\sqrt{3}g}{\Delta}|e, 2\rangle - \frac{2g}{\Sigma}|e, 4\rangle \\ & - \frac{\sqrt{6}g^2}{2\omega_a\Delta}|g, 1\rangle + \frac{\sqrt{5}g^2}{\omega_a\Sigma}|g, 5\rangle \\ & + \mathcal{O}\left[(g/\omega_a)^3\right].\end{aligned}\quad (\text{S45b})$$

The first-order terms reproduce Eq. (S31). The second-order terms show explicitly how

amplitude propagates along the eliminated chain before returning to the retained sector at third order.

The projected vectors $|\psi_P\rangle$ are not normalized with the ordinary model-space inner product. Their physical norm is

$$\begin{aligned}\langle\Psi|\Psi\rangle = & \langle\psi_P|\hat{M}(E)|\psi_P\rangle, \\ \hat{M}(E) = & \hat{P}_3\hat{\Omega}_{\text{BH}}^\dagger(E)\hat{\Omega}_{\text{BH}}(E)\hat{P}_3.\end{aligned}\quad (\text{S46})$$

Through second order, the metric is diagonal in the bare retained basis,

$$\begin{aligned}\hat{M} = & \left(1 + \frac{g^2}{\Delta^2}\right)|A\rangle\langle A| \\ & + \left(1 + \frac{3g^2}{\Delta^2} + \frac{4g^2}{\Sigma^2}\right)|B\rangle\langle B| + \mathcal{O}\left(\frac{g^3}{\omega_a^3}\right).\end{aligned}$$

Symmetric orthonormalization by $\hat{M}^{-1/2}$ produces the normalized reconstructed states in Eqs. (S32). Thus, the normalization corrections that appear there are not optional adjustments; they are the model-space metric associated with eliminating physical amplitudes.

For an observable $\hat{O}$, a projection-based complete prediction can be written in the orthonormal model space as

$$\hat{O}_{\text{P,eff}} = \hat{M}^{-1/2}\hat{P}_3\hat{\Omega}_{\text{BH}}^\dagger\hat{O}\hat{\Omega}_{\text{BH}}\hat{P}_3\hat{M}^{-1/2}.$$

Using only $\hat{P}_3\hat{O}\hat{P}_3$ omits the virtual contributions carried by the wave operator. This is why the projected population of $|B\rangle$, the physical photon number, and the true eliminated-sector weight answer different questions even when they are evaluated from the same two retained amplitudes.

Result of the projection derivation.

The projection result consists of the energy-dependent Bloch–Horowitz matrix, its energy-independent Hermitian reduction, the wave operator, and the model-space metric. Both resonant endpoints must remain in P_3 , and the $|e, 4\rangle$ self-energy contribution must be retained at second order. Equations (S43), (S44), and (S46) are the objects used in the cross-method alignment below.

S3.4 Schrieffer–Wolff derivation

The projection construction eliminates the remote sector through an energy-dependent Green operator. The Schrieffer–Wolff (SW)

construction reaches the same retained-space physics through a near-identity unitary transformation. Its main advantage in the present problem is conceptual: the two resonant endpoint states remain together inside one block, whereas the virtual intermediate states are absorbed into a dressed basis. The small three-photon splitting is therefore retained rather than placed in a perturbative denominator.

We use the transformation convention of the Schrieffer–Wolff construction in the article,

$$\hat{U}_{\text{SW}} = e^{\hat{S}}, \quad \tilde{\hat{H}} = e^{-\hat{S}} \hat{H}_{\text{R}} e^{\hat{S}}, \quad \hat{S}^\dagger = -\hat{S},$$

and impose the canonical block gauge

$$\hat{P}_3 \hat{S}_k \hat{P}_3 = \hat{Q}_3 \hat{S}_k \hat{Q}_3 = 0 \quad (k \geq 1).$$

Thus, every perturbative generator is block off diagonal with respect to $\hat{P}_3 \oplus \hat{Q}_3$. A block-diagonal anti-Hermitian operator could be added at any order, but that would merely rotate the effective basis inside the retained and eliminated sectors. The canonical choice provides a fixed gauge for comparison with the Hermitian model-space Hamiltonian in Eq. (S43).

For compactness, define the eliminated states

$$\begin{aligned} |C\rangle &\equiv |g, 1\rangle, & |D\rangle &\equiv |e, 2\rangle, \\ |F\rangle &\equiv |e, 4\rangle, & |G\rangle &\equiv |g, 5\rangle. \end{aligned}$$

Only $|C\rangle$ and $|D\rangle$ lie on the shortest path connecting $|A\rangle$ and $|B\rangle$. The state $|F\rangle$ contributes to the second-order shift of $|B\rangle$, while $|G\rangle$ first appears in the second-order reconstruction of that state.

S3.4.1 Block decomposition and BCH expansion

Relative to $\hat{P}_3 \oplus \hat{Q}_3$, write

$$\hat{V}_{\text{od}} = \hat{P}_3 \hat{V} \hat{Q}_3 + \hat{Q}_3 \hat{V} \hat{P}_3, \quad (\text{S47a})$$

$$\hat{V}_{\text{d}} = \hat{P}_3 \hat{V} \hat{P}_3 + \hat{Q}_3 \hat{V} \hat{Q}_3. \quad (\text{S47b})$$

Because $\hat{P}_3 \hat{V} \hat{P}_3 = 0$, the relevant pieces are

$$\hat{V}_{\text{od}} = g |C\rangle \langle A| + \sqrt{3}g |D\rangle \langle B| + 2g |F\rangle \langle B| + \text{H.c.}, \quad (\text{S48a})$$

$$\hat{V}_{\text{d}} = \sqrt{2}g |D\rangle \langle C| + \sqrt{5}g |G\rangle \langle F| + \text{H.c.} + \dots \quad (\text{S48b})$$

The first term in $\hat{V}_{\text{d}}$ is the counter-rotating step inside the eliminated sector that makes the

three-photon process possible. Treating the full interaction as purely block off diagonal would remove this internal propagation and would miss the cubic coupling.

Introduce the bookkeeping expansion

$$\hat{S} = \lambda \hat{S}_1 + \lambda^2 \hat{S}_2 + \lambda^3 \hat{S}_3 + \dots, \quad g \mapsto \lambda g.$$

The first two cancellation equations are

$$[\hat{H}_0, \hat{S}_1] = -\hat{V}_{\text{od}}, \quad (\text{S49a})$$

$$[\hat{H}_0, \hat{S}_2] = -[\hat{V}_{\text{d}}, \hat{S}_1]. \quad (\text{S49b})$$

Expanding $e^{-\hat{S}} \hat{H} e^{\hat{S}}$ through third order and using Eqs. (S49) gives

$$\begin{aligned} \tilde{\hat{H}} &= \hat{H}_0 + \lambda \hat{V}_{\text{d}} + \frac{\lambda^2}{2} [\hat{V}_{\text{od}}, \hat{S}_1] \\ &+ \lambda^3 \left\{ [\hat{H}_0, \hat{S}_3] + [\hat{V}_{\text{d}}, \hat{S}_2] + \frac{1}{2} [\hat{V}_{\text{od}}, \hat{S}_2] \right. \\ &\quad \left. + \frac{1}{3} [[\hat{V}_{\text{od}}, \hat{S}_1], \hat{S}_1] \right\} + \mathcal{O}(\lambda^4). \end{aligned}$$

The first, second, and fourth terms inside braces are block off diagonal. They determine $\hat{S}_3$ but do not contribute to the retained block at this order. The block-diagonal Hamiltonian relevant to $\hat{P}_3$ is therefore

$$\begin{aligned} \hat{P}_3 \tilde{\hat{H}} \hat{P}_3 &= \hat{P}_3 \hat{H}_0 \hat{P}_3 + \frac{\lambda^2}{2} \hat{P}_3 [\hat{V}_{\text{od}}, \hat{S}_1] \hat{P}_3 \\ &+ \frac{\lambda^3}{2} \hat{P}_3 [\hat{V}_{\text{od}}, \hat{S}_2] \hat{P}_3 + \mathcal{O}(\lambda^4). \end{aligned} \quad (\text{S50})$$

The second-order shifts are generated by the first dressing rotation, whereas the third-order resonant interaction requires the second rotation.

S3.4.2 First-order generator and second-order shifts

For a retained state $|i\rangle$ and an eliminated state $|\alpha\rangle$, Eq. (S49a) gives

$$\langle i | \hat{S}_1 | \alpha \rangle = \frac{\langle i | \hat{V} | \alpha \rangle}{E_{\alpha}^{(0)} - E_i^{(0)}}.$$

Using the denominators in Eq. (S21), the relevant generator is

$$\begin{aligned} \hat{S}_1 &= \frac{g}{\Delta} (|C\rangle \langle A| - |A\rangle \langle C|) \\ &+ \frac{\sqrt{3}g}{\Delta} (|B\rangle \langle D| - |D\rangle \langle B|) \\ &+ \frac{2g}{\Sigma} (|B\rangle \langle F| - |F\rangle \langle B|) + \dots \end{aligned}$$

The associated local rotation amplitudes are

$$\frac{g}{|\Delta|}, \quad \frac{\sqrt{3}g}{|\Delta|}, \quad \frac{2g}{\Sigma}.$$

They must be small in the occupied manifold. The small retained-space splitting δ_3 does not occur in these denominators because both nearly degenerate states belong to $\hat{P}_3$.

The second-order retained block is

$$\begin{aligned} \frac{1}{2}\hat{P}_3[\hat{V}_{\text{od}}, \hat{S}_1]\hat{P}_3 &= \frac{g^2}{\Delta} |A\rangle\langle A| \\ &\quad - \left(\frac{3g^2}{\Delta} + \frac{4g^2}{\Sigma} \right) |B\rangle\langle B|. \end{aligned}$$

Thus, the canonical SW rotation reproduces the shifts in Eqs. (S25) and (S26). In particular, the coupling to $|F\rangle = |e, 4\rangle$ is required even though that state does not lie on the shortest path between the retained endpoints.

S3.4.3 Second-order generator and the cubic coupling

The commutator $[\hat{V}_{\text{d}}, \hat{S}_1]$ contains the remaining second-order couplings between the retained and eliminated sectors. Solving Eq. (S49b) gives

$$\begin{aligned} \hat{S}_2 &= \frac{\sqrt{2}g^2}{2\omega_a\Delta} (|A\rangle\langle D| - |D\rangle\langle A|) \\ &\quad + \frac{\sqrt{6}g^2}{2\omega_a\Delta} (|B\rangle\langle C| - |C\rangle\langle B|) \\ &\quad - \frac{\sqrt{5}g^2}{\omega_a\Sigma} (|B\rangle\langle G| - |G\rangle\langle B|) + \dots \end{aligned}$$

The first two terms dress opposite endpoints through the same virtual chain. The last term contributes to state reconstruction but not to the retained Hamiltonian through third order.

The cubic matrix element follows from the final term in Eq. (S50). Explicitly,

$$\begin{aligned} \langle B| \frac{1}{2}[\hat{V}_{\text{od}}, \hat{S}_2] |A\rangle &= \frac{1}{2} \left[\langle B| \hat{V} |D\rangle \langle D| \hat{S}_2 |A\rangle \right. \\ &\quad \left. - \langle B| \hat{S}_2 |C\rangle \langle C| \hat{V} |A\rangle \right] \\ &= -\frac{\sqrt{6}g^3}{2\omega_a\Delta}. \end{aligned}$$

The two equal contributions correspond to dressing the initial and final endpoints of the virtual path. Their sum is the Hermitian,

symmetric model-space coupling. The third-order retained block is therefore

$$\hat{H}_{\text{SW},3}^{(3)} = -\frac{\sqrt{6}g^3}{2\omega_a\Delta} (|B\rangle\langle A| + |A\rangle\langle B|).$$

At the leading resonance, this becomes

$$G_3^{\text{SW}} = -\frac{\sqrt{6}g^3}{4\omega_a^2} = -\frac{9\sqrt{6}g^3}{4\omega_0^2},$$

in agreement with the reference and projection results.

Combining the retained terms gives

$$\begin{aligned} \hat{H}_{\text{SW},3} &= \left(E_A^{(0)} + \frac{g^2}{\Delta} \right) |A\rangle\langle A| \\ &\quad + \left(E_B^{(0)} - \frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma} \right) |B\rangle\langle B| \\ &\quad - \frac{\sqrt{6}g^3}{2\omega_a\Delta} (|B\rangle\langle A| + |A\rangle\langle B|) \\ &\quad + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right)_{\text{diag}} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right)_{\text{off}}. \end{aligned} \quad (\text{S51})$$

After subtracting the mean energy, this is the canonical Hamiltonian in Eq. (S24), with the effective detuning in Eq. (S27). The canonical SW and symmetrically orthonormalized projection Hamiltonians are therefore identical through the displayed orders.

S3.4.4 Unitary reconstruction of states and observables

In the SW representation, retained-space vectors evolve under Eq. (S51). The corresponding laboratory state is obtained with the inverse dressing,

$$|\psi_{\text{rec}}(t)\rangle = e^{\hat{S}} [c_A(t) |A\rangle + c_B(t) |B\rangle], \quad (\text{S52})$$

with $\hat{S}$ truncated consistently. Expanding $e^{\hat{S}} = 1 + \hat{S}_1 + \hat{S}_2 + \hat{S}_1^2/2 + \dots$ gives the dressed retained

basis

$$|A\rangle_{\text{rec}}^{\text{SW}} = \left(1 - \frac{g^2}{2\Delta^2}\right) |A\rangle + \frac{g}{\Delta} |C\rangle - \frac{\sqrt{2}g^2}{2\omega_a\Delta} |D\rangle + \mathcal{O}\left[(g/\omega_a)^3\right], \quad (\text{S53a})$$

$$|B\rangle_{\text{rec}}^{\text{SW}} = \left[1 - \frac{1}{2}\left(\frac{3g^2}{\Delta^2} + \frac{4g^2}{\Sigma^2}\right)\right] |B\rangle - \frac{\sqrt{3}g}{\Delta} |D\rangle - \frac{2g}{\Sigma} |F\rangle - \frac{\sqrt{6}g^2}{2\omega_a\Delta} |C\rangle + \frac{\sqrt{5}g^2}{\omega_a\Sigma} |G\rangle + \mathcal{O}\left[(g/\omega_a)^3\right]. \quad (\text{S53b})$$

These expressions coincide with the normalized projection-wave-operator states through second order. The agreement is structural: symmetric orthonormalization converts the nonunitary Bloch–Horowitz wave operator into the same canonical unitary dressing generated by the SW rotation. Unlike the projection representation, the SW basis carries the ordinary inner product automatically because $e^{\hat{S}}$ is unitary before perturbative truncation.

A laboratory observable $\hat{O}$ is represented in the retained SW space by

$$\hat{O}_{\text{SW,eff}} = \hat{P}_3 e^{-\hat{S}} \hat{O} e^{\hat{S}} \hat{P}_3. \quad (\text{S54})$$

Using only $\hat{P}_3 \hat{O} \hat{P}_3$ discards the same virtual contributions that appear explicitly in Eqs. (S53). For example, the physical photon number and the bare population of $|B\rangle$ receive different dressing corrections even when they are calculated from the same two retained amplitudes.

Result of the Schrieffer–Wolff derivation.

The canonical transformation removes only P_3 – Q_3 couplings. The generated $A \leftrightarrow B$ matrix element belongs to the retained block and must not be eliminated by a generator containing the small denominator δ_3 . The effective Hamiltonian and inverse unitary reconstruction are given by Eqs. (S51) and (S52).

S3.5 Method-of-small-rotations derivation

The method of small rotations (MSR) reorganizes perturbation theory as a sequence

of near-identity unitary rotations generated by the operator algebra of the interaction [5–7]. For the dispersive Jaynes–Cummings model, one deformed $\mathfrak{su}(2)$ ladder is sufficient. The full Rabi interaction contains both co-rotating and counter-rotating ladders, so the relevant algebra is enlarged. This enlargement is useful rather than merely technical: nested commutators between the two one-photon ladders generate the two-photon dressing and the resonant three-photon operator that are hidden in the microscopic Hamiltonian.

We use the same transformation convention as in the higher-order MSR construction in the main text,

$$\tilde{H} = e^{\hat{A}} \hat{H} e^{-\hat{A}}, \quad \hat{A}^\dagger = -\hat{A}.$$

The rotations below are global operator rotations rather than state-by-state transformations. The resonant two-state manifold is selected only after the generated operator content has been separated into slow and nonresonant sectors.

S3.5.1 Two one-photon ladders and the first rotation

Define the co-rotating and counter-rotating ladder operators

$$\hat{X}_{\Delta,+} = \hat{a} \hat{\sigma}_+, \quad \hat{X}_{\Delta,-} = \hat{a}^\dagger \hat{\sigma}_-, \quad (\text{S55a})$$

$$\hat{X}_{\Sigma,+} = \hat{a}^\dagger \hat{\sigma}_+, \quad \hat{X}_{\Sigma,-} = \hat{a} \hat{\sigma}_-. \quad (\text{S55b})$$

They obey

$$[\hat{H}_0, \hat{X}_{\Delta,\pm}] = \pm \Delta \hat{X}_{\Delta,\pm}, \quad (\text{S56a})$$

$$[\hat{H}_0, \hat{X}_{\Sigma,\pm}] = \pm \Sigma \hat{X}_{\Sigma,\pm}. \quad (\text{S56b})$$

The interaction is

$$\hat{V} = g \left(\hat{X}_{\Delta,+} + \hat{X}_{\Delta,-} + \hat{X}_{\Sigma,+} + \hat{X}_{\Sigma,-} \right).$$

The first small rotation is generated by

$$\hat{A}_1 = \frac{g}{\Delta} \left(\hat{X}_{\Delta,+} - \hat{X}_{\Delta,-} \right) + \frac{g}{\Sigma} \left(\hat{X}_{\Sigma,+} - \hat{X}_{\Sigma,-} \right).$$

With this choice,

$$[\hat{A}_1, \hat{H}_0] = -\hat{V}.$$

The collected Baker–Campbell–Hausdorff series therefore reads

$$e^{\hat{A}_1} \hat{H}_R e^{-\hat{A}_1} = \hat{H}_0 + \frac{1}{2} [\hat{A}_1, \hat{V}] + \frac{1}{3} [\hat{A}_1, [\hat{A}_1, \hat{V}]] + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right).$$

The numerical coefficients in this expression follow from combining the BCH terms containing $\hat{H}_0$ with those containing $\hat{V}$; they are not phenomenological factors.

S3.5.2 Second-order shifts and the two-photon residual

The second-order term separates into a diagonal contribution and an off-diagonal two-photon contribution,

$$\frac{1}{2}[\hat{A}_1, \hat{V}] = \hat{D}_2 + \hat{W}_2,$$

where

$$\begin{aligned} \hat{D}_2 = & \frac{g^2}{2} \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) (2\hat{n} + 1) \hat{\sigma}_z \\ & + \frac{g^2}{2} \left(\frac{1}{\Delta} - \frac{1}{\Sigma} \right) \hat{I}, \end{aligned} \quad (\text{S57})$$

and

$$\hat{W}_2 = \frac{g^2 \omega_0}{\Delta \Sigma} (\hat{a}^2 + \hat{a}^{\dagger 2}) \hat{\sigma}_z.$$

The diagonal term already contains the complete second-order shifts of the two target states,

$$\langle A | \hat{D}_2 | A \rangle = \frac{g^2}{\Delta}, \quad (\text{S58a})$$

$$\langle B | \hat{D}_2 | B \rangle = -\frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma}. \quad (\text{S58b})$$

Thus the first rotation reproduces the shifted resonance before the three-photon interaction is identified.

The operator $\hat{W}_2$ changes the oscillator occupation by two while leaving the qubit state unchanged. It is nonresonant near $\omega_0 \simeq 3\omega_a$ and is removed by the second small rotation

$$\hat{A}_2 = \frac{g^2 \omega_0}{2\omega_a \Delta \Sigma} (\hat{a}^{\dagger 2} - \hat{a}^2) \hat{\sigma}_z,$$

which satisfies

$$[\hat{A}_2, \hat{H}_0] = -\hat{W}_2.$$

On the occupied finite-energy manifold, the dimensionless rotation amplitudes in $\hat{A}_2$ scale as $\mathcal{O}[(g/\omega_a)^2]$. This second rotation therefore does not alter the third-order terms generated by the first rotation. It does, however, contribute at second order to the laboratory-frame state reconstruction.

S3.5.3 Third-order algebra and the resonant three-photon operator

To display the operator content of the third-order commutator, define

$$\hat{\mathcal{B}}_\Delta = (\hat{n} + 1) \hat{a} \hat{\sigma}_+ + \hat{a}^\dagger \hat{\sigma}_- (\hat{n} + 1), \quad (\text{S59a})$$

$$\hat{\mathcal{B}}_\Sigma = \hat{n} \hat{a}^\dagger \hat{\sigma}_+ + \hat{a} \hat{\sigma}_- \hat{n}, \quad (\text{S59b})$$

$$\hat{\mathcal{T}}_3 = \hat{a}^{\dagger 3} \hat{\sigma}_- + \hat{a}^3 \hat{\sigma}_+, \quad (\text{S59c})$$

$$\hat{\mathcal{T}}_3 = \hat{a}^{\dagger 3} \hat{\sigma}_+ + \hat{a}^3 \hat{\sigma}_-. \quad (\text{S59d})$$

Direct evaluation of the nested commutator gives

$$\begin{aligned} \frac{1}{3}[\hat{A}_1, [\hat{A}_1, \hat{V}]] = & c_\Delta \hat{\mathcal{B}}_\Delta + c_\Sigma \hat{\mathcal{B}}_\Sigma \\ & + \lambda_3 \hat{\mathcal{T}}_3 + \bar{\lambda}_3 \hat{\mathcal{T}}_3, \end{aligned}$$

with

$$c_\Delta = -\frac{4g^3 \omega_0 (3\omega_0 + \omega_a)}{3\Delta^2 \Sigma^2}, \quad (\text{S60a})$$

$$c_\Sigma = -\frac{4g^3 \omega_0 (3\omega_0 - \omega_a)}{3\Delta^2 \Sigma^2}, \quad (\text{S60b})$$

$$\lambda_3 = -\frac{4g^3 \omega_0}{3\Delta^2 \Sigma}, \quad (\text{S60c})$$

$$\bar{\lambda}_3 = -\frac{4g^3 \omega_0}{3\Delta \Sigma^2}. \quad (\text{S60d})$$

The two operators $\hat{\mathcal{B}}_\Delta$ and $\hat{\mathcal{B}}_\Sigma$ are number-dependent residual one-photon couplings. The operator $\hat{\mathcal{T}}_3$ simultaneously creates or annihilates three photons and a qubit excitation and oscillates at $\omega_0 + 3\omega_a$. All three are nonresonant near the positive three-photon condition. By contrast, $\hat{\mathcal{T}}_3$ couples $|e, n\rangle$ to $|g, n+3\rangle$ and becomes slow when $\omega_0 \simeq 3\omega_a$.

The target retained-space matrix element is

$$\begin{aligned} G_3^{\text{MSR}} = & \langle B | \lambda_3 \hat{\mathcal{T}}_3 | A \rangle \\ = & -\frac{4\sqrt{6}g^3 \omega_0}{3\Delta^2 \Sigma}. \end{aligned} \quad (\text{S61})$$

At the leading resonance $\omega_0 = 3\omega_a$,

$$G_3^{\text{MSR}} = -\frac{\sqrt{6}g^3}{4\omega_a^2} = -\frac{9\sqrt{6}g^3}{4\omega_0^2},$$

which is the reference result.

Away from the exact bare crossing, the global MSR coefficient in Eq. (S61) and the canonical model-space coefficient in Eq. (S28) are not literally identical. At the operator level their difference is

$$\lambda_3 + \frac{g^3}{2\omega_a \Delta} = \frac{g^3 (\omega_0 - 3\omega_a) (3\omega_0 + \omega_a)}{6\omega_a \Delta^2 \Sigma}.$$

Under the resonant power counting $\omega_0 - 3\omega_a = \mathcal{O}(g^2/\omega_a)$, this difference is $\mathcal{O}(g^5/\omega_a^4)$. The two constructions therefore agree through the retained third order, while assigning higher-order terms to different effective gauges.

S3.5.4 Removing nonresonant residuals while retaining the target interaction

The nonresonant third-order terms can be removed by another small rotation. Define the anti-Hermitian combinations

$$\hat{\mathcal{Y}}_{\Delta}^{(3)} = (\hat{n} + 1)\hat{a}\hat{\sigma}_+ - \hat{a}^\dagger\hat{\sigma}_-(\hat{n} + 1), \quad (\text{S62a})$$

$$\hat{\mathcal{Y}}_{\Sigma}^{(3)} = \hat{n}\hat{a}^\dagger\hat{\sigma}_+ - \hat{a}\hat{\sigma}_-\hat{n}, \quad (\text{S62b})$$

$$\hat{\mathcal{Y}}_{+3}^{(3)} = \hat{a}^{\dagger 3}\hat{\sigma}_+ - \hat{a}^3\hat{\sigma}_-. \quad (\text{S62c})$$

Then

$$\begin{aligned} \hat{A}_{3,\text{nr}} &= \frac{c_{\Delta}}{\Delta}\hat{\mathcal{Y}}_{\Delta}^{(3)} + \frac{c_{\Sigma}}{\Sigma}\hat{\mathcal{Y}}_{\Sigma}^{(3)} \\ &\quad + \frac{\bar{\lambda}_3}{\omega_0 + 3\omega_a}\hat{\mathcal{Y}}_{+3}^{(3)} \end{aligned}$$

satisfies

$$\begin{aligned} [\hat{A}_{3,\text{nr}}, \hat{H}_0] &= -c_{\Delta}\hat{\mathcal{B}}_{\Delta} - c_{\Sigma}\hat{\mathcal{B}}_{\Sigma} \\ &\quad - \bar{\lambda}_3\hat{\mathcal{T}}_3. \end{aligned}$$

No analogous perturbatively small rotation should be used to remove $\lambda_3\hat{\mathcal{T}}_3$. Formally, such a rotation would contain

$$\frac{\lambda_3}{\omega_0 - 3\omega_a}(\hat{a}^3\hat{\sigma}_+ - \hat{a}^{\dagger 3}\hat{\sigma}_-).$$

After the second-order shifts are included, the relevant denominator is the residual effective detuning δ_3 . In the target resonance window $|\delta_3| \lesssim |G_3|$, the corresponding rotation angle is not small. Attempting this rotation would therefore eliminate the physical endpoint dynamics and introduce precisely the small denominator that the quasi-degenerate effective theory is designed to avoid.

With the nonresonant rotations applied, the transformed Hamiltonian is

$$\hat{H}_{\text{MSR}}^{(3\gamma)} = \hat{H}_0 + \hat{D}_2 + \lambda_3\hat{\mathcal{T}}_3 + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right), \quad (\text{S63})$$

where the displayed remainder refers to the full transformed operator. Inside the retained space, the first omitted diagonal correction is of order

g^4/ω_a^3 , while the next correction to the parity-allowed off-diagonal coupling is of order g^5/ω_a^4 . Projecting Eq. (S63) onto $\hat{P}_3$ gives

$$\begin{aligned} \hat{H}_{\text{MSR},3} &= \left(E_A^{(0)} + \frac{g^2}{\Delta}\right)|A\rangle\langle A| \\ &\quad + \left(E_B^{(0)} - \frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma}\right)|B\rangle\langle B| \\ &\quad + G_3^{\text{MSR}}(|B\rangle\langle A| + |A\rangle\langle B|) + \dots \end{aligned} \quad (\text{S64})$$

At the retained order, this Hamiltonian has the same shifted resonance and avoided-crossing gap as the projection and canonical SW constructions.

S3.5.5 Ordered reconstruction and equivalence with the canonical dressing

Let

$$\hat{R}_{\text{MSR}} = e^{\hat{A}_{3,\text{nr}}}e^{\hat{A}_2}e^{\hat{A}_1}. \quad (\text{S65})$$

The effective state is encoded with $\hat{R}_{\text{MSR}}$, evolved under Eq. (S63), and reconstructed as

$$|\psi_{\text{rec}}(t)\rangle = \hat{R}_{\text{MSR}}^\dagger [c_A(t)|A\rangle + c_B(t)|B\rangle].$$

On the occupied finite-energy manifold, the dimensionless rotation amplitudes in $\hat{A}_{3,\text{nr}}$ scale as $\mathcal{O}[(g/\omega_a)^3]$. The reconstructed basis through second order therefore follows from $e^{-\hat{A}_1}e^{-\hat{A}_2}$. One obtains

$$\begin{aligned} |A\rangle_{\text{rec}}^{\text{MSR}} &= \left(1 - \frac{g^2}{2\Delta^2}\right)|A\rangle + \frac{g}{\Delta}|C\rangle \\ &\quad - \frac{\sqrt{2}g^2}{2\omega_a\Delta}|D\rangle + \mathcal{O}[(g/\omega_a)^3], \end{aligned} \quad (\text{S66a})$$

$$\begin{aligned} |B\rangle_{\text{rec}}^{\text{MSR}} &= \left[1 - \frac{1}{2}\left(\frac{3g^2}{\Delta^2} + \frac{4g^2}{\Sigma^2}\right)\right]|B\rangle \\ &\quad - \frac{\sqrt{3}g}{\Delta}|D\rangle - \frac{2g}{\Sigma}|F\rangle \\ &\quad - \frac{\sqrt{6}g^2}{2\omega_a\Delta}|C\rangle + \frac{\sqrt{5}g^2}{\omega_a\Sigma}|G\rangle \\ &\quad + \mathcal{O}[(g/\omega_a)^3]. \end{aligned} \quad (\text{S66b})$$

These are the same normalized laboratory states obtained from the projection metric and the canonical SW inverse rotation. The agreement is nontrivial: the global MSR sequence first removes both one-photon ladders and then the induced two-photon residual, whereas the

canonical SW calculation is organized directly by the $\hat{P}_3 \oplus \hat{Q}_3$ partition. Through the displayed order, the ordered products represent the same physical dressing in different algebraic coordinates.

A laboratory observable is represented in the MSR retained space by

$$\hat{O}_{\text{MSR,eff}} = \hat{P}_3 \hat{R}_{\text{MSR}} \hat{O} \hat{R}_{\text{MSR}}^\dagger \hat{P}_3. \quad (\text{S67})$$

The same ordered rotation must be used for the initial state and for every observable. In particular, the ideal two-state population of $|B\rangle$, the physical bare-state population, and the oscillator photon number are not identical quantities after reconstruction.

Result of the small-rotation derivation.

The ordered rotations generate the sequence of second-order diagonal and two-photon terms followed by the resonant cubic operator. Nonresonant third-order residuals may be removed, but the component proportional to $\hat{a}^{\dagger 3} \hat{\sigma}_- + \hat{a}^3 \hat{\sigma}_+$ remains in the normal form. The final generator and reconstruction are summarized by Eqs. (S63)–(S67).

S3.6 Magnus derivation

The Magnus construction organizes the interaction-picture propagator as a single exponential. For the present problem, it makes two complementary features especially transparent. First, the one-photon virtual motion appears in bounded first-order terms and in the micromotion associated with the slow model. Second, the three-photon interaction appears as the part of the third-order exponent that grows linearly when the frequency combination $\Sigma - 2\Delta = 3\omega_a - \omega_0$ becomes slow. The resulting secular term is not a divergence to be discarded: it is the generator of the resolved resonant dynamics.

We use the convention of the Magnus construction in the article,

$$\hat{U}_I(t, 0) = \exp \left[-i \sum_{k=1}^{\infty} \hat{\Omega}_k(t) \right],$$

where every $\hat{\Omega}_k$ is Hermitian. The first three

terms are

$$\hat{\Omega}_1(t) = \int_0^t dt_1 \hat{V}_I(t_1), \quad (\text{S68a})$$

$$\hat{\Omega}_2(t) = -\frac{i}{2} \int_0^t dt_1 \int_0^{t_1} dt_2 [\hat{V}_I(t_1), \hat{V}_I(t_2)], \quad (\text{S68b})$$

$$\begin{aligned} \hat{\Omega}_3(t) = & -\frac{1}{6} \int_0^t dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \\ & \times \left([\hat{V}_I(t_1), [\hat{V}_I(t_2), \hat{V}_I(t_3)]] \right. \\ & \left. + [\hat{V}_I(t_3), [\hat{V}_I(t_2), \hat{V}_I(t_1)]] \right). \end{aligned} \quad (\text{S68c})$$

S3.6.1 Interaction harmonics and resonant bookkeeping

In the interaction picture generated by $\hat{H}_0$,

$$\begin{aligned} \hat{V}_I(t) = & g \hat{a} \hat{\sigma}_+ e^{i\Delta t} + g \hat{a}^\dagger \hat{\sigma}_- e^{-i\Delta t} \\ & + g \hat{a}^\dagger \hat{\sigma}_+ e^{i\Sigma t} + g \hat{a} \hat{\sigma}_- e^{-i\Sigma t}. \end{aligned} \quad (\text{S69})$$

The ordered pathway

$$|A\rangle \xrightarrow{\hat{a}^\dagger \hat{\sigma}_-} |g, 1\rangle \xrightarrow{\hat{a}^\dagger \hat{\sigma}_+} |e, 2\rangle \xrightarrow{\hat{a}^\dagger \hat{\sigma}_-} |B\rangle \quad (\text{S70})$$

therefore carries the total phase

$$\begin{aligned} e^{-i\Delta t} e^{i\Sigma t} e^{-i\Delta t} &= e^{-i\delta_0 t}, \\ \delta_0 &= 2\Delta - \Sigma = \omega_0 - 3\omega_a. \end{aligned} \quad (\text{S71})$$

The selected harmonic becomes stationary at the bare three-photon condition. In the resolved resonance window, $\delta_0 = \mathcal{O}(g^2/\omega_a)$, it varies slowly compared with the individual detuning and sum-frequency harmonics. The second-order level shifts must then be included before the remaining phase mismatch is compared with the cubic coupling.

S3.6.2 First order: bounded virtual motion

Direct integration gives

$$\begin{aligned} \hat{\Omega}_1(t) = & \frac{g}{i\Delta} (e^{i\Delta t} - 1) \hat{a} \hat{\sigma}_+ \\ & + \frac{g}{i\Sigma} (e^{i\Sigma t} - 1) \hat{a}^\dagger \hat{\sigma}_+ + \text{H.c.} \end{aligned}$$

No part of $\hat{\Omega}_1$ grows linearly in time. Its matrix elements are bounded by the same matrix-element-to-gap ratios that control the first SW or MSR rotation. Thus, the first-order interaction does not generate a slow

Hamiltonian, but it cannot be discarded from a continuous-time prediction: it describes the leading virtual motion and contributes to the micromotion/reconstruction map.

Introduce the first-order kick operator

$$\hat{K}_1(t) = \frac{g}{i\Delta} e^{i\Delta t} \hat{a} \hat{\sigma}_+ + \frac{g}{i\Sigma} e^{i\Sigma t} \hat{a}^\dagger \hat{\sigma}_+ + \text{H.c.}, \quad (\text{S72})$$

for which

$$\hat{\Omega}_1(t) = \hat{K}_1(t) - \hat{K}_1(0).$$

The endpoint factors generated by $\hat{K}_1$ are the time-dependent counterpart of the static dressing represented by the projection wave operator and the SW/MSR rotations.

S3.6.3 Second order: secular shifts and bounded two-photon motion

The second-order exponent separates into a linearly accumulating diagonal part and bounded oscillatory terms,

$$\hat{\Omega}_2(t) = t\hat{D}_2 + \hat{\Omega}_{2,\text{bnd}}(t),$$

where the slow operator is

$$\hat{D}_2 = \frac{g^2}{2} \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) (2\hat{n} + 1) \hat{\sigma}_z + \frac{g^2}{2} \left(\frac{1}{\Delta} - \frac{1}{\Sigma} \right) \hat{I}.$$

The bounded remainder contains sinusoidal corrections to the diagonal phases and an off-resonant two-photon term proportional to $(\hat{a}^2 + \hat{a}^{\dagger 2})\hat{\sigma}_z$. Its magnitude is controlled by $g^2/(|\Delta|\Sigma)$, while its oscillation frequency remains of order ω_a near the three-photon resonance.

For the two target states, the diagonal matrix elements can be displayed without suppressing the bounded pieces:

$$\langle A | \hat{\Omega}_2(t) | A \rangle = \frac{g^2}{\Delta} \left[t - \frac{\sin(\Delta t)}{\Delta} \right], \quad (\text{S73a})$$

$$\begin{aligned} \langle B | \hat{\Omega}_2(t) | B \rangle = & -\frac{3g^2}{\Delta} \left[t - \frac{\sin(\Delta t)}{\Delta} \right] \\ & - \frac{4g^2}{\Sigma} \left[t - \frac{\sin(\Sigma t)}{\Sigma} \right]. \end{aligned} \quad (\text{S73b})$$

The linearly growing pieces reproduce

$$\delta E_A^{(2)} = \frac{g^2}{\Delta}, \quad \delta E_B^{(2)} = -\frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma},$$

and hence the shifted detuning in Eq. (S27). The $|e, 4\rangle$ contribution appears here through the term proportional to $4g^2/\Sigma$; omitting that virtual state changes the resonance before the cubic interaction is even considered.

S3.6.4 Third order: extraction of the slow three-photon generator

Only the chain in Eq. (S70) contributes to the shortest third-order matrix element between the retained states. The nested-commutator integrand is

$$\begin{aligned} \mathcal{N}_{BA}(t_1, t_2, t_3) &\equiv \langle B | [\hat{V}_I(t_1), [\hat{V}_I(t_2), \hat{V}_I(t_3)]] | A \rangle \\ &\quad + \langle B | [\hat{V}_I(t_3), [\hat{V}_I(t_2), \hat{V}_I(t_1)]] | A \rangle \\ &= -2\sqrt{6}g^3 e^{i\Sigma t_1} e^{-i\Delta(t_2+t_3)} \\ &\quad + 4\sqrt{6}g^3 e^{i\Sigma t_2} e^{-i\Delta(t_1+t_3)} \\ &\quad - 2\sqrt{6}g^3 e^{i\Sigma t_3} e^{-i\Delta(t_1+t_2)}. \end{aligned}$$

At the leading crossing, $\Sigma = 2\Delta$ and $\Delta = 2\omega_a$. The ordered time integrals then give

$$\langle B | \hat{\Omega}_3(t) | A \rangle = -\frac{\sqrt{6}g^3}{\Delta^2} t + \mathcal{B}_3(t),$$

where the bounded contribution is

$$\begin{aligned} \mathcal{B}_3(t) &= \frac{i\sqrt{6}g^3}{12\Delta^3} e^{-2i\Delta t} (e^{2i\Delta t} - 1) \\ &\quad \times (e^{2i\Delta t} - 8e^{i\Delta t} + 1). \end{aligned}$$

Thus, the third-order exponent contains the secular operator

$$\hat{\Omega}_{3,\text{sec}}(t) = tG_3^{\text{M}} (\hat{\tau}_+ + \hat{\tau}_-), \quad G_3^{\text{M}} = -\frac{\sqrt{6}g^3}{\Delta^2}$$

at the leading crossing. Since $\Delta = 2\omega_a$ there,

$$G_3^{\text{M}} = -\frac{\sqrt{6}g^3}{4\omega_a^2} = -\frac{9\sqrt{6}g^3}{4\omega_0^2},$$

which agrees with the reference, projection, SW, and MSR results.

Across the resonant window, the fast denominators may be kept in the canonical form

$$G_3^{\text{M}} = -\frac{\sqrt{6}g^3}{2\omega_a\Delta} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right).$$

Indeed, the difference between $-\sqrt{6}g^3/\Delta^2$ and $-\sqrt{6}g^3/(2\omega_a\Delta)$ is proportional to $(\omega_0 - 3\omega_a)g^3$, and is therefore fifth order under the power counting in Eq. (S23). The literal off-resonant coefficient is representation dependent at that order, whereas the resonance position and the avoided-crossing gap are unchanged within the controlled approximation.

S3.6.5 Resonant frame and the canonical retained Hamiltonian

Away from the exact bare crossing, the cubic harmonic retains the slow phase in Eq. (S71). In the retained space, the rotation

$$\hat{R}_3(t) = \exp\left(\frac{i\delta_0 t}{2}\hat{\tau}_z\right) \quad (\text{S74})$$

removes this phase and contributes $\delta_0\hat{\tau}_z/2$ to the rotating-frame generator. Up to an irrelevant mean phase (and with the free Hamiltonian restored when returning to the Schrödinger picture), adding the secular second-order shifts gives

$$\begin{aligned} \hat{H}_{M,3} = & \bar{E}_3\hat{I}_3 + \frac{\delta_3}{2}\hat{\tau}_z \\ & + G_3^M\hat{\tau}_+ + \left(G_3^M\right)^*\hat{\tau}_- + \dots, \end{aligned} \quad (\text{S75})$$

with

$$\delta_3 = \delta_0 + 4g^2\left(\frac{1}{\Delta} + \frac{1}{\Sigma}\right) + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right).$$

The Magnus construction therefore reproduces both ingredients required by the resolved high-order process: the second-order displacement of the crossing and the third-order opening of the gap.

S3.6.6 Micromotion, reconstruction, and stroboscopic agreement

The slow Hamiltonian in Eq. (S75) does not by itself reproduce continuous-time laboratory observables. An order-consistent factorization has the form

$$\hat{U}_I(t, 0) \simeq e^{-i\hat{K}_M(t)}e^{-i\hat{H}_{M,3}t}e^{i\hat{K}_M(0)}, \quad (\text{S76})$$

where

$$\hat{K}_M(t) = \hat{K}_1(t) + \hat{K}_2(t) + \hat{K}_{3,\text{bnd}}(t) + \dots$$

collects the bounded parts of the first three Magnus orders together with the commutator corrections required by the factorization. The first term is given explicitly in Eq. (S72); the next terms contain the two-photon motion and the bounded remainder $\mathcal{B}_3(t)$.

For an initial laboratory state $|\psi_0\rangle$, the corresponding complete interaction-picture prediction is

$$|\psi_I^{\text{rec}}(t)\rangle = e^{-i\hat{K}_M(t)}e^{-i\hat{H}_{M,3}t}e^{i\hat{K}_M(0)}|\psi_0\rangle. \quad (\text{S77})$$

Returning to the laboratory frame adds the free evolution generated by $\hat{H}_0$. Similarly, an observable must be dressed by the endpoint kick rather than projected directly onto $\hat{P}_3$.

At the exact leading crossing, $\Sigma = 2\Delta$, the interaction is periodic with the fast period $T_f = \frac{2\pi}{|\Delta|} = \frac{\pi}{\omega_a}$. At the stroboscopic times $t_m = mT_f$,

$$\hat{K}_1(t_m) = \hat{K}_1(0), \quad \mathcal{B}_3(t_m) = 0,$$

and the leading bounded second-order harmonics also return to their initial values. The endpoint micromotion then cancels, leaving the slow three-photon evolution directly visible. Away from these times, or away from exact commensurability, the kick operators are part of the physical prediction and cannot be omitted.

Result of the Magnus derivation. The slow cubic term is extracted from the part of $\hat{\Omega}_3(t)$ that grows linearly in time under the resonant power counting. Bounded terms remain in the kick operator. The resulting approximation is the factorization in Eq. (S76); it is used as a resonant secular reorganization on the retained manifold, not as a claim of global convergence for the full unbounded Rabi operator.

S3.7 James–Jerke derivation

The Magnus derivation identifies the three-photon interaction through the third-order harmonic whose accumulated phase becomes slow. The James–Jerke construction makes the corresponding resolution assumption explicit: a temporal filter suppresses the fast one- and two-photon carriers while retaining the slow three-photon beat. This representation is especially useful for separating three distinct statements that are often conflated: a fast carrier is unresolved by the chosen coarse graining, its virtual action can still modify the slow generator, and its bounded motion must still be restored when continuous-time laboratory observables are required.

Throughout this subsection we work in the positive-detuning neighborhood of the three-photon crossing, so that $\Delta > 0$, $\Sigma > 0$. The final physical expressions are analytic continuations of the same matrix-element construction whenever the ordering of the carrier frequencies is changed consistently.

S3.7.1 Temporal filter and the three-photon passband

For an interaction-picture operator $\hat{O}(t)$, define

$$\begin{aligned}\bar{\hat{O}}(t) &= \int_{-\infty}^{\infty} ds f_{\tau}(s) \hat{O}(t-s), \\ \int_{-\infty}^{\infty} ds f_{\tau}(s) &= 1,\end{aligned}$$

with transfer function

$$F_{\tau}(\nu) = \int_{-\infty}^{\infty} ds f_{\tau}(s) e^{i\nu s}.$$

For a real filter,

$$F_{\tau}(-\nu) = F_{\tau}(\nu)^*,$$

so that filtering a Hermitian operator preserves Hermiticity when the positive- and negative-frequency terms are retained together.

Near $\omega_0 \simeq 3\omega_a$, the relevant frequency hierarchy is

$$\begin{aligned}|\delta_0| &\ll \min\{\Delta, \Sigma, 2\omega_a\}, \\ \delta_0 &\equiv \omega_0 - 3\omega_a = 2\Delta - \Sigma.\end{aligned}$$

A filter designed to retain the target process should satisfy

$$F_{\tau}(0) = 1, \quad (\text{S78a})$$

$$F_{\tau}(\delta_0) \simeq 1, \quad (\text{S78b})$$

$$F_{\tau}(\Delta), F_{\tau}(\Sigma), F_{\tau}(2\omega_a), F_{\tau}(\omega_0 + 3\omega_a) \simeq 0, \quad (\text{S78c})$$

with analogous suppression of the remaining nonzero harmonics generated through the retained perturbative order. A controlled coarse-graining window is therefore

$$\max\{\Delta^{-1}, \Sigma^{-1}, (2\omega_a)^{-1}\} \ll \tau \ll |\delta_0|^{-1}, \quad (\text{S79})$$

when $\delta_0 \neq 0$. At the shifted resonance the right-hand bound is understood through the slow envelope and ultimately through the retained scales $|\delta_3|$ and $|G_3|$. The inequalities in Eq. (S79) describe a band of acceptable filters, not a unique kernel.

The filter error relevant to the three-photon reduction can be summarized by

$$\epsilon_f^{(3\gamma)} = \max\left\{|1 - F_{\tau}(\delta_0)|, |F_{\tau}(\Delta)|, |F_{\tau}(\Sigma)|, |F_{\tau}(2\omega_a)|, |F_{\tau}(\omega_0 + 3\omega_a)|\right\}.$$

A small value of this quantity is necessary for the idealized James–Jerke selection used below, but it does not replace the matrix-element control conditions in the populated manifold.

S3.7.2 Harmonic decomposition and first-order boundary operator

The interaction-picture Rabi coupling from Eq. (S69) can be written in the harmonic form

$$\begin{aligned}\hat{V}_I(t) &= \hat{h}_{\Delta} e^{-i\Delta t} + \hat{h}_{\Delta}^{\dagger} e^{i\Delta t} \\ &\quad + \hat{h}_{\Sigma} e^{-i\Sigma t} + \hat{h}_{\Sigma}^{\dagger} e^{i\Sigma t},\end{aligned}$$

where

$$\hat{h}_{\Delta} = g\hat{a}^{\dagger}\hat{\sigma}_{-}, \quad \hat{h}_{\Sigma} = g\hat{a}\hat{\sigma}_{-}.$$

The leading anti-Hermitian primitive is

$$\begin{aligned}\hat{V}_1(t) &= \frac{g}{\Delta} \left(\hat{a}^{\dagger}\hat{\sigma}_{-} e^{-i\Delta t} - \hat{a}\hat{\sigma}_{+} e^{i\Delta t} \right) \\ &\quad + \frac{g}{\Sigma} \left(\hat{a}\hat{\sigma}_{-} e^{-i\Sigma t} - \hat{a}^{\dagger}\hat{\sigma}_{+} e^{i\Sigma t} \right),\end{aligned}$$

which satisfies

$$\dot{\hat{V}}_1(t) = -i\hat{V}_I(t). \quad (\text{S80})$$

For a finite switch-on time t_0 , the first Dyson correction is

$$\hat{U}_1(t, t_0) = \hat{V}_1(t) - \hat{V}_1(t_0).$$

The constant endpoint operator $\hat{V}_1(t_0)$ cancels from the slow Hamiltonian when the filtered-generator expansion is applied consistently, but it remains part of the encoding map. Thus, suppressing the carrier components does not justify setting the initial-time operator to zero.

A convenient higher-order James–Jerke gauge is obtained by defining

$$\hat{W}_1(t, t_0) = e^{-\hat{V}_1(t)} \hat{U}_I(t, t_0) e^{\hat{V}_1(t_0)}.$$

Its exact generator is

$$\begin{aligned}\hat{H}_{W_1}(t) &= e^{-\hat{V}_1(t)} \hat{V}_I(t) e^{\hat{V}_1(t)} \\ &\quad - i e^{-\hat{V}_1(t)} \frac{d}{dt} e^{\hat{V}_1(t)}.\end{aligned}$$

Using Eq. (S80), its expansion through third order is

$$\begin{aligned}\hat{H}_{W_1}(t) &= \frac{1}{2} [\hat{V}_I(t), \hat{V}_1(t)] \\ &\quad + \frac{1}{3} [[\hat{V}_I(t), \hat{V}_1(t)], \hat{V}_1(t)] \quad (\text{S81}) \\ &\quad + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right).\end{aligned}$$

Filtering Eq. (S81) selects the coherent slow generator in a reconstruction gauge aligned with the Magnus kick and the successive MSR rotations. Other filter or kick gauges can redistribute bounded terms without changing the complete prediction through the controlled order.

S3.7.3 Second order: filtered level shifts and rejected two-photon motion

The second-order transformed generator separates as

$$\frac{1}{2}[\hat{V}_I(t), \hat{V}_1(t)] = \hat{D}_2 + \hat{W}_{2,I}(t),$$

where $\hat{D}_2$ is the diagonal operator defined in Eq. (S57), and

$$\hat{W}_{2,I}(t) = \frac{g^2\omega_0}{\Delta\Sigma} \left(\hat{a}^{\dagger 2} e^{2i\omega_a t} + \hat{a}^2 e^{-2i\omega_a t} \right) \hat{\sigma}_z.$$

The filter therefore gives

$$\begin{aligned} \hat{H}_{JJ}^{(2)}(t) &= \hat{D}_2 \\ &+ F_\tau(2\omega_a) \frac{g^2\omega_0}{\Delta\Sigma} \hat{a}^{\dagger 2} e^{2i\omega_a t} \hat{\sigma}_z \\ &+ F_\tau(-2\omega_a) \frac{g^2\omega_0}{\Delta\Sigma} \hat{a}^2 e^{-2i\omega_a t} \hat{\sigma}_z. \end{aligned}$$

In the ideal passband of Eq. (S78),

$$\hat{H}_{JJ}^{(2)} = \hat{D}_2.$$

Consequently,

$$\begin{aligned} \langle A | \hat{H}_{JJ}^{(2)} | A \rangle &= \frac{g^2}{\Delta}, \\ \langle B | \hat{H}_{JJ}^{(2)} | B \rangle &= -\frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma}. \end{aligned}$$

Thus, the filter removes the explicit two-photon oscillation but retains its virtual contribution to the diagonal shifts. This is the temporal coarse-graining counterpart of the second MSR rotation: MSR removes the nonresonant operator by a static normal-form rotation, whereas James–Jerke assigns it to rejected micromotion.

For continuous-time reconstruction through second order, introduce

$$\hat{V}_2(t) = -\frac{g^2\omega_0}{2\omega_a\Delta\Sigma} \left(\hat{a}^{\dagger 2} e^{2i\omega_a t} - \hat{a}^2 e^{-2i\omega_a t} \right) \hat{\sigma}_z,$$

which obeys

$$\dot{\hat{V}}_2(t) = -i\hat{W}_{2,I}(t).$$

At the reference time,

$$\hat{V}_1(0) = -\hat{A}_1, \quad \hat{V}_2(0) = -\hat{A}_2,$$

so the endpoint encoding agrees with the ordered MSR rotations through second order.

S3.7.4 Third order: filtering the slow three-photon harmonic

Let the interaction-picture version of an operator be denoted by

$$\hat{O}_I(t) = e^{i\hat{H}_0 t} \hat{O} e^{-i\hat{H}_0 t}.$$

Using the operator families and coefficients defined in Eqs. (S59) and (S60), the cubic part of Eq. (S81) is

$$\begin{aligned} \hat{H}_{W1}^{(3)}(t) &= c_\Delta \hat{\mathcal{B}}_{\Delta,I}(t) + c_\Sigma \hat{\mathcal{B}}_{\Sigma,I}(t) \\ &+ \lambda_3 \hat{\mathcal{T}}_{3,I}(t) + \bar{\lambda}_3 \hat{\mathcal{T}}_{3,I}(t). \end{aligned}$$

The four operator families carry the characteristic frequencies

$$\hat{\mathcal{B}}_{\Delta,I} : \quad \pm\Delta, \quad (\text{S82a})$$

$$\hat{\mathcal{B}}_{\Sigma,I} : \quad \pm\Sigma, \quad (\text{S82b})$$

$$\hat{\mathcal{T}}_{3,I} : \quad \pm\delta_0, \quad (\text{S82c})$$

$$\hat{\mathcal{T}}_{3,I} : \quad \pm(\omega_0 + 3\omega_a). \quad (\text{S82d})$$

In particular,

$$\begin{aligned} \hat{\mathcal{T}}_{3,I}(t) &= \hat{a}^{\dagger 3} \hat{\sigma}_- e^{-i\delta_0 t} \\ &+ \hat{a}^3 \hat{\sigma}_+ e^{i\delta_0 t}. \end{aligned}$$

The filtered cubic generator is therefore

$$\begin{aligned} \hat{H}_{JJ}^{(3)}(t) &= c_\Delta \overline{\hat{\mathcal{B}}_{\Delta,I}}(t) + c_\Sigma \overline{\hat{\mathcal{B}}_{\Sigma,I}}(t) \\ &+ \lambda_3 F_\tau(\delta_0) \hat{a}^{\dagger 3} \hat{\sigma}_- e^{-i\delta_0 t} \\ &+ \lambda_3^* F_\tau(-\delta_0) \hat{a}^3 \hat{\sigma}_+ e^{i\delta_0 t} \\ &+ \bar{\lambda}_3 \overline{\hat{\mathcal{T}}_{3,I}}(t). \end{aligned}$$

For the ideal filter, all terms except the target three-photon harmonic are suppressed, giving

$$\begin{aligned} \hat{H}_{JJ}^{(3\gamma)}(t) &= \lambda_3 F_\tau(\delta_0) \hat{a}^{\dagger 3} \hat{\sigma}_- e^{-i\delta_0 t} \\ &+ \lambda_3^* F_\tau(-\delta_0) \hat{a}^3 \hat{\sigma}_+ e^{i\delta_0 t}. \end{aligned} \quad (\text{S83})$$

The retained-space matrix element is

$$\begin{aligned} G_3^{JJ}(\tau) &= \sqrt{6} \lambda_3 F_\tau(\delta_0) \\ &= -\frac{4\sqrt{6}g^3\omega_0}{3\Delta^2\Sigma} F_\tau(\delta_0). \end{aligned} \quad (\text{S84})$$

For an ideal passband, $F_\tau(\delta_0) \simeq 1$. At the leading crossing,

$$G_3^{JJ} = -\frac{\sqrt{6}g^3}{4\omega_a^2} = -\frac{9\sqrt{6}g^3}{4\omega_0^2},$$

in agreement with projection, SW, MSR, and Magnus.

The finite filter response has a direct physical interpretation. Its magnitude attenuates the resolved coupling, while its phase changes the phase convention of the effective endpoint coupling. For a symmetric real filter, $F_\tau(\delta_0)$ is real and even. For a causal filter, a nonzero phase delay is possible and must be retained together with the conjugate term in Eq. (S83).

As in the MSR derivation, the global coefficient in Eq. (S84) differs away from the crossing from the canonical model-space coefficient by terms proportional to $g^3\delta_0$. Under the resonant power counting $\delta_0 = \mathcal{O}(g^2/\omega_a)$, this difference is fifth order. Thus, after the filter and effective gauges are aligned,

$$G_3^{\text{JJ}} = -\frac{\sqrt{6}g^3}{2\omega_a\Delta} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right)$$

for an ideal passband.

S3.7.5 Resonant retained-space Hamiltonian

Combining the zero-frequency second-order term with the filtered cubic harmonic gives, in the interaction picture,

$$\begin{aligned}\hat{H}_{\text{JJ},I}(t) &= \hat{D}_2 \\ &+ G_3^{\text{JJ}}(\tau)e^{-i\delta_0 t}|B\rangle\langle A| \\ &+ G_3^{\text{JJ}}(\tau)^*e^{i\delta_0 t}|A\rangle\langle B| \\ &+ \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right).\end{aligned}$$

Applying the retained-space resonant-frame transformation from Eq. (S74) yields

$$\begin{aligned}\hat{H}_{\text{JJ},3} &= \bar{E}_3\hat{I}_3 + \frac{\delta_3}{2}\hat{\tau}_z \\ &+ G_3^{\text{JJ}}(\tau)\hat{\tau}_+ + G_3^{\text{JJ}}(\tau)^*\hat{\tau}_- + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right),\end{aligned}\tag{S85}$$

with

$$\delta_3 = \delta_0 + 4g^2\left(\frac{1}{\Delta} + \frac{1}{\Sigma}\right) + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right).$$

The James–Jerke construction therefore retains the same order-consistent combination as every preceding method: the second-order displacement of the resonance and the third-order avoided-crossing coupling.

S3.7.6 Encoding, micromotion, and complete prediction

The leading reconstructed interaction-picture propagator through the orders required for the present comparison is

$$\begin{aligned}\hat{U}_{\text{rec}}^{(I)}(t, t_0) &= e^{\hat{V}_1(t)}e^{\hat{V}_2(t)} \\ &\times \mathcal{T} \exp\left[-i\int_{t_0}^t ds \hat{H}_{\text{JJ},I}(s)\right] \\ &\times e^{-\hat{V}_2(t_0)}e^{-\hat{V}_1(t_0)} \\ &+ \mathcal{O}(g_{\text{fast}}^3, g^4),\end{aligned}\tag{S86}$$

where g_{fast}^3 denotes the bounded third-order harmonics suppressed by the filter. They can be restored by an additional bounded operator $\hat{V}_{3,f}(t)$ satisfying

$$\dot{\hat{V}}_{3,f}(t) = -i\hat{H}_{3,f}(t), \quad \overline{\hat{H}_{3,f}} \simeq 0,$$

without moving the slow target term into the micromotion. The first- and second-order operators in Eq. (S86) are sufficient to reconstruct the virtual state components displayed in the reference, projection, SW, and MSR subsections through second order.

The encoded initial state is

$$\begin{aligned}\tilde{\rho}_{\text{JJ}}(t_0) &= e^{-\hat{V}_2(t_0)}e^{-\hat{V}_1(t_0)}\rho(t_0) \\ &\times e^{\hat{V}_1(t_0)}e^{\hat{V}_2(t_0)},\end{aligned}\tag{S87}$$

and a laboratory observable is represented in the slow frame as

$$\begin{aligned}\tilde{O}_{\text{JJ}}(t) &= e^{-\hat{V}_2(t)}e^{-\hat{V}_1(t)}O \\ &\times e^{\hat{V}_1(t)}e^{\hat{V}_2(t)}.\end{aligned}\tag{S88}$$

The filtered Hamiltonian alone therefore predicts the slow envelope, but not the full continuous-time bare populations. In particular, the virtual occupations of $|g, 1\rangle$, $|e, 2\rangle$, and $|e, 4\rangle$ reside in the endpoint transformations.

The coarse-grained state itself need not evolve unitarily even though the microscopic Rabi dynamics is closed. Temporal filtering discards time resolution and is not a partial trace over a physical environment. The Hermitian operator in Eq. (S85) represents the coherent slow component; the complete filtered map can contain a nonunitary covariance contribution when information about the fast motion is removed.

Result of the James–Jerke derivation. The filtered endpoint coupling carries the transfer function of the slow three-photon beat, whereas the rejected carrier harmonics are stored in boundary and kick operators. The complete prediction is therefore the retained Hamiltonian in Eq. (S85) together with the propagator and encoding in Eqs. (S86) and (S87).

S3.8 Systematic adiabatic-elimination derivation

Adiabatic elimination provides a complementary description of the same three-photon process. Here, the word *adiabatic* does not refer to a slow external parameter ramp. It denotes a separation between the slow amplitudes in the nearly degenerate endpoint manifold and the rapidly adjusting amplitudes of the off-resonant intermediate states. The eliminated amplitudes are expressed as functionals of the retained ones, and their substitution generates the slow endpoint dynamics. The construction follows the systematic slow-manifold formulation developed in the adiabatic-elimination construction in the article; simply setting every fast time derivative to zero is only its lowest approximation.

The example is especially instructive because a one-step elimination recovers the second-order level shifts but does not yet generate the third-order endpoint coupling. The internal propagation $|C\rangle \leftrightarrow |D\rangle$ must be inserted once more into the slaving relations. The state $|F\rangle = |e, 4\rangle$, although absent from the shortest endpoint path, must also be retained in the fast sector because it contributes to the shifted resonance.

S3.8.1 Centered rotating frame and slow–fast amplitude equations

Remove the mean bare energy of the two retained states,

$$\bar{E}_3^{(0)} = \frac{E_A^{(0)} + E_B^{(0)}}{2} = \frac{3\omega_a}{2},$$

and expand the state as

$$\begin{aligned} |\psi(t)\rangle = e^{-i\bar{E}_3^{(0)}t} [& c_A(t) |A\rangle + c_B(t) |B\rangle \\ & + c_C(t) |C\rangle + c_D(t) |D\rangle \\ & + c_F(t) |F\rangle + \dots]. \end{aligned}$$

In this centered frame, define

$$\Lambda \equiv \frac{\Sigma}{2}, \quad (\text{S89a})$$

$$\Lambda_F \equiv \Sigma - \frac{\delta_0}{2} = \frac{\omega_0 + 5\omega_a}{2}. \quad (\text{S89b})$$

The amplitudes obey

$$i\dot{c}_A = \frac{\delta_0}{2}c_A + gc_C, \quad (\text{S90a})$$

$$i\dot{c}_B = -\frac{\delta_0}{2}c_B + \sqrt{3}gc_D + 2gc_F, \quad (\text{S90b})$$

$$i\dot{c}_C = -\Lambda c_C + gc_A + \sqrt{2}gc_D, \quad (\text{S90c})$$

$$i\dot{c}_D = \Lambda c_D + \sqrt{2}gc_C + \sqrt{3}gc_B, \quad (\text{S90d})$$

$$i\dot{c}_F = \Lambda_F c_F + 2gc_B. \quad (\text{S90e})$$

The endpoint amplitudes c_A and c_B evolve on the shifted three-photon scale, whereas the intermediate amplitudes rotate at rates of order Λ and Λ_F . The centered frame treats the two retained states symmetrically and avoids assigning the small endpoint splitting to a perturbative denominator.

Introduce the slow and fast vectors

$$\mathbf{p} = \begin{pmatrix} c_A \\ c_B \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} c_C \\ c_D \\ c_F \end{pmatrix}.$$

Equations (S90) become

$$i\dot{\mathbf{p}} = \mathbf{H}_P \mathbf{p} + \mathbf{V}_{PQ} \mathbf{q}, \quad (\text{S91a})$$

$$i\dot{\mathbf{q}} = \mathbf{H}_Q \mathbf{q} + \mathbf{V}_{QP} \mathbf{p}, \quad (\text{S91b})$$

where

$$\mathbf{H}_P = \begin{pmatrix} \delta_0/2 & 0 \\ 0 & -\delta_0/2 \end{pmatrix},$$

$$\mathbf{H}_Q = \begin{pmatrix} -\Lambda & \sqrt{2}g & 0 \\ \sqrt{2}g & \Lambda & 0 \\ 0 & 0 & \Lambda_F \end{pmatrix},$$

and

$$\mathbf{V}_{QP} = \begin{pmatrix} g & 0 \\ 0 & \sqrt{3}g \\ 0 & 2g \end{pmatrix}, \quad \mathbf{V}_{PQ} = \mathbf{V}_{QP}^\dagger.$$

The off-diagonal element inside $\mathbf{H}_Q$ is the counter-rotating propagation between the two virtual states. It must remain in the fast block: deleting it makes $\mathbf{H}_Q$ diagonal and removes the cubic endpoint interaction.

S3.8.2 Systematic slaving expansion and effective metric

The fast equation can be written formally as

$$\begin{aligned} \mathbf{q} = & -\mathbf{H}_Q^{-1}\mathbf{V}_{QP}\mathbf{p} \\ & -i\mathbf{H}_Q^{-2}\mathbf{V}_{QP}\dot{\mathbf{p}} \\ & +\mathbf{H}_Q^{-3}\mathbf{V}_{QP}\ddot{\mathbf{p}} + \dots \end{aligned}$$

Substitution into the slow equation yields

$$\begin{aligned} i\mathbf{M}_{\text{AE}}\dot{\mathbf{p}} = & \mathbf{H}_{\text{AE}}^{\text{raw}}\mathbf{p} \\ & +\mathbf{V}_{PQ}\mathbf{H}_Q^{-3}\mathbf{V}_{QP}\ddot{\mathbf{p}} + \dots \end{aligned}$$

with

$$\mathbf{H}_{\text{AE}}^{\text{raw}} = \mathbf{H}_P - \mathbf{V}_{PQ}\mathbf{H}_Q^{-1}\mathbf{V}_{QP}, \quad (\text{S92a})$$

$$\mathbf{M}_{\text{AE}} = \mathbf{I}_P + \mathbf{V}_{PQ}\mathbf{H}_Q^{-2}\mathbf{V}_{QP}. \quad (\text{S92b})$$

The matrix $\mathbf{M}_{\text{AE}}$ is the slow-manifold metric generated by the norm carried by the slaved amplitudes. Defining the orthonormal slow amplitude $\phi = \mathbf{M}_{\text{AE}}^{1/2}\mathbf{p}$ gives the Hermitian generator

$$\mathbf{H}_{\text{AE}} = \mathbf{M}_{\text{AE}}^{-1/2}\mathbf{H}_{\text{AE}}^{\text{raw}}\mathbf{M}_{\text{AE}}^{-1/2}. \quad (\text{S93})$$

This normalization is the adiabatic-elimination counterpart of the projection metric and the unitary SW reconstruction. Through the second-order diagonal and third-order off-diagonal terms considered here, the metric changes the generator only at fourth and fifth order, respectively. It is nevertheless required for normalized states and observables.

S3.8.3 Explicit elimination and the cubic endpoint coupling

Define $\mathcal{D}_\Lambda = \Lambda^2 + 2g^2$. The inverse fast block is

$$\mathbf{H}_Q^{-1} = \begin{pmatrix} -\Lambda/\mathcal{D}_\Lambda & \sqrt{2}g/\mathcal{D}_\Lambda & 0 \\ \sqrt{2}g/\mathcal{D}_\Lambda & \Lambda/\mathcal{D}_\Lambda & 0 \\ 0 & 0 & 1/\Lambda_F \end{pmatrix}.$$

Equation (S92a) therefore gives

$$\begin{aligned} \mathbf{H}_{\text{AE}}^{\text{raw}} = & \begin{pmatrix} \frac{\delta_0}{2} + \frac{g^2\Lambda}{\mathcal{D}_\Lambda} & -\frac{\sqrt{6}g^3}{\mathcal{D}_\Lambda} \\ -\frac{\sqrt{6}g^3}{\mathcal{D}_\Lambda} & -\frac{\delta_0}{2} - \frac{3g^2\Lambda}{\mathcal{D}_\Lambda} - \frac{4g^2}{\Lambda_F} \end{pmatrix} \\ & + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right)_{\text{diag}} + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right)_{\text{off}}, \end{aligned}$$

where the indicated remainder includes processes through states outside the local eliminated

set. The internal C - D propagation has been resummed in $\mathcal{D}_\Lambda$; expanding it to the controlled orders gives

$$\delta E_A^{\text{AE}} = \frac{g^2}{\Lambda} + \mathcal{O}(g^4/\Lambda^3), \quad (\text{S94a})$$

$$\delta E_B^{\text{AE}} = -\frac{3g^2}{\Lambda} - \frac{4g^2}{\Lambda_F} + \mathcal{O}(g^4/\Lambda^3), \quad (\text{S94b})$$

$$G_3^{\text{AE}} = -\frac{\sqrt{6}g^3}{\Lambda^2} + \mathcal{O}(g^5/\Lambda^4). \quad (\text{S94c})$$

Since

$$\Lambda = \Delta - \frac{\delta_0}{2} = \frac{\Sigma}{2}, \quad \Lambda_F = \Sigma - \frac{\delta_0}{2},$$

the diagonal terms align with the canonical shifts as

$$\frac{g^2}{\Lambda} = \frac{g^2}{\Delta} + \mathcal{O}(g^4/\omega_a^3), \quad (\text{S95a})$$

$$\frac{g^2}{\Lambda_F} = \frac{g^2}{\Sigma} + \mathcal{O}(g^4/\omega_a^3), \quad (\text{S95b})$$

under the resonant scaling $\delta_0 = \mathcal{O}(g^2/\omega_a)$. The effective detuning is consequently

$$\begin{aligned} \delta_3^{\text{AE}} = & \delta_0 + \frac{4g^2\Lambda}{\mathcal{D}_\Lambda} + \frac{4g^2}{\Lambda_F} \\ = & \delta_0 + 4g^2\left(\frac{1}{\Delta} + \frac{1}{\Sigma}\right) + \mathcal{O}(g^4/\omega_a^3). \end{aligned}$$

For the cubic coefficient,

$$\begin{aligned} G_3^{\text{AE}} = & -\frac{4\sqrt{6}g^3}{\Sigma^2} + \mathcal{O}(g^5/\omega_a^4) \\ = & -\frac{\sqrt{6}g^3}{2\omega_a\Delta} + \mathcal{O}(g^5/\omega_a^4). \end{aligned} \quad (\text{S96})$$

The replacement of the centered leading denominator by the canonical model-space denominator is itself unusually accurate. Their difference follows from the identity

$$\frac{4}{\Sigma^2} - \frac{1}{2\omega_a\Delta} = -\frac{\delta_0^2}{2\omega_a\Delta\Sigma^2}. \quad (\text{S97})$$

Under the resonant power counting, the denominator-choice difference in Eq. (S97) contributes only at seventh order. The fifth-order remainder shown in Eq. (S96) instead represents the expansion of $\mathcal{D}_\Lambda$ and other higher virtual paths. At the leading crossing, all third-order forms coincide exactly:

$$G_3^{\text{AE}} = -\frac{\sqrt{6}g^3}{4\omega_a^2} = -\frac{9\sqrt{6}g^3}{4\omega_0^2}.$$

Thus, systematic adiabatic elimination reproduces both the second-order resonance displacement and the third-order avoided-crossing coupling.

S3.8.4 Why the slaving relation must be iterated

The same result can be read directly from the component equations. At leading static order, Eqs. (S90c)–(S90e) give

$$c_C^{[1]} = \frac{g}{\Lambda} c_A, \quad (\text{S98a})$$

$$c_D^{[1]} = -\frac{\sqrt{3}g}{\Lambda} c_B, \quad (\text{S98b})$$

$$c_F^{[1]} = -\frac{2g}{\Lambda_F} c_B. \quad (\text{S98c})$$

Substituting these expressions once into the internal $C \leftrightarrow D$ coupling gives

$$c_C^{[2]} = \frac{g}{\Lambda} c_A - \frac{\sqrt{6}g^2}{\Lambda^2} c_B, \quad (\text{S99a})$$

$$c_D^{[2]} = -\frac{\sqrt{3}g}{\Lambda} c_B - \frac{\sqrt{2}g^2}{\Lambda^2} c_A. \quad (\text{S99b})$$

The first iteration produces only the diagonal shifts. The second iteration creates the cross terms in which the amplitude of one endpoint appears in the virtual state adjacent to the other endpoint. Substitution into the slow equations then yields

$$i\dot{c}_A = \left(\frac{\delta_0}{2} + \frac{g^2}{\Lambda} \right) c_A - \frac{\sqrt{6}g^3}{\Lambda^2} c_B, \quad (\text{S100a})$$

$$i\dot{c}_B = -\frac{\sqrt{6}g^3}{\Lambda^2} c_A \quad (\text{S100b})$$

$$+ \left(-\frac{\delta_0}{2} - \frac{3g^2}{\Lambda} - \frac{4g^2}{\Lambda_F} \right) c_B. \quad (\text{S100c})$$

This calculation makes the principal pitfall especially clear: the prescription $\dot{c}_C = \dot{c}_D = 0$, followed by only one substitution, misses the target interaction even though each eliminated amplitude looks locally correct to first order. A third-order endpoint process requires the slaving expansion to be iterated through the full virtual chain.

S3.8.5 Reconstruction, normalization, and initial slip

The static slaving matrix is

$$\mathbf{S}_{\text{AE}} = -\mathbf{H}_Q^{-1} \mathbf{V}_{QP} = \begin{pmatrix} \frac{g\Lambda}{\mathcal{D}_\Lambda} & -\frac{\sqrt{6}g^2}{\mathcal{D}_\Lambda} \\ -\frac{\sqrt{2}g^2}{\mathcal{D}_\Lambda} & -\frac{\sqrt{3}g\Lambda}{\mathcal{D}_\Lambda} \\ 0 & -\frac{2g}{\Lambda_F} \end{pmatrix}. \quad (\text{S101})$$

Because $\mathbf{H}_Q$ is Hermitian,

$$\mathbf{M}_{\text{AE}} = \mathbf{I}_P + \mathbf{S}_{\text{AE}}^\dagger \mathbf{S}_{\text{AE}}.$$

Using $\mathbf{H}_Q^2 = \mathcal{D}_\Lambda \mathbf{I}$ in the C, D block gives

$$\mathbf{M}_{\text{AE}} = \begin{pmatrix} 1 + \frac{g^2}{\mathcal{D}_\Lambda} & 0 \\ 0 & 1 + \frac{3g^2}{\mathcal{D}_\Lambda} + \frac{4g^2}{\Lambda_F^2} \end{pmatrix}.$$

The normalized embedding of the slow state into the laboratory Hilbert space is therefore

$$\hat{\Omega}_{\text{AE}} = \begin{pmatrix} \mathbf{I}_P \\ \mathbf{S}_{\text{AE}} \end{pmatrix} \mathbf{M}_{\text{AE}}^{-1/2}. \quad (\text{S102})$$

Through second order, its action on the retained basis is

$$|A\rangle_{\text{rec}}^{\text{AE}} = \left(1 - \frac{g^2}{2\Lambda^2} \right) |A\rangle + \frac{g}{\Lambda} |C\rangle - \frac{\sqrt{2}g^2}{\Lambda^2} |D\rangle + \mathcal{O}(g^3/\omega_a^3),$$

and

$$|B\rangle_{\text{rec}}^{\text{AE}} = \left[1 - \frac{1}{2} \left(\frac{3g^2}{\Lambda^2} + \frac{4g^2}{\Lambda_F^2} \right) \right] |B\rangle - \frac{\sqrt{3}g}{\Lambda} |D\rangle - \frac{2g}{\Lambda_F} |F\rangle - \frac{\sqrt{6}g^2}{\Lambda^2} |C\rangle + \mathcal{O}(g^3/\omega_a^3).$$

At the leading crossing, these expressions coincide with the normalized projection, SW, and MSR reconstructed states. Away from the crossing, the centered-frame denominators differ from the endpoint-energy denominators only beyond the displayed resonant order.

For complete second-order reconstruction of $|B\rangle$, the next state on the F -ladder is

$$|G\rangle = |g, 5\rangle, \quad \Lambda_G = \frac{7\omega_a - \omega_0}{2}.$$

One additional iteration gives

$$c_G = \frac{2\sqrt{5}g^2}{\Lambda_F \Lambda_G} c_B + \mathcal{O}(g^3/\omega_a^3).$$

At $\omega_0 = 3\omega_a$, this becomes $\sqrt{5}g^2/(\omega_a \Sigma)$, reproducing the reference reconstruction. Its feedback on the retained Hamiltonian begins beyond third order, which is why $|G\rangle$ was not required in $\mathbf{H}_Q$ above.

The effective observable associated with a laboratory operator $\hat{O}$ is

$$\hat{O}_{\text{AE}} = \hat{\Omega}_{\text{AE}}^\dagger \hat{O} \hat{\Omega}_{\text{AE}}.$$

Thus, the ideal slow-space population $|B\rangle\langle B|$ is not identical to the laboratory bare population after virtual dressing is included.

A bare endpoint preparation is generally not exactly on the slow manifold. For an arbitrary laboratory initial state, write

$$|\psi(0)\rangle = \hat{\Omega}_{\text{AE}} |\phi(0)\rangle + |\chi_f(0)\rangle, \quad \hat{\Omega}_{\text{AE}}^\dagger |\chi_f(0)\rangle = 0.$$

The second term produces a small fast transient or *initial slip*. Dropping it corresponds to assuming preparation of the dressed slow state, not the bare state. Conversely, encoding with $\hat{\Omega}_{\text{AE}}^\dagger$ and retaining the fast residual gives a fair comparison with the full initial-value problem. This is the amplitude-space version of the endpoint corrections encountered in the Magnus and James–Jerke descriptions.

Result of the adiabatic-elimination derivation. The one-step inverse produces the endpoint self-energies, while the cubic coupling requires propagation through the internal fast-sector link. The derivative expansion generates the model-space metric and the normalized Hermitian generator. The technical outputs are Eqs. (S93), (S100), and (S102).

S3.9 Floquet–Sambe derivation

Floquet theory enters the three-photon problem in two related but physically distinct ways. First, at the bare static resonance the interaction-picture Rabi Hamiltonian is periodic because its two carrier frequencies become commensurate. Sambe space then provides an alternative bookkeeping of the same intrinsic virtual path derived above; its Fourier indices are not physical drive quanta. Second, a genuine longitudinal modulation of the qubit transition generates controllable sidebands. A Floquet reduction of the driven laboratory Hamiltonian produces an anisotropic Rabi model whose effective frequencies can satisfy a three-photon condition even when the physical qubit and cavity are only weakly detuned [13]. The three-photon reduction is then a second, quasi-degenerate effective theory built on top of the first Floquet reduction.

S3.9.1 The static resonance as a commensurate quasienergy problem

At the bare crossing $\omega_0 = 3\omega_a$, $\Delta = 2\omega_a$, $\Sigma = 4\omega_a$, define the fundamental interaction-picture frequency $\nu = 2\omega_a = \Delta$, $\Sigma = 2\nu$. The interaction-picture Hamiltonian becomes

$$\begin{aligned} \hat{V}_I(t) = & g\hat{a}^\dagger\hat{\sigma}_-e^{-i\nu t} + g\hat{a}\hat{\sigma}_+e^{i\nu t} \\ & + g\hat{a}^\dagger\hat{\sigma}_+e^{2i\nu t} + g\hat{a}\hat{\sigma}_-e^{-2i\nu t}. \end{aligned}$$

It is periodic with period $T_\nu = \frac{2\pi}{\nu} = \frac{\pi}{\omega_a}$. Using the Fourier convention of the Floquet–Sambe construction in the article,

$$\hat{V}_I(t) = \sum_{m \in \mathbb{Z}} e^{-im\nu t} \hat{V}^{(m)},$$

the nonzero Fourier operators are

$$\hat{V}^{(1)} = g\hat{a}^\dagger\hat{\sigma}_-, \quad \hat{V}^{(-1)} = g\hat{a}\hat{\sigma}_+, \quad (\text{S103a})$$

$$\hat{V}^{(2)} = g\hat{a}\hat{\sigma}_-, \quad \hat{V}^{(-2)} = g\hat{a}^\dagger\hat{\sigma}_+. \quad (\text{S103b})$$

Let $|\mu, m\rangle_S$ denote the physical state $|\mu\rangle$ in Fourier sector m . In the interaction picture, the unperturbed Sambe operator is

$$\hat{Q}_0 = -i\partial_t, \quad \hat{Q}_0 |\mu, m\rangle_S = -m\nu |\mu, m\rangle_S.$$

The resonant Sambe-space projector is

$$\begin{aligned} \hat{\mathcal{P}}_3 = & |A, 0\rangle_S \langle A, 0| \\ & + |B, 0\rangle_S \langle B, 0|, \end{aligned}$$

where $|A\rangle = |e, 0\rangle$ and $|B\rangle = |g, 3\rangle$. Both retained states have zero unperturbed quasienergy. The shortest connected path in Sambe space is

$$\begin{aligned} |A, 0\rangle_S & \xrightarrow{g} |C, 1\rangle_S \xrightarrow{\sqrt{2}g} |D, -1\rangle_S \\ & \xrightarrow{\sqrt{3}g} |B, 0\rangle_S, \end{aligned} \quad (\text{S104})$$

with intermediate quasienergies $\varepsilon_{C,1}^{(0)} = -\nu$, $\varepsilon_{D,-1}^{(0)} = +\nu$. The two denominators therefore have opposite signs. Degenerate perturbation theory in Sambe space gives

$$\begin{aligned} G_3^{\text{S,stat}} &= \frac{g(\sqrt{2}g)(\sqrt{3}g)}{[0 - (-\nu)][0 - (+\nu)]} \\ &= -\frac{\sqrt{6}g^3}{\nu^2} = -\frac{\sqrt{6}g^3}{4\omega_a^2}. \end{aligned} \quad (\text{S105})$$

This is the same invariant coupling obtained from the real-energy projection, SW, MSR,

Magnus, James–Jerke, and adiabatic-elimination constructions.

The diagonal Sambe-space paths also reproduce the complete second-order shift. The state $|A, 0\rangle_S$ couples to $|C, 1\rangle_S$, while $|B, 0\rangle_S$ couples to $|D, -1\rangle_S$ and $|F, -2\rangle_S = |e, 4; -2\rangle_S$. Consequently,

$$\delta\varepsilon_A^{(2)} = \frac{g^2}{\nu}, \quad (\text{S106a})$$

$$\delta\varepsilon_B^{(2)} = -\frac{3g^2}{\nu} - \frac{4g^2}{2\nu}. \quad (\text{S106b})$$

Using $\nu = \Delta$ and $2\nu = \Sigma$, these are exactly

$$\delta E_A^{(2)} = \frac{g^2}{\Delta}, \quad \delta E_B^{(2)} = -\frac{3g^2}{\Delta} - \frac{4g^2}{\Sigma}.$$

Thus the Sambe formulation contains both the cubic avoided-crossing gap and the lower-order displacement of the crossing.

The exact interaction-picture periodicity used above holds at the bare commensurability. Inside the physical resonant window, $\delta_0 = \omega_0 - 3\omega_a = \mathcal{O}(g^2/\omega_a)$, the target harmonic is slowly detuned rather than exactly periodic with the carrier period. One may treat this situation using a generalized multifrequency Sambe construction or, equivalently, retain $\delta_0\hat{\tau}_z/2$ in the quasi-degenerate block. Corrections to Eq. (S105) generated by inserting δ_0 into the fast denominators are fifth order under the resonant power counting. The second-order diagonal shifts must still be included explicitly, leading to the same effective detuning δ_3 derived above.

This static Sambe representation is a frequency-space description of an undriven Hamiltonian. Its Fourier labels organize interaction-picture harmonics and must not be interpreted as absorption or emission of physical drive quanta. The next construction introduces an actual periodic control, for which the Sambe replicas do represent drive sidebands.

S3.9.2 Longitudinal modulation and the Floquet sideband model

Consider a laboratory Hamiltonian with a sinusoidally modulated qubit transition,

$$\hat{H}_d(t) = \Omega_c \hat{a}^\dagger \hat{a} + \frac{1}{2} [\Omega_0 + A \cos(\omega_f t)] \hat{\sigma}_z + \lambda (\hat{a} + \hat{a}^\dagger)(\hat{\sigma}_+ + \hat{\sigma}_-).$$

Here Ω_0 and Ω_c are the physical qubit and cavity frequencies, A and ω_f are the

modulation amplitude and frequency, and λ is the microscopic light–matter coupling. Define $x = \frac{A}{\omega_f}$ and transform with

$$\hat{U}_d(t) = \exp \left\{ -i\Omega_c t \hat{a}^\dagger \hat{a} - \frac{i}{2} [\Omega_0 t + x \sin(\omega_f t)] \hat{\sigma}_z \right\}. \quad (\text{S107})$$

Using

$$e^{ix \sin(\omega_f t)} = \sum_{k \in \mathbb{Z}} J_k(x) e^{ik\omega_f t},$$

the transformed interaction is

$$\begin{aligned} \tilde{H}_d(t) &= \lambda \sum_{k \in \mathbb{Z}} J_k(x) \hat{a} \hat{\sigma}_+ e^{i(\Omega_0 - \Omega_c + k\omega_f)t} \\ &\quad + \lambda \sum_{k \in \mathbb{Z}} J_k(x) \hat{a}^\dagger \hat{\sigma}_+ e^{i(\Omega_0 + \Omega_c + k\omega_f)t} \\ &\quad + \text{H.c.} \end{aligned}$$

The modulation therefore redistributes the original coupling among Floquet sidebands whose amplitudes are Bessel dressed.

Following Ref. [13], retain the co-rotating $k = 0$ sideband and the counter-rotating $k = -1$ sideband. Define

$$\delta = \Omega_0 - \Omega_c, \quad (\text{S108a})$$

$$\Delta_f = \Omega_0 + \Omega_c - \omega_f, \quad (\text{S108b})$$

$$\lambda_2 = \lambda J_0(x), \quad (\text{S108c})$$

$$\lambda_1 = \lambda J_{-1}(x). \quad (\text{S108d})$$

The selected-sideband Hamiltonian is

$$\begin{aligned} \hat{H}_{\text{sb}}(t) &= \lambda_2 \hat{a} \hat{\sigma}_+ e^{i\delta t} + \lambda_1 \hat{a}^\dagger \hat{\sigma}_+ e^{i\Delta_f t} \\ &\quad + \text{H.c.} \end{aligned} \quad (\text{S109})$$

For a populated oscillator manifold $n \leq n_*$, a local diagnostic for this first Floquet truncation is

$$\begin{aligned} \epsilon_{\text{sb}} &= \max_{\substack{0 \leq n \leq n_* \\ k \in \mathbb{Z}}} \left\{ \frac{|\lambda J_k(x)| \sqrt{n+1}}{|\Omega_0 - \Omega_c + k\omega_f|} \quad (k \neq 0), \right. \\ &\quad \left. \frac{|\lambda J_k(x)| \sqrt{n+1}}{|\Omega_0 + \Omega_c + k\omega_f|} \quad (k \neq -1) \right\} \ll 1. \end{aligned}$$

This condition must be supplemented by convergence when additional sidebands are restored. A small Bessel coefficient does not compensate for a sideband denominator that is itself near resonance.

Equation (S109) is the interaction-picture form of the static anisotropic Rabi Hamiltonian

$$\hat{H}_{\text{aR}} = \bar{\omega}_a \hat{a}^\dagger \hat{a} + \frac{\bar{\omega}_0}{2} \hat{\sigma}_z + \lambda_1 \hat{a}^\dagger \hat{\sigma}_+ + \lambda_2 \hat{a} \hat{\sigma}_+ + \text{H.c.}, \quad (\text{S110})$$

where

$$\bar{\omega}_a = \frac{\Delta_f - \delta}{2}, \quad \bar{\omega}_0 = \frac{\Delta_f + \delta}{2}.$$

The effective detuning and sum frequency are therefore

$$\bar{\Delta} = \bar{\omega}_0 - \bar{\omega}_a = \delta, \quad \bar{\Sigma} = \bar{\omega}_0 + \bar{\omega}_a = \Delta_f.$$

The leading three-photon condition of the effective anisotropic model is

$$\bar{\omega}_0 = 3\bar{\omega}_a \iff \Delta_f = 2\delta.$$

Because Δ_f is controlled by ω_f , this condition can be satisfied even when the physical qubit and cavity frequencies obey $|\Omega_0 - \Omega_c| \ll \Omega_0$, rather than the static physical condition $\Omega_0 \simeq 3\Omega_c$. This relocation of the effective frequency hierarchy is the principal advantage of the modulation-based protocol for weakly anharmonic artificial atoms [13].

S3.9.3 Three-photon reduction of the anisotropic Floquet model

The second reduction retains the endpoint states $|A\rangle = |e, 0\rangle$ and $|B\rangle = |g, 3\rangle$ of Eq. (S110). The virtual path is

$$|e, 0\rangle \xrightarrow{\lambda_2^*} |g, 1\rangle \xrightarrow{\sqrt{2}\lambda_1} |e, 2\rangle \xrightarrow{\sqrt{3}\lambda_2^*} |g, 3\rangle.$$

The first and third vertices are co-rotating in the effective model, whereas the middle vertex is the modulation-selected counter-rotating sideband. The second-order endpoint shifts are

$$\delta E_{A,F}^{(2)} = \frac{|\lambda_2|^2}{\delta}, \quad (\text{S111a})$$

$$\delta E_{B,F}^{(2)} = -\frac{3|\lambda_2|^2}{\delta} - \frac{4|\lambda_1|^2}{\Delta_f}. \quad (\text{S111b})$$

The corrected effective detuning is consequently

$$\delta_{3,F} = 2\delta - \Delta_f + \frac{4|\lambda_2|^2}{\delta} + \frac{4|\lambda_1|^2}{\Delta_f} + \mathcal{O}(\lambda^4/\bar{\omega}_a^3).$$

Thus, the drive frequency that satisfies the bare sideband condition $\Delta_f = 2\delta$ must itself be corrected by the virtual level shifts before the cubic gap is compared with the full driven spectrum.

The third-order endpoint coupling is

$$G_{3,F} = -\frac{\sqrt{6}\lambda_1(\lambda_2^*)^2}{2\bar{\omega}_a\delta} + \mathcal{O}(\lambda^5/\bar{\omega}_a^4).$$

At the leading effective resonance, $\delta = 2\bar{\omega}_a$ and $\Delta_f = 4\bar{\omega}_a$, so

$$G_{3,F} = -\frac{\sqrt{6}\lambda_1(\lambda_2^*)^2}{4\bar{\omega}_a^2} + \mathcal{O}(\lambda^5/\bar{\omega}_a^4). \quad (\text{S112})$$

For a real sinusoidal modulation with the phase convention of Eq. (S107),

$$G_{3,F} = -\frac{\sqrt{6}\lambda^3 J_{-1}(x) J_0^2(x)}{4\bar{\omega}_a^2} + \mathcal{O}(\lambda^5/\bar{\omega}_a^4).$$

The avoided-crossing gap is $\Delta_{\text{gap},F}^{(3\gamma)} = 2|G_{3,F}|$.

It is important not to identify the elementary sideband amplitude $\lambda_1 = \lambda J_{-1}(x)$ with the three-photon coupling. The sideband amplitude supplies one vertex of the virtual pathway; the endpoint matrix element contains one counter-rotating and two co-rotating vertices and therefore remains cubic in the microscopic coupling, $G_{3,F} \propto \lambda^3 J_{-1}(x) J_0^2(x)$. The modulation makes the effective frequencies, anisotropy, magnitude, and phase of the vertices tunable. It can remove the need for a large *physical* qubit-cavity detuning and thereby reduce unwanted access to higher levels of a weakly anharmonic artificial atom, but it does not change the perturbative order of the endpoint process.

At $\Delta_f = 2\delta$, Eq. (S109) is periodic with fundamental frequency $|\delta|$. The corresponding Sambe path has the same Fourier-sector pattern as Eq. (S104), with

$$g \rightarrow \lambda_2, \quad \sqrt{2}g \rightarrow \sqrt{2}\lambda_1, \quad \sqrt{3}g \rightarrow \sqrt{3}\lambda_2.$$

The quasienergy denominators are $+\delta$ and $-\delta$, yielding $G_{3,F} = -\frac{\sqrt{6}\lambda_1(\lambda_2^*)^2}{\delta^2}$, which is equivalent to Eq. (S112) because $\delta = 2\bar{\omega}_a$.

S3.9.4 Two-stage reconstruction and Floquet gauge

The driven protocol contains two distinct layers of eliminated motion. The first Floquet

reduction removes the off-resonant modulation sidebands and produces a periodic micromotion operator $\hat{P}_{\text{sb}}(t)$. The second reduction eliminates the virtual states of the anisotropic Rabi model and produces a dressing or wave operator $\hat{\Omega}_{3,\text{F}}$. A complete closed-system prediction therefore has the schematic propagator

$$\hat{U}_{\text{rec}}(t, t_0) \simeq \hat{U}_{\text{d}}(t) \hat{P}_{\text{sb}}(t) \hat{\Omega}_{3,\text{F}} \times e^{-i\hat{H}_{3,\text{F}}(t-t_0)} \hat{\Omega}_{3,\text{F}}^\dagger \hat{P}_{\text{sb}}^\dagger(t_0) \hat{U}_{\text{d}}^\dagger(t_0), \quad (\text{S113})$$

where

$$\hat{H}_{3,\text{F}} = \bar{E}_{3,\text{F}} \hat{I}_3 + \frac{\delta_{3,\text{F}}}{2} \hat{\tau}_z + G_{3,\text{F}} \hat{\tau}_+ + G_{3,\text{F}}^* \hat{\tau}_-. \quad (\text{S114})$$

For a nonunitary embedding, the adjoint pair in Eq. (S113) is replaced by the corresponding encoding and reconstruction maps. Laboratory observables must be transformed through both layers. Comparing the bare two-state evolution directly with continuous-time driven dynamics omits both virtual-state dressing and Floquet micromotion.

Changing the initial drive phase changes the phases of the selected sidebands and hence the phase of $G_{3,\text{F}}$. Changing the Floquet Brillouin zone shifts quasienergy replicas by integer multiples of ω_f . Neither operation changes the physical avoided-crossing gap. Comparisons must therefore use a common drive phase, quasienergy zone, and stroboscopic origin, or else compare gauge-invariant quantities such as the gap and reconstructed laboratory observables.

Result of the Floquet–Sambe derivation. The static Sambe construction and the modulation-assisted construction share the same retained two-state structure after the quasienergy gauge and micromotion are fixed. The driven endpoint interaction contains the complete product of three microscopic sideband vertices. Its numerical test requires a direct simulation of the driven laboratory Hamiltonian; the static benchmark tested in Notebook N05 does not provide that test.

S3.10 Alignment of generators and reconstruction maps

The main text compares the physical pictures supplied by the methods. Here we record only the technical correspondence needed to check the algebra.

S3.10.1 Common invariant content and effective-gauge alignment

For the static Rabi problem, every controlled construction reduces the near-degenerate pair to the form

$$\hat{H}_3^{(m)} = \bar{E}_3^{(m)} \hat{I}_3 + \frac{\delta_3^{(m)}}{2} \hat{\tau}_z + G_3^{(m)} \hat{\tau}_+ + G_3^{(m)*} \hat{\tau}_- + \hat{R}_3^{(m)}, \quad (\text{S115})$$

where m labels the method and $\hat{R}_3^{(m)}$ contains terms beyond the displayed perturbative order. Under the resonant power counting

$$\delta_0 = \omega_0 - 3\omega_a = \mathcal{O}\left(\frac{g^2}{\omega_a}\right), \quad (\text{S116})$$

the aligned coefficients satisfy

$$\delta_3^{(m)} = \delta_0 + 4g^2 \left(\frac{1}{\Delta} + \frac{1}{\Sigma} \right) + \mathcal{O}\left(\frac{g^4}{\omega_a^3}\right), \quad (\text{S117a})$$

$$G_3^{(m)} = e^{i\phi_m} \left[-\frac{\sqrt{6}g^3}{2\omega_a\Delta} \right] + \mathcal{O}\left(\frac{g^5}{\omega_a^4}\right). \quad (\text{S117b})$$

The phase ϕ_m records the retained-basis, transformation, or Floquet-gauge convention. At the leading crossing, the common invariant magnitude is

$$|G_3| = \frac{\sqrt{6}g^3}{4\omega_a^2} = \frac{9\sqrt{6}g^3}{4\omega_0^2}, \quad \Delta_{\text{gap}}^{(3\gamma)} = 2|G_3|.$$

Literal equality of the matrices in Eq. (S115) is not required. A near-identity unitary acting only inside the retained space can redistribute higher-order diagonal and off-diagonal terms without changing the spectrum. Energy-dependent projection operators, nonunitary slow-manifold embeddings, and different choices of micromotion gauge can produce further representational differences. A meaningful comparison therefore aligns the retained basis and phase and then tests the following quantities:

$$\delta_3, \quad |G_3|, \quad 2|G_3|, \quad \rho_{\text{rec}}(t), \quad \langle \hat{\mathcal{O}} \rangle_{\text{rec}}(t).$$

The last two quantities distinguish the physical predictions. Two effective Hamiltonians can have the same avoided-crossing gap while assigning different meanings to their retained amplitudes. They become physically equivalent only after the corresponding encoding and reconstruction maps are included. The agreement of the exact and reconstructed observables in Fig. S1 is the numerical test of this requirement for the reference construction.

S3.10.2 Correspondence of method-specific outputs

The supplemental correspondence is:

Projection. Reduced blocks: Eqs. (S42) and (S43). Reconstruction: Eqs. (S44) and (S46).

Schrieffer–Wolff. Reduced block: Eq. (S51). Reconstruction: Eqs. (S52) and (S54).

Small rotations. Reduced normal form: Eqs. (S63) and (S64). Reconstruction: Eqs. (S65)–(S67).

Magnus. Slow generator: Eq. (S75). Reconstruction: Eqs. (S76) and (S77).

James–Jerke. Filtered retained block: Eq. (S85). Reconstruction: Eqs. (S86)–(S88).

Adiabatic elimination. Raw and normalized generators: Eqs. (S92a) and (S93). Reconstruction: Eqs. (S101)–(S102).

Floquet–Sambe. Retained two-state block: Eq. (S114). Two-stage reconstruction: Eq. (S113).

The raw coefficients produced before the final gauge alignment need not be identical away from the resonance. Terms that differ by the product of the second-order detuning displacement and a nominal third-order coefficient are of fifth order under Eq. (S116). The aligned comparison must therefore use the same resonant power counting before a difference is interpreted as physical.

Further reading. Ma and Law analyze the static large-detuning three-photon resonance and its adiabatic use in Ref. [12]. Multiphoton Rabi oscillations in the ultrastrong-coupling regime are discussed by Garziano *et al.* [14]. A modulation-assisted route that relaxes the static one-third-frequency condition appears in Ref. [13]. For the general perturbative structures used here, Kato and Bravyi *et al.* provide complementary operator-theory and block-diagonalization viewpoints [15, 16].

S4 Liouville-Space, Langevin, and Map Conventions

Open-system reductions are often written in three complementary languages. A master equation evolves density operators, a quantum Langevin equation evolves system operators together with their input noise, and a finite-time map acts directly on the initial state. The same eliminated mode can appear as a block of a Liouvillian, a frequency-dependent susceptibility, or a channel. This section fixes the conventions needed to move between these descriptions and uses a damped bosonic mediator as the common example.

S4.1 Column-stacking vectorization

For an operator $\hat{X} = \sum_{m,n} X_{mn} |m\rangle\langle n|$, we define

$$|\hat{X}\rangle\rangle = \text{vec}(\hat{X})$$

by stacking the columns of its matrix representation. The basic identity is

$$\text{vec}(\hat{A}\hat{X}\hat{B}) = (\hat{B}^\top \otimes \hat{A}) |\hat{X}\rangle\rangle.$$

The transpose is taken in the basis used to define the vectorization. Changing this convention changes the matrix representation of the Liouvillian but not the physical superoperator.

For the convention used throughout the article,

$$\dot{\hat{\rho}} = -i[\hat{H}, \hat{\rho}] + \sum_{\ell} \mathcal{D}[\hat{J}_{\ell}] \hat{\rho},$$

with

$$\mathcal{D}[\hat{J}] \hat{\rho} = \hat{J} \hat{\rho} \hat{J}^\dagger - \frac{1}{2} \{ \hat{J}^\dagger \hat{J}, \hat{\rho} \},$$

the vectorized equation is

$$\frac{d}{dt} |\hat{\rho}\rangle\rangle = \mathbb{L} |\hat{\rho}\rangle\rangle,$$

where

$$\begin{aligned} \mathbb{L} = & -i \left(\hat{I} \otimes \hat{H} - \hat{H}^\top \otimes \hat{I} \right) \\ & + \sum_{\ell} \left[\hat{J}_{\ell}^* \otimes \hat{J}_{\ell} \right. \\ & \left. - \frac{1}{2} \hat{I} \otimes \hat{J}_{\ell}^\dagger \hat{J}_{\ell} - \frac{1}{2} \left(\hat{J}_{\ell}^\dagger \hat{J}_{\ell} \right)^\top \otimes \hat{I} \right]. \end{aligned}$$

If rates are written separately as $\gamma_{\ell} \mathcal{D}[\hat{L}_{\ell}]$, then $\hat{J}_{\ell} = \sqrt{\gamma_{\ell}} \hat{L}_{\ell}$.

S4.2 Trace preservation and steady states

The Hilbert–Schmidt inner product becomes

$$\langle\langle \hat{A}|\hat{B} \rangle\rangle = \text{Tr}(\hat{A}^\dagger \hat{B}).$$

The trace functional is represented by $\langle\langle \hat{I} |$, so trace preservation requires

$$\langle\langle \hat{I} | \mathbb{L} = 0.$$

A stationary state satisfies

$$\mathbb{L}|\hat{\rho}_{\text{ss}}\rangle\rangle = 0, \quad \langle\langle \hat{I} | \hat{\rho}_{\text{ss}} \rangle\rangle = 1.$$

The null vector must additionally be Hermitian and positive semidefinite to represent a physical state.

S4.3 Quantum Langevin equations and a damped mediator

The Langevin description is particularly convenient when the eliminated subsystem is a linear bosonic mode and the observables of interest include response functions, output spectra, or propagating fields. Consider a mode $\hat{a}$ coupled to a slow-system operator $\hat{F}$. In a frame rotating at the relevant slow frequency, take

$$\hat{H} = -\Delta \hat{a}^\dagger \hat{a} + \hat{H}_s + \hat{a}^\dagger \hat{F} + \hat{F}^\dagger \hat{a}, \quad (\text{S118})$$

and let the mode lose energy through one Markovian port at rate κ . The corresponding Heisenberg–Langevin equation is

$$\dot{\hat{a}}(t) = \left(i\Delta - \frac{\kappa}{2}\right) \hat{a}(t) - i\hat{F}(t) - \sqrt{\kappa} \hat{a}_{\text{in}}(t). \quad (\text{S119})$$

For vacuum input,

$$\langle\hat{a}_{\text{in}}(t) \hat{a}_{\text{in}}^\dagger(t')\rangle = \delta(t - t'),$$

and the normally ordered correlation with the operators reversed vanishes. Thermal or squeezed inputs change the noise correlations, but not the linear response derived below.

Equation (S119) can be integrated exactly from an initial time t_0 :

$$\begin{aligned} \hat{a}(t) = & e^{(i\Delta - \kappa/2)(t-t_0)} \hat{a}(t_0) \\ & - i \int_{t_0}^t d\tau e^{(i\Delta - \kappa/2)(t-\tau)} \hat{F}(\tau) \\ & - \sqrt{\kappa} \int_{t_0}^t d\tau e^{(i\Delta - \kappa/2)(t-\tau)} \hat{a}_{\text{in}}(\tau). \end{aligned} \quad (\text{S120})$$

The first line is the initial layer. It decays on the cavity timescale $2/\kappa$. The remaining terms are a retarded response to the slow system and a filtered input noise. After Fourier transforming with $\hat{A}[\nu] = \int dt e^{i\nu t} \hat{A}(t)$, one obtains

$$\hat{a}[\nu] = -i\chi_c(\nu)\hat{F}[\nu] - \sqrt{\kappa}\chi_c(\nu)\hat{a}_{\text{in}}[\nu] + \hat{a}_{\text{hom}}[\nu], \quad (\text{S121})$$

with the retarded cavity susceptibility

$$\chi_c(\nu) = \frac{1}{\kappa/2 - i(\Delta + \nu)}. \quad (\text{S122})$$

The homogeneous term carries the decaying initial condition. Equation (S121) is already an effective description of the mediator: it reconstructs the eliminated amplitude from the slow operator and the input field at every frequency.

S4.3.1 Markov limit and induced open-system dynamics

Suppose that $\hat{F}(t)$ varies over a bandwidth B around $\nu = 0$, with

$$B \ll \sqrt{\Delta^2 + \kappa^2/4}.$$

The mediator must also remain weakly occupied. In a populated slow manifold $\mathcal{M}$, a local condition is

$$\eta_{\mathbb{L}}(\mathcal{M}) = \max_{|\mu\rangle, |\nu\rangle \in \mathcal{M}} \frac{|\langle\nu|\hat{F}|\mu\rangle|}{\sqrt{\Delta^2 + \kappa^2/4}} \ll 1.$$

The induced slow frequencies and decay rates must also remain below the response scale of the eliminated mode. These two requirements distinguish a controlled slow-manifold reduction from the algebraic substitution obtained by setting $\hat{a} = 0$ without a timescale estimate. The susceptibility may then be replaced by $\chi_c(0)$ over the occupied band. Once the initial layer has decayed,

$$\hat{a}(t) \simeq -i\chi_c(0)\hat{F}(t) - \sqrt{\kappa}\chi_c(0)\hat{a}_{\text{in}}(t). \quad (\text{S123})$$

For a transition operator $\hat{F} = g\hat{S}_-$, substitution into the slow equation gives a complex self-energy. Its real and imaginary parts define the familiar frequency shift and decay rate,

$$\delta\omega = \frac{g^2\Delta}{\Delta^2 + \kappa^2/4}, \quad (\text{S124a})$$

$$\Gamma = \frac{g^2\kappa}{\Delta^2 + \kappa^2/4}. \quad (\text{S124b})$$

The reduced density operator consequently obeys

$$\dot{\hat{\rho}}_s = -i \left[\hat{H}_s + \delta\omega \hat{S}_+ \hat{S}_-, \hat{\rho}_s \right] + \Gamma \mathcal{D}[\hat{S}_-] \hat{\rho}_s, \quad (\text{S125})$$

for vacuum input and within the weak-coupling, narrow-band approximation. The same susceptibility therefore produces both the coherent shift and the irreversible channel.

The noise term in Eq. (S123) cannot be omitted. Keeping the damping proportional to Γ while discarding the associated noise would not preserve the operator commutators and would not reproduce the unconditional master equation. In a multiport device each port contributes its own response and noise operator; in a thermal or squeezed bath the same elimination produces the corresponding absorption or phase-sensitive channels.

S4.3.2 Output fields and reconstruction

With the phase convention used in Eq. (S119), the input–output boundary condition is

$$\hat{a}_{\text{out}}(t) = \hat{a}_{\text{in}}(t) + \sqrt{\kappa} \hat{a}(t). \quad (\text{S126})$$

Combining it with Eq. (S121) yields

$$\begin{aligned} \hat{a}_{\text{out}}[\nu] = & [1 - \kappa\chi_c(\nu)] \hat{a}_{\text{in}}[\nu] \\ & - i\sqrt{\kappa} \chi_c(\nu) \hat{F}[\nu] + \sqrt{\kappa} \hat{a}_{\text{hom}}[\nu]. \end{aligned} \quad (\text{S127})$$

The eliminated mode can thus be absent from the reduced state space while remaining visible in a measurable output field. Equation (S127) is the open-system analogue of an observable reconstruction map.

S4.3.3 What the Langevin and Liouvillian routes reveal

For a linear mediator coupled to a Markov bath, the Langevin equation and the microscopic GKLS master equation are two representations of the same model. They give the same susceptibility, shift, damping rate, and output correlations when the frame, bath state, and linewidth convention are matched. Their practical strengths differ:

- Langevin equations expose response functions, noise spectra, input–output fields, and frequency-dependent memory directly;

- Liouvillian projection treats arbitrary density operators, nonlinear slow systems, several dissipative channels, and complete positivity tests more systematically;

- finite-time maps provide the most direct test of a claimed channel, without assuming that its logarithm has GKLS form.

If $\chi_c(\nu)$ varies appreciably across the slow bandwidth, replacing it by $\chi_c(0)$ loses retardation. The correct reduced description then contains a memory kernel or auxiliary modes. Likewise, strong nonlinear coupling, appreciable mediator occupation, or non-Gaussian reservoir states can require a larger retained system or a full Liouvillian calculation. Setting $\dot{\hat{a}} = 0$ is therefore a shorthand for the controlled narrow-band limit of Eq. (S120), not an independent derivation.

S4.4 Finite-time maps and Choi operators

For a time-independent matrix representation,

$$\hat{\mathbb{L}}(t) = e^{t\mathbb{L}}.$$

The physical map Φ can be tested through the Choi operator

$$\hat{C}_\Phi = \sum_{m,n} |m\rangle\langle n| \otimes \Phi(|m\rangle\langle n|).$$

With this unnormalized convention,

$$\Phi \text{ is completely positive} \iff \hat{C}_\Phi \succeq 0,$$

and trace preservation requires

$$\text{Tr}_{\text{out}} \hat{C}_\Phi = \hat{I}_{\text{in}}.$$

For a perturbatively truncated map, the smallest Choi eigenvalue should be reported together with the perturbative order and numerical cutoff. A negative eigenvalue at the first omitted order signals a structural defect of the truncation at that accuracy; it does not imply that the untruncated reduced dynamics is unphysical.

S4.5 Generator-level Kossakowski test

A finite-dimensional, time-local, trace-preserving generator can be expanded in a traceless orthonormal operator basis $\{\hat{F}_\alpha\}$,

$$\begin{aligned} \mathcal{L}\hat{\rho} = & -i[\hat{H}, \hat{\rho}] \\ & + \sum_{\alpha,\beta} C_{\alpha\beta} \left[\hat{F}_\alpha \hat{\rho} \hat{F}_\beta^\dagger - \frac{1}{2} \{ \hat{F}_\beta^\dagger \hat{F}_\alpha, \hat{\rho} \} \right]. \end{aligned}$$

The generator is of GKLS form when the Kossakowski matrix $\mathbf{C}$ is positive semidefinite. This is a generator-level condition. A finite-time CPTP map can exist even when a selected time-local logarithm is not of GKLS form.

S4.6 Sparse implementation

For a Hilbert-space dimension D , the Liouville-space dimension is D^2 . Dense storage therefore scales as D^4 , and dense diagonalization has a worst-case cost of order D^6 . Practical calculations should exploit sparse Kronecker products, symmetry sectors, matrix-free actions of $\mathbb{L}$, Krylov propagation, and direct steady-state solvers. Numerical convergence in Liouville space does not replace convergence of the underlying Hilbert-space or bosonic cutoff.

Further reading. Gardiner and Collett develop the input–output Langevin formalism and its connection with master equations in Ref. [17]. The books by Gardiner and Zoller and by Walls and Milburn give complementary quantum-optics treatments of damping, noise, and output fields [18, 19]. Clerk *et al.* emphasize quantum noise spectra, susceptibilities, and measurement [20]; cavity optomechanics supplies many explicit examples of eliminating a driven, lossy mediator [21]. For density-operator and map viewpoints, see Refs. [22–24].

S5 Companion Computational Modules

The companion notebooks extend selected calculations through symbolic checks, numerical benchmarks, parameter scans, and exercises. Each one begins with a microscopic model and a physical question, derives one primary effective description, reconstructs laboratory quantities, and compares the result with a converged reference. The slower computational route is useful when a compact derivation in the article leaves an implementation choice implicit.

S5.1 Relation between the Article and the Computational Modules

The article and the computational modules use the same conventions for frames, detunings, dissipators, perturbative order, and state

reconstruction. Apparently similar effective Hamiltonians can represent different physical predictions when their encoding maps, micromotion, eliminated amplitudes, or jump operators are not treated consistently.

S5.2 Common Structure of the Computational Modules

Although the applications require different analytical and numerical tools, the computational modules follow a common sequence:

physical question $\longrightarrow$ relevant structure
 $\longrightarrow$ effective model,
 effective model $\longrightarrow$ reconstruction
 $\longrightarrow$ reference,
 reference $\longrightarrow$ validity analysis.

Each module follows six common stages:

1. state the physical task, microscopic model, preparation, frame, and operator conventions;
2. identify the retained and eliminated structures and the local matrix-element-to-gap or timescale criterion controlling the reduction;
3. derive the principal effective interaction and add a complementary viewpoint only when it exposes independent physical information;
4. encode and reconstruct states, observables, eliminated amplitudes, micromotion, controls, or dissipative channels;
5. compare exact, effective, reconstructed, and intentionally incomplete predictions, including symbolic, convergence, and validity checks;
6. provide adaptation guidance, exercises, extensions, and the source data needed to reproduce every quantitative figure.

The computational archive stores numerical data separately from the plotting routines so that figures can be regenerated without repeating every calculation. The implementations use explicit operator constructions and record the truncations, tolerances, and parameters needed to reproduce the results.

Table S1: Complementary roles of the article and its computational materials.

Resource	Principal role
Main text	Physical motivation, retained-space choice, derivational spine, complete effective prediction, experimentally relevant interpretation, selected validation, and the first failure or redesign criterion
Computational module	Detailed analytical calculation, symbolic checks, reference simulation, state and observable reconstruction, validity maps, convergence studies, common failure modes, and extensions of the physical model
This supplement	Organization of the computational materials, suggested learning routes, common numerical conventions, and the relation between calculations, figures, and numerical data

S5.3 Guide to the individual notebooks

Notebook N00 is the release index. The remaining notebooks can be opened in any order because each restates its model and conventions before the first calculation.

N01: Beyond-RWA multiqubit interactions. N01 follows the virtual paths that generate exchange, pair processes, and conditional bosonic squeezing. It compares full Rabi dynamics with RWA and non-RWA effective models. Its failure example shows that an approximation can recover dispersive shifts while deleting a physically allowed anomalous operator sector.

N02: Spin-chain quantum bus. N02 resolves the bus into normal modes and compares dispersive endpoint transfer with a protocol that retains one resonant mode. It studies polarized, thermally occupied, and noisy buses. The main lesson is that a small denominator calls for an enlarged retained space, not a more aggressive elimination.

N03: Floquet-controlled tunnelling. N03 derives the driven frame, the Bessel-renormalized tunnelling, and the micromotion that restores laboratory dynamics. Continuous and stroboscopic trajectories are compared over drive scans. The incomplete calculation uses the stroboscopic generator as if it predicted all intermediate times.

N04: Lossy bosonic mediator. N04 compares the full mediator master equation with Hamiltonian-only and complete effective master equations. The complex susceptibility

is evaluated in Langevin, projection, and effective-operator forms. Initial slip, mediator reconstruction, induced jumps, and the onset of memory are tested separately.

N05: Three-photon Rabi resonance. N05 is the higher-order benchmark used in Sec. S3. It locates the shifted resonance, verifies the cubic avoided-crossing gap, reconstructs virtual populations, and compares exact and two-state dynamics over the perturbative time scale. Bare resonance tuning and omitted endpoint dressing provide two distinct failure tests.

N06: Three-level quantum mixers. N06 derives Raman conversion, Stark shifts, bright and dark states, and the reconstruction of the auxiliary level. It then adds spontaneous scattering. The numerical comparison makes clear why the cross coupling, diagonal shifts, and effective jumps belong to the same reduction.

N07: Langevin–Liouville maps. N07 implements the conventions of Sec. S4. It reconstructs a frequency-dependent susceptibility, vectorizes a Liouvillian, propagates finite-time maps, and evaluates Choi and trace-preservation tests. The examples distinguish structural physicality from quantitative agreement with the microscopic model.

N08: Quantum Rabi regime map. N08 compares near-resonant, dispersive, displaced-oscillator, and large-frequency-ratio descriptions using parity-resolved spectra and subspace fidelities. It also compares bare occupation with detector-defined excitations. Distinct scale separations appear as different useful descriptions, not as successive corrections to one

approximation.

S5.4 Suggested Learning Routes

The computational materials may be studied in the order of the applications or through routes organized around a specific physical or methodological question.

Quantum-optics route. The Raman module introduces virtual transitions, light shifts, and auxiliary-state reconstruction. The beyond-RWA multi-qubit module then shows how counter-rotating paths enlarge the generated operator algebra. The lossy-mediator module extends these ideas to coherent and dissipative interaction engineering.

Many-body and transport route. The spin-chain module develops mode-resolved mediation, finite-size spectral structure, and adaptive retained spaces. The Floquet module then introduces driven transport, micromotion, and the distinction between stroboscopic and continuous-time predictions. Collective extensions of the multi-qubit interaction provide a bridge to spin squeezing and multipartite dynamics.

Open-system route. The article and Supplemental Material derive the spontaneous-scattering extension of the Raman problem; N06 retains its numerical implementation as an exercise. N02 compares Hamiltonian, master-equation, and finite-time-map descriptions for different bus preparations. N04 provides the most detailed study of fast Liouvillian elimination, induced dissipation, and memory, while N07 isolates the Langevin, Liouville-space, and map conventions.

Reconstruction route. The Raman calculation reconstructs an eliminated auxiliary amplitude. The spin-chain module reconstructs the spatial distribution of virtual bus excitations. The lossy-mediator module reconstructs mediator occupation and induced dissipative channels. The Floquet calculation reconstructs laboratory-frame micromotion.

Cross-method route. The dispersive Jaynes–Cummings benchmark establishes the common second-order conventions. The Raman problem compares harmonic, projection, and slow-manifold viewpoints on a simple virtual

transition. The three-photon resonance then provides the higher-order cross-method benchmark.

S5.5 Exercises and Extensions

The exercises are organized according to the type of reasoning they require:

Core calculations derive the principal effective interaction and reproduce the corresponding reference comparison;

Diagnostic studies change the observable, reconstruction, state preparation, or parameter domain used to assess the approximation;

Failure analysis implement an incomplete or inconsistent reduction and identify the first physical quantity that exposes the error;

Model extensions modify the topology, excitation sector, bath state, drive protocol, or noise model;

Research directions investigate a platform-dependent question or a regime not claimed in the article.

The failure-analysis exercises are particularly important. A reduced model may reproduce one energy shift or population curve while giving an incorrect phase, eliminated-sector population, squeezing signal, dissipative channel, or long-time trajectory. The modules therefore use several observables and error measures rather than treating agreement in one quantity as a global validation.

S5.6 Figures and Numerical Data

The quantitative figures accompanying the applications are generated from the same calculations used for the validation analysis. For every panel, the archive records the model and parameters, preparation, Hilbert-space or harmonic truncations, reference calculation, reconstructed effective prediction, plotted observable and error metric, and the numerical data used to draw the figure. The correspondence is tabulated in `FIGURE_DATA_MAP.csv` at the release root.

Schematic illustrations are used only to identify the physical components, spectral pathways, or retained spaces. Quantitative

claims are supported by independently reproducible numerical data rather than by schematic diagrams.

S5.7 Computational Environment and Archival Record

The computational materials accompany the article in executable and rendered forms. The record contains:

1. the computational modules;
2. the numerical data used in the article;
3. the scripts used to generate the figures;
4. the software environment and package versions;
5. the numerical tolerances and convergence settings;
6. release metadata connecting the computational record to the corresponding version of the article.

The notebooks avoid machine-specific paths and record their package versions, truncations, tolerances, and solver settings. The archive includes Conda and pip specifications, a Windows launcher for each notebook, rendered HTML and PDF copies, machine-readable data, and vector and high-resolution figure exports. A versioned release links these files to the corresponding article version.

Further reading. The original QuTiP paper describes the operator and solver framework used by the notebooks [25]. Kluyver *et al.* discuss the notebook format as a way to publish code, explanations, and results together [26]. The numerical checks in these notes follow the scientific rule developed in the article: convergence of the software output is necessary, but the physical approximation must also be tested against its stated claim and control parameter.

S6 Selective Glossary of Recurrent Concepts

This glossary collects terms that recur across several methods and are needed to interpret the core framework. Concepts used only in the

advanced structural review are defined locally there rather than repeated here.

S6.1 Effective Hamiltonian, Generator, and Dynamical Map

An *effective Hamiltonian* generates a reduced unitary description, possibly in a transformed frame or retained subspace. A *generator* is the infinitesimal operator in an equation such as $\dot{\rho} = \mathcal{L}(t)\rho$. A *dynamical map* $\Phi(t, 0)$ propagates the state over a finite interval. These objects are related but not interchangeable: a CPTP map need not possess a time-independent logarithm of GKLS form.

S6.2 Retained Space, Eliminated Sector, and Slow Manifold

A retained or model space is the set of states, modes, bands, or operators represented explicitly in the effective description. Its complement is the eliminated sector. In dynamical reductions, the analogous object is a slow manifold whose fast coordinates are slaved to the retained variables. A near-resonant state should be moved into the retained space rather than represented through a small denominator.

S6.3 Encoding, Dressing, and Reconstruction

Encoding maps the physical initial condition into the effective description. Dressing refers to the admixture generated by a transformation or virtual process. Reconstruction restores the physical state or observable after reduced evolution:

$$\rho_{\text{rec}}(t) = [\mathcal{R}_t \circ \Phi_{\text{eff}}(t, 0) \circ \mathcal{E}_0] [\rho(0)].$$

Using an effective Hamiltonian with bare states or bare observables generally omits part of the approximation.

S6.4 Control Parameter and Local, Collective, and Global Scales

A perturbative control parameter compares the matrix element of an eliminated process with the energy or dynamical gap that suppresses it. A retained-manifold form is

$$\eta_{\mathcal{M}} = \sup_{\substack{\mu \in \mathcal{M} \\ \nu \notin \mathcal{M}}} \frac{|\langle \nu | \hat{V} | \mu \rangle|}{|E_{\mu} - E_{\nu}|}.$$

A global operator norm can be overly conservative in an extended system; local norms, linked-cluster bounds, collective enhancements, and symmetry-sector matrix elements may provide the relevant scale.

S6.5 Leakage, Residual, and Validity Domain

Leakage is population or amplitude outside the retained sector. A residual measures how accurately the reconstructed prediction satisfies the full equation of motion or block-decoupling condition. The validity domain is the set of parameters, states, observables, system sizes, and times for which the declared error and convergence criteria are satisfied.

S6.6 Coarse-Graining and Temporal Filtering

Temporal coarse-graining replaces a rapidly varying object by a filtered one,

$$\overline{X}_{\tau_{\text{cg}}}(t) = \int ds f_{\tau_{\text{cg}}}(s) X(t-s),$$

with a window

$$\tau_{\text{fast}} \ll \tau_{\text{cg}} \ll \tau_{\text{slow}}.$$

Fast motion is unresolved by the selected description, not physically absent. A complete prediction may require a kick or micromotion reconstruction.

S6.7 Secular Terms, the RWA, and the Open-System Secular Approximation

A secular contribution is stationary or slowly varying in the chosen interaction picture and can accumulate over the observation time. The Hamiltonian RWA discards terms oscillating rapidly relative to the retained dynamics. In a weak-coupling open-system derivation, the secular approximation more specifically discards terms oscillating at differences between distinct Bohr frequencies. These operations have related time-averaging logic but different assumptions and consequences.

S6.8 Floquet Mode, Quasienergy, Gauge, and Micromotion

For a periodic Hamiltonian,

$$U(t, 0) = P(t)e^{-iH_F t}, \quad P(t+T) = P(t).$$

Eigenvalues of H_F are quasienergies, defined modulo $\omega = 2\pi/T$. Changing the quasienergy branch or the stroboscopic origin changes the Floquet representation without changing the exact propagator. The periodic operator $P(t)$ contains micromotion.

S6.9 Adiabatic Elimination and Initial Slip

Adiabatic elimination expresses fast variables in terms of slow ones after a separation of dynamical scales. The resulting slaving relation also supplies reconstruction of the eliminated sector. If the physical initial state does not lie on the slow manifold, a short fast transient produces an initial-slip correction before the reduced trajectory becomes accurate.

S6.10 GKLS Generator, Kossakowski Matrix, CPTP Map, and Choi Matrix

A generator is of GKLS form when its dissipative coefficient or Kossakowski matrix is positive semidefinite in a canonical operator basis. Complete positivity is a property of a finite-time map. In finite dimension, a map is completely positive exactly when its Choi matrix is positive semidefinite, and trace preservation is expressed by the appropriate partial trace of the Choi matrix.

S6.11 Memory Kernel, Time-Local Generator, and Markovianity

A memory-kernel equation has the form

$$\dot{\rho}(t) = \int_0^t ds K(t, s) \rho(s),$$

whereas a time-local equation is

$$\dot{\rho}(t) = \mathcal{L}_{\text{loc}}(t) \rho(t).$$

The two can represent the same reduced map under suitable conditions. “Markovian” can refer to a semigroup, CP divisibility, a short-memory approximation, or a stochastic property; the intended definition should be stated.

S6.12 Representation and Gauge Freedom

A representation change is an exact invertible transformation when all states, operators,

and couplings are transformed consistently. Block gauge refers to unitary freedom inside a retained subspace. Floquet gauge refers to quasienergy branches and the stroboscopic origin. Electromagnetic gauge changes the light-matter representation and subsystem partition. These notions express different forms of nonuniqueness and should not be conflated.

S6.13 Interaction Vertex, Virtual Path, and Self-Energy

Write $\hat{H} = \hat{H}_0 + \hat{V}$, and separate $\hat{V}$ into elementary interaction terms. In these notes, a *vertex* means one occurrence of one such term in a perturbative transition. For example, $g\hat{a}\hat{\sigma}_+$ is a vertex that connects two bare basis states when its matrix element is nonzero. A *virtual path* is an ordered sequence of these transitions through intermediate states outside the retained manifold. Its amplitude contains the matrix element of every step and the energy or frequency mismatch of every intermediate state. A *self-energy* is a path that leaves a retained state or sector and returns to it. It can shift the energy and, when the excursion reaches a dissipative or continuum sector, can also produce broadening or decay.

S6.14 Perturbative Order, Asymptotic Control, and Accumulated Error

An order- N approximation retains terms through a specified power of the control parameter. A convergent series approaches a limit as $N \rightarrow \infty$; an asymptotic series can improve only up to an optimal order at fixed parameters. Even when the instantaneous generator error is small, a frequency error can accumulate into a large phase difference over long times.

Further reading. The method-specific references at the ends of Secs. S1–S5 are the best route from these definitions back to complete derivations. For the open-system terms, Refs. [22–24] compare generators, maps, memory, and complete positivity. Floquet gauge and micromotion are reviewed in Refs. [27, 28], while gauge consistency after light-matter truncation is discussed in Ref. [29].

S7 Summary Equations and Diagnostic Sheets

This section collects formulas that are scattered across the derivations. It is meant for retrieval after the assumptions of the relevant method have been read.

S7.1 Exact changes of representation

For $|\tilde{\psi}\rangle = \hat{U}^\dagger(t)|\psi\rangle$,

$$\tilde{\hat{H}}(t) = \hat{U}^\dagger \hat{H} \hat{U} - i\hat{U}^\dagger \dot{\hat{U}}, \quad (\text{S128a})$$

$$\tilde{\hat{\rho}} = \hat{U}^\dagger \hat{\rho} \hat{U}, \quad (\text{S128b})$$

$$\tilde{\hat{O}} = \hat{U}^\dagger \hat{O} \hat{U}, \quad (\text{S128c})$$

$$\tilde{\hat{J}}_\ell = \hat{U}^\dagger \hat{J}_\ell \hat{U}. \quad (\text{S128d})$$

For sequential frames $\hat{U}_{\text{tot}} = \hat{U}_1 \hat{U}_2 \cdots \hat{U}_r$, the physical state is reconstructed by the inverse sequence,

$$\hat{\rho} = \hat{U}_{\text{tot}} \tilde{\hat{\rho}} \hat{U}_{\text{tot}}^\dagger.$$

S7.2 Complete effective prediction

The common object compared throughout the article is

$$\Phi_{\text{pred}}(t, 0) = \mathcal{R}_t \circ \Phi_{\text{eff}}(t, 0) \circ \mathcal{E}_0. \quad (\text{S129})$$

Here $\mathcal{E}_0$ encodes the physical initial state, Φ_{eff} evolves the retained variables, and $\mathcal{R}_t$ restores dressing, eliminated modes, initial layers, or micromotion. At the final endpoint, one may reconstruct the physical state and use the bare observable, or leave the state in effective coordinates and use the corresponding effective observable. Applying both alternatives in the same expectation value counts the same endpoint dressing twice.

S7.3 Recurring second-order benchmarks

For the dispersive Jaynes–Cummings model with $\Delta = \omega_0 - \omega_a$,

$$\hat{H}_{\text{disp}} = \omega_a \hat{a}^\dagger \hat{a} + \frac{\omega_0}{2} \hat{\sigma}_z + \frac{g^2}{\Delta} \left(\hat{a}^\dagger \hat{a} \hat{\sigma}_z + \hat{\sigma}_+ \hat{\sigma}_- \right),$$

and the populated-sector parameter is

$$\eta_{\text{JC}}(n_\star) = \frac{|g| \sqrt{n_\star + 1}}{|\Delta|}.$$

Table S2: The reduced object and the information needed to return to physical observables.

Method	Reduced object	Required reconstruction
Projection	projected resolvent or model-space Hamiltonian	wave or embedding operator
Schrieffer–Wolff	block-diagonal Hamiltonian	inverse unitary dressing of states, observables, and jumps
Magnus	exponential generator	bounded kick or micromotion terms
Adiabatic elimination	slow-manifold generator	slaving relation and initial layer
Small rotations	algebraic normal form	inverse sequence of rotations
James–Jerke	temporally filtered generator or map	boundary terms and filtered micromotion
Floquet	quasienergy block or stroboscopic generator	Floquet gauge and periodic micromotion

For a lossy mediator,

$$\chi_c(\nu) = \frac{1}{\kappa/2 - i(\Delta + \nu)}, \quad (\text{S130a})$$

$$\delta\omega = \frac{g^2\Delta}{\Delta^2 + \kappa^2/4}, \quad (\text{S130b})$$

$$\Gamma = \frac{g^2\kappa}{\Delta^2 + \kappa^2/4}. \quad (\text{S130c})$$

The shift and decay rate are the reactive and dissipative parts of one response. The Markov reduction also requires the slow bandwidth to be small over the scale on which $\chi_c(\nu)$ varies.

S7.4 Method outputs and endpoint data

S7.5 Resonance-aware control parameters

The useful denominator is attached to a matrix element that is actually eliminated. Representative local ratios are

$$\eta_{\mu\nu} = \frac{|V_{\mu\nu}|}{|E_\mu - E_\nu|}, \quad \eta_{\mu\nu}^{(m)} = \frac{|H_{\mu\nu}^{(m)}|}{|E_\mu - E_\nu - m\omega|}.$$

A small denominator is normally repaired by retaining the associated state, mode, or Floquet replica. It is not repaired by raising the order of a nondegenerate expansion.

S7.6 Claim-dependent validation

Spectrum or resonance: compare aligned eigenvalues, gaps, and resonance positions in the declared sector.

Continuous dynamics: compare reconstructed observables over the declared

time interval, including micromotion or initial slip.

State transfer or gate: evaluate the quantum channel or process fidelity, not a single population alone.

Open-system reduction: compare finite-time maps or representative states, check trace and Hermiticity, and test Choi positivity when a CPTP map is claimed.

Block construction: verify the off-diagonal residual and its perturbative scaling.

Numerical evidence: vary every Hilbert-space, Sambe, lattice, and solver cutoff that can affect the stated observable.

When a test fails structurally, change the model before increasing numerical precision: enlarge the retained space near a resonance, restore a missing endpoint map, keep the relevant frequency dependence when memory appears, or transform the dissipative channels with the Hamiltonian.

Further reading. The most direct entry points are the pedagogical open-system notes in Refs. [22–24], the block-diagonalization review in Ref. [16], and the Floquet reviews in Refs. [27, 28]. The method-specific reading notes at the ends of Secs. S1–S5 indicate where to continue from each summary formula.

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